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A Penalty Method for Almost Surely Constrained Stochastic Optimization

Master Thesis

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Abstract

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Introduction

1.1. Problem statement and objective

Mathematical optimization is concerned with problems of the form

$$\min_{x \in \mathcal{X}} f(x),$$

where $f : \mathbb{R}^n \to \mathbb{R} \cup \{\infty\}$ is a function and $\mathcal{X} \subset \operatorname{dom}(f)$ is a set. Typically, the set $\mathcal{X}$ is called the *feasible set*, elements $x \in \mathcal{X}$ are *feasible points*, and f is called the *objective function*, or simply *objective*. Oftentimes, a point $x \in \mathbb{R}^n$ is referred to as a *decision variable*. If $\mathcal{X}$ is nonempty, then the problem is called *feasible*. In that case, if there exists $x^\star \in \mathcal{X}$ such that $f(x^\star) \leq f(x)$ for all $x \in \mathcal{X}$, then $x^\star$ is called *solution*. The value $f(x^\star)$ is called *optimal value* or *minimal value*. In practice, the feasible set $\mathcal{X}$ is often defined implicitly through the use of auxillary functions, which yields to the formulation

$$\begin{aligned} &\min_{x \in \mathbb{R}^n} f(x) \\ &\text{subject to (s. t.)} \quad g(x) \leq 0 \\ &\qquad\qquad\qquad\quad\; h(x) = 0 \end{aligned}$$

for functions $g : \mathbb{R}^n \to \mathbb{R}^m$, $h : \mathbb{R}^n \to \mathbb{R}^\ell$. The feasible set is then given by $\mathcal{X} = \{\, x \in \mathbb{R}^n \mid g(x) \leq 0 \text{ and } h(x) = 0 \,\}$. An important special case of optimization problems are *convex optimization* problems, in which the objective f and the map g are convex functions, and h is affine. Convex problems have (among other things) the highly desirable property that every local minimum is a global minimum, which makes optimization algorithms that only use local information (like gradients) work effectively.

In this work, we will condsider convex problems that are subject to randomness. Such *stochastic optimization* problems arise in many applications (TODO: add refs). The problems we will analyze take the following general form

$$\begin{aligned} &\min_{x\in\mathbb{R}^n} \{ f(x) := \mathbb{E}(F_\zeta(x)) + r(x) \} \\ &\text{s. t.} \quad A(\xi)x - b(\xi) \leq 0 \quad \text{almost surely (a. s.)}, \end{aligned} \tag{P}$$

where ζ and ξ are a random variables that capture uncertainty in the objective and the constraints. The objective f is composed of two functions: The expectation functional $\mathbb{E}(F_\zeta)\colon \mathbb{R}^n \to \mathbb{R}$, which we assume to be smooth, and the potentially nonsmooth function $r\colon \mathbb{R}^n \to \mathbb{R} \cup \{\infty\}$, which we will refer to as the *regularizer*. The constraints are affine inequalities with matrices $A(z) \in \mathbb{R}^{m\times n}$ and $b(z) \in \mathbb{R}^m$ for $z \in \mathbb{R}^p$.

Examples of these problems appear in numerous areas of applied mathematics. One such domain is optimal control, where the randomness often arises from some continuous uncertainty in some variables, which leads to an infinite amount of constraints. For example, such uncertainty could come from an unknown future demand that is subject to gaussian noise (see **??**). In that case, optimization algorithms for solving (P) need a way to deal with the constraints one-by-one or in batches, as a simlutaneous treatment of all constraints, like in the classical projected gradient method [1], would be impossible to implement. This holds true even if the number of constraints is not infinite but merely very large, as is the case in modern machine learning, where the random variables ζ and ξ model data points in a data set of size $N \in \mathbb{N}$, and the decision variable x represents parameters of some statistical model. Problem (P) then takes on the specific form

$$\begin{aligned} &\min_{x\in\mathbb{R}^n} \left\{ f(x) = \frac{1}{N}\sum_{i=1}^{N} F_i(x) + r(x) \right\} \\ &\text{s. t.} \quad A_i x - b_i \leq 0 \quad \text{for all } i \in \{1, \dots, N\}. \end{aligned}$$

Example problems that can be formulated in this way are support vector machines and logistic regression with fairness constraints (TODO).

A classical approach to solve (P) is stochastic subgradient descent (SGD) [2]: We start from an initial point $x_0 \in \mathbb{R}^n$. In iteration $k \in \mathbb{N}$, we pick a *step size* $\eta_k \in (0, \infty)$ and a *stochastic subgradient* g_k of f at x_k. Then we set

$$x_{k+1} = \Pi_{\mathcal{X}}(x_k - \eta_k g_k)$$

and repeat. The map $\Pi_{\mathcal{X}} \colon \mathbb{R}^n \to \mathbb{R}^n$ is the *projection map onto* $\mathcal{X}$, which is defined as $\Pi_{\mathcal{X}}(x) := \arg\min_{y \in \mathcal{X}} \|x - y\|$, and ensures that the iterates stay in the feasible set $\mathcal{X}$. If $x^\star \in \mathcal{X}$ solves (P), then one can show that, under suitable choice of step sizes, $\|x_k - x^\star\|^2$ converges to the solution of (P) with rate $\mathcal{O}(1/k)$ for *strongly convex* objectives (TODO: add ref for strongly convex). However, Nemirovski et al. [1] showed that this convergence is highly dependent on knowing the strong convexity constant, and proposed a more robust version of the algorithm that utilizes suitable averages of the original iterates. They proved that the resulting iterates $(\bar{x}_k)_{k \in \mathbb{N}}$ yield convergence of the function values $f(\bar{x}_k)$ to that of $f(x^\star)$ with rate $\mathcal{O}(1/\sqrt{k})$ for objectives f which need only be convex. Unfortunately, we can not apply SGD to our problem, because the complexity of $\mathcal{X}$ makes computing the projection $\Pi_{\mathcal{X}}(x_k - \eta_K g_k)$ infeasible in our case.

A classic idea to deal with complex feasibility sets is to use *penalty functions*. In this approach the constrained problem (P) gets approximated by an unconstrained problem, by introducing a convex function $\pi \colon \mathbb{R}^n \to \mathbb{R}$ that penalizes points that are not feasible: $\pi(x) > 0$ for infeasible x and $\pi(x) = 0$ for feasible x. The resulting approximation to (P) then takes the form

$$\min_{x \in \mathbb{R}^n} \left\{ f_\gamma(x) := f(x) + \gamma \pi(x) \right\}, \tag{1.1}$$

where $\gamma \in (0, \infty)$ is a constant that is used to control the influence of the penalty function π on the objective. For this unconstrained problem, the projection map is simply the identiy map, so one can easily apply SGD to solve (1.1). The larger γ, the closer the solution to (1.1) is to being feasible. Under suitable choice for π, one can show that in the limit $\gamma \to \infty$ the sequence of solutions $(x_\gamma^\star)_{\gamma \in (0,\infty)}$ to (1.1) converges to the solution $x^\star$ of (P): $\lim_{\gamma \to \infty} x_\gamma^\star = x^\star$. For certain choices of π, one can even show that there exists some finite $\gamma \in (0, \infty)$, such that $x_\gamma^\star = x^\star$. Such penalties are called *exact penalties*. A standard example is the Hinge penalty π_{ℓ_1}, defined by $\pi_{\ell_1}(x) := \mathbb{E}(\|(A(\xi)x - b(\xi))_+\|_1)$, where $\|\cdot\|_1$ is the ℓ_1-norm, and $(y)_+$ applies $\mathbb{R} \ni t \mapsto \max(t, 0)$ to every element of $y \in \mathbb{R}^m$. While the defining property of exact penalties is very desirable, they, like the Hinge penalty, all suffer from necessarily being nonsmooth, which is known to slow down stochastic subgradient descent. On the other hand, smooth penalties like the squared hinge penalty $\pi_{\ell_2}(x) := \mathbb{E}(\|(A(\xi)x - b(\xi))_+\|_2^2)$ with $\|\cdot\|_2$ the ℓ_2-norm, often need γ to grow very large to get solutions that are reasonably close to the feasible set. This has the unfortunate side effect that the gradient norm $\|\nabla f_\gamma\|_2$ can grow very large, which makes stochastic gradient descent iterates often very unstable in practice. Additionally, one is forced to use step sizes that decay to 0 quickly to counter the large gradient norms, which slows down convergence.

A solution to the drawbacks of classical penalty methods was introduced by Nedić et al. [3], where the authors analyzed problems similar to ours of the form

$$\begin{aligned} &\min_{x\in\mathbb{R}^n} f(x) \\ &\text{s. t.} \quad a_i^\top x - b_i \leq 0 \quad \text{for all } i \in \{1, \dots, m\}. \end{aligned} \tag{1.2}$$

Instead of a fixed penalty function π however, the authors introduced a sequence of smooth *inexact* penalties $(\pi_k^{\text{hub}})_{k\in\mathbb{N}}$, defined as follows: Let $(\delta_k)_{k\in\mathbb{N}}$ be a sequence of positive real numbers. For $k \in \mathbb{N}$, we define

$$\pi_k^{\text{hub}}(x) := \frac{1}{m}\sum_{i=1}^{m} h_k^{\text{hub}}(x; a_i, b_i),$$

where

$$h_k^{\text{hub}}(x; a, b) := \begin{cases} \frac{\langle a,x\rangle - b}{\|a\|} & \text{if } \langle a, x\rangle - b > \delta_k, \\ \frac{(\langle a,x\rangle - b + \delta_k)^2}{4\delta_k\|a\|} & \text{if } -\delta_k \leq \langle a, x\rangle - b \leq \delta_k, \\ 0, & \text{if } \langle a, x\rangle - b < -\delta_k, \end{cases}$$

for $x, a \in \mathbb{R}^n$, and $b \in \mathbb{R}$. The authors then considered the sequence of unconstrained problems

$$\min_{x\in\mathbb{R}^n} \left\{ f_k(x) := f(x) + \gamma_k \pi_k^{\text{hub}}(x) \right\}, \tag{1.3}$$

for $k \in \mathbb{N}$ and $\gamma_k \in (0, \infty)$. Whereas before we required $\pi(x) = 0$ for all feasible points, the inexact penalties only satisfy $\pi_k^{\text{hub}}(x) \geq 0$ for feasible $x \in \mathbb{R}^n$. The crucial properties of the particular penalty sequence that the authors introduced were that the sequence a) majorizes the hinge penalty, $\pi_k^{\text{hub}}(x) \geq \pi_{\ell_1}(x)$ for all $x \in \mathbb{R}^n$, b) converges to π_{ℓ_1} pointwis, $\lim_{k\to\infty} \pi_k^{\text{hub}}(x) = \pi_{\ell_1}(x)$ for all $x \in \mathbb{R}^n$, and c) has uniformly bounded gradients, $\sup_{k\in\mathbb{N}} \sup_{x\in\mathbb{R}^n} \left\|\nabla \pi_k^{\text{hub}}(x)\right\|_2 < \infty$. By carefully choosing the sequence of parameters that control the convergence to π_{ℓ_1}, denoted by $(\delta_k)_{k\in\mathbb{N}}$, as well as the sequence $(\gamma_k)_{k\in\mathbb{N}}$, the authors were able to show that there exists a $\gamma_K \in (0, \infty)$ large enough such that the distance-to-feasibility of the solution $x_K^\star$ to the corresponding problem (1.3), is independent of γ_K and only controlled by δ_K. Written in mathematical notation, this means that $\operatorname{dist}(x_K^\star, \mathcal{X}) := \inf_{x\in\mathcal{X}} \left\|x - x_K^\star\right\| = \mathcal{O}(\delta_K)$. Thus, this approach manages to combine the highly desirable properties of smooth unconstrained problems with that of exact penalties. The authors then go on to present an iterative stochastic gradient algorithm (see algorithm 1), which proceeds as follows: Start from an initial point $x_0 \in \mathbb{R}^n$. Then, in iteration $k \in \mathbb{N}$, we compute a subgradient of f at x_k, denoted by $\tilde{\nabla} f(x_k)$. Then we sample a random

Algorithm 1 (Nedić et al. [3])

Require: Initial point $x_0 \in \mathbb{R}^n$, step sizes $(\eta_k)_{k\in\mathbb{N}_0}$, penalty weights $(\gamma_k)_{k\in\mathbb{N}_0}$

```
for k = 0 to K − 1 do
    Uniformly sample random index i ∈ { 1, ..., m }
    g ← ∇̃f(x_k) + γ_k ∇h_k^hub(x_k; a_i, b_i)
    x_{k+1} ← x_k − η_k g
end for
S_K ← Σ_{k=1}^K η_k^{-1}
x̄_K ← S_K^{-1} Σ_{k=1}^K η_k^{-1} x_k
return x̄_K
```

index $i \in \{1, \dots, m\}$ and calculate the gradient $\nabla h_k^{\text{hub}}(x_k; a_i, b_i)$. Finally, update

$$x_{k+1} := x_k - \eta_k(\tilde{\nabla} f(x_k) + \gamma_k \nabla h_k^{\text{hub}}(x_k; a_i, b_i)).$$

After a set amount of $K \in \mathbb{N}$ iterations, one then computes the weighted average

$$\bar{x}_K := \sum_{k=0}^{K} \eta_k^{-1} x_k.$$

For strongly convex objectives (that satisfy certain assumptions on the gradients), the authors show that, for any $\epsilon \in (0, \infty)$, one can choose $(\eta_k)_{k\in\mathbb{N}}, (\gamma_k)_{k\in\mathbb{N}}, (\delta_k)_{k\in\mathbb{N}}$ such that the sequece $\bar{x}_K$ satisfies

$$\operatorname{dist}(\bar{x}_K, \mathcal{X}) = O\Big(\frac{\log^{\epsilon} K}{K}\Big) \text{ and } |f(\bar{x}_K) - f(x^\star)| = O\Big(\frac{\log^{2\epsilon} K}{K}\Big).$$

Note that this asymptotic rate is essentially as good as it gets, as we have seen earlier that the projected gradient algorithm achieves the rate $O(1/K)$.

In this work, we aim to build on the incremental gradient method (algorithm 1). First, we will extend the method to the more general situation of (P). Namely, our version of (1.3) has the form

$$\min_{x\in\mathbb{R}^n} \left\{ f_k(x) := \mathbb{E}(F_\zeta(x)) + r(x) + \gamma_k \pi_k(x) \right\},$$

for $k \in \mathbb{N}$, where, as before, $\gamma_k \in (0, \infty)$. We keep the penalties $\pi_k(x)$ more generic, but assume that there exist smooth real-valued functions $(h_k)_{k\in\mathbb{N}}$ defined on $\mathbb{R}^m$, such that

$$\pi_k(x) = \mathbb{E}(h_k(x_k; A(\xi), b(\xi))) \text{ and } \pi_k(x) \downarrow_{k\to\infty} \pi_{\ell_1}(x)$$

for all $x \in \mathbb{R}^n$. This more general treatment allows for more flexibility in the design

of the method, and includes the softplus penalty introduced in [4].

As opposed to the setting of algorithm 1, we may not be able to calculate the gradient of our objective, because r may be nonsmooth or because calculating the gradient of the expectation functional, $\mathbb{E}(F_\zeta)$, may be infeasible. Say, for example, because the distribution of ζ is unknown or, in the case of large-scale machine learning, because the size of the dataset is too large to feasibly compute the full gradient of $\mathbb{E}(F_\zeta(x))$ (which would require the evaluation of a very large sum) for multiple iterations. Nedić et al. deal with nonsmoothness by using *subgradients* instead of gradients, which are also defined at nondifferentiable points. However, in our approach we choose to instead use a *proximal operator* [5] to deal with nonsmooth objectives: For $\eta \in (0, \infty)$ and $r : \mathbb{R}^n \to \mathbb{R} \cup \{\infty\}$, the proximal operator $\mathrm{prox}_{\eta r} : \mathbb{R}^n \to \mathbb{R}^n$ is defined as

$$\mathrm{prox}_{\eta r}(x) := \arg\min_{u \in \mathbb{R}^n} \left\{ r(u) + \frac{1}{2\eta} \|u - x\|_2^2 \right\}.$$

Under mild conditions on r, the proximal operator is well-defined and always yields a point in the domain of f. The proximal operator allows us to only work with gradients of the differentiable terms in our objective. To calculate these gradients for objectives involving intractable expectation functionals/large sums, we work with *stochastic gradients* instead of regular (full) gradients. A stochastic gradient of a differentiable function f at some point $x \in \mathbb{R}^n$ is any random variable g such that $\mathbb{E}(g) = \nabla f(x)$. In our case, we can calculate a stochastic gradient of $\mathbb{E}(F_\zeta) + \gamma_k \pi_k$ at $x \in \mathbb{R}^n$, based on a sample (s, z) from the joint distribution of ζ and ξ, as

$$\nabla F_s(x) + \gamma_k \nabla h_k(x; A(z), b(z)).$$

To deal with the variance introduced by using a stochastic gradient instead of the full gradient, we will use a minibatch of samples and average over the resulting stochastic gradients to get a variance-reduced estimate of $\mathbb{E}(\nabla F_\zeta(x)) + \gamma_k \nabla \pi_k(x)$. The pseudo-code for the full algorithm is presented in algorithm 2.

Our objective is to analyze algorithm 2 theoretically and support our theoretical findings with numerical examples. TODO: Mention here the outline. Difficulties: No bounded gradient assumption for f, infinite constraints, non-feasibility of iterates.

Algorithm 2 Stochastic Inexact Penalty Method

Require: Initial point $x_0 \in \mathbb{R}^n$, step sizes $(\eta_k)_{k\in\mathbb{N}_0}$, penalty weights $(\gamma_k)_{k\in\mathbb{N}_0}$, penalty functions $(h_k)_{k\in\mathbb{N}_0}$, iterate weights $(w_k)_{k\in\mathbb{N}}$

```
for k = 0 to K − 1 do
    Sample (ζ_k, ξ_k) from their joint distribution
    g_k ← ∇F_{ζ_k}(x_k) + γ_k ∇h_k(x_k; A(ξ_k), b(ξ_k))
    x_{k+1} ← prox_{η_k r}(x_k − η_k g_k)
end for
S_K ← Σ_{k=1}^{K} w_k
x̄_K ← S_K^{-1} Σ_{k=1}^{K} w_k x_k
return x̄_K
```

1.2. Contributions

1.3. Related literature

1.4. Example applications

1.5. Notation

Since we only ever work with functions, which are proper, closed, and convex, the subdifferentials are nonempty and we may always select a subgradient at a point $x \in \mathbb{R}^n$, which we will generically denote by $\tilde{\nabla} f(x) \in \partial f(x)$.

2

Theory Background

Throughout, we let $\|\cdot\|$ denote the standard Euclidian norm and $\langle\cdot,\cdot\rangle$ the standard inner product on $\mathbb{R}^d$.

2.1. Probability theory

2.1.1. Measure and integration

Definition 2.1. Let Ω be a set and let 2^Ω denote its power set. A subset $\mathcal{F} \subset 2^\Omega$ is called a σ-**algebra over** Ω if it satisfies the following three conditions:

1. $\emptyset \in \mathcal{F}$.
2. If $A, B \in \mathcal{F}$, then $B \backslash A \in \mathcal{F}$.
3. For any countable sequence $A_1, A_2, \dots \in \mathcal{F}$, we have $\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$.

If $\mathcal{F}$ is a σ-algebra over Ω, then the tuple $(\Omega, \mathcal{F})$ is called a **measurable space**. For any subset $\mathcal{G} \subset 2^\Omega$, we define the σ-**algebra generated by** $\mathcal{G}$ as the intersection over all σ-algebras that contain $\mathcal{G}$ as an element, and we denote this σ-algebra by $\sigma(\mathcal{G})$.

Example 2.2. An important example of a σ-algebra over $\mathbb{R}^d$ is the **Borel** σ-**algebra** $\mathcal{B}(\mathbb{R}^d)$, which is defined to be the σ-algebra generated by the subset of all open sets on $\mathbb{R}^d$. Functions that are $\mathcal{B}(\mathbb{R}^d)$-measurable are called **Borel measurable**.

Unless otherwise stated, we will always consider $\mathbb{R}^d$ to be endowed with the Borel σ-algebra.

Definition 2.3. Let $(\Omega, \mathcal{F})$ and $(E, \mathcal{G})$ be measurable spaces. A map $f\colon \Omega \to E$ is called $\mathcal{F}, \mathcal{G}$**-measurable** if $f^{-1}(G) := \{\omega \in \Omega \mid f(\omega) \in G\} \in \mathcal{F}$ for all $G \in \mathcal{G}$. We may say f is $\mathcal{F}$**-measurable** or simply **measurable** if one or both of the σ-algebras are either clear from the context or not relevant.

Definition 2.4. Let $(\Omega, \mathcal{F})$ be a measurable space. A map $\mu : \mathcal{F} \to [0, \infty]$ is called a **measure on** $(\Omega, \mathcal{F})$ if it satisfies the following two conditions:

1. $\mu(\emptyset) = 0$.
2. For any countable sequence $A_1, A_2, \dots \in \mathcal{F}$, we have $\mu(\bigcup_{n=1}^{\infty} A_n) = \sum_{n=1}^{\infty} \mu(A_n)$.

If μ is a measure on $(\Omega, \mathcal{F})$, then the triplet $(\Omega, \mathcal{F}, \mu)$ is called a **measure space**.

Definition 2.5. Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and let $f\colon \Omega \to \mathbb{R}$ be measurable. We define the (μ-)**integral** of f, denoted $\int f \, \mathrm{d}\mu$, in three steps:

1. If $f(\omega) = \sum_{i=1}^{n} c_i 1_{A_i}(\omega)$ for some $n \in \mathbb{N}$, $c_1, \dots, c_n \in (0, \infty)$, and disjoint measurable sets $A_1, \dots, A_n \in \mathcal{F}$, then we define

$$\int f \, \mathrm{d}\mu := \sum_{i=1}^{n} c_i \mu(A_i).$$

   In this case f is called a **simple function**. The set of all simple functions on Ω is denoted by $\mathcal{S}(\Omega)$.

2. If f is nonnegative, i. e. $f(\omega) \geq 0$ of all $\omega \in \Omega$, then we define

$$\int f \, \mathrm{d}\mu := \sup_{g \in \mathcal{S}(\Omega),\, g \leq f} \int g \, \mathrm{d}\mu.$$

3. If f is neither a simple function, nor nonnegative, but $\int |f| \, \mathrm{d}\mu < \infty$, then we define

$$\int f \, \mathrm{d}\mu := \int \max(0, f) \, \mathrm{d}\mu - \int \max(0, -f) \, \mathrm{d}\mu.$$

Otherwise, we say that the (μ-)integral of f does not exist. If any of these three conditions apply to f, we say that f is (μ-)**integrable**.

Apart from notation used above, we will also sometimes use

$$\int f(x) \, \mu(\mathrm{d}x)$$

to denote the μ-integral of f.

Proposition 2.6. Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and let $f\colon \Omega \to \mathbb{R}$ and $g\colon \Omega \to \mathbb{R}$ be integrable. Then, the following hold:

(i) $\int f + g \,\mathrm{d}\mu = \int f \,\mathrm{d}\mu + \int g \,\mathrm{d}\mu$.

(ii) $\int cf \,\mathrm{d}\mu = c \int f \,\mathrm{d}\mu$ for all $c \in \mathbb{R}$.

(iii) If $f \leq g$, then $\int f \,\mathrm{d}\mu \leq \int g \,\mathrm{d}\mu$. If additionally $f < g$ on some set $A \in \mathcal{F}$ with $\mu(A) > 0$, then $\int f \,\mathrm{d}\mu < \int g \,\mathrm{d}\mu$.

Proposition 2.7. Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and let $f\colon \Omega \to \mathbb{R}$ and $g\colon \mathbb{R} \to \mathbb{R}$ be integrable. Then, it holds that

$$\int g \circ f \,\mathrm{d}\mu = \int g \,\mathrm{d}\mu^f,$$

where μ^f is the measure defined as

$$\mu^f(B) := \mu(f^{-1}(B)),$$

for all $B \in \mathcal{B}(\mathbb{R})$.

2.1.2. Random variables, expected value, and variance

Definition 2.8. Let $(\Omega, \mathcal{F})$ be a measurable space. We say that a measure $\mathbb{P} : \mathcal{F} \to [0, \infty)$ is a **probability measure** measure on $(\Omega, \mathcal{F})$, if

1. $\mathbb{P}(A) \in [0, 1]$ for all $A \in \mathcal{F}$, and
2. $\mathbb{P}(\Omega) = 1$.

If $\mathbb{P}$ is a probability measure on $(\Omega, \mathcal{F})$, then we call the triplet $(\Omega, \mathcal{F}, \mathbb{P})$ a **probability space**. In this context, elements of $\mathcal{F}$ are called **events**.

Definition 2.9. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. An event $A \in \mathcal{F}$ is said to hold **almost surely** (a. s. for short) if $\mathbb{P}(A) = 1$.

Definition 2.10 (Random variable)**.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A map $X\colon \Omega \to \mathbb{R}^n$ is called a **random variable** on $(\Omega, \mathcal{F}, \mathbb{P})$ if X is measurable. In the case $n > 1$, we may call X a **random vector**. Further, we define the notation $\mathbb{P}(X \in B) := \mathbb{P}(X^{-1}(B))$ for $B \in \mathcal{B}(\mathbb{R}^n)$. We define the **distribution of** X to be the probability measure $\mathbb{P}^X = \mathbb{P} \circ X^{-1}$ on $\mathbb{R}^n$. Finally, we define $\sigma(X) := \sigma(\{\, X^{-1}(B) \mid B \in \mathcal{B}(\mathbb{R}^n) \,\})$ and call this the **σ-algebra generated by** X.

Definition 2.11. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. Two random variables X and Y on $(\Omega, \mathcal{F}, \mathbb{P})$ are called **independent** if $\mathbb{P}(X \in A,\ Y \in B) = \mathbb{P}(X \in A) \cdot \mathbb{P}(Y \in B)$ for all $A, B \in \mathcal{F}$. X and Y are called **identically distributed** if $\mathbb{P}^X = \mathbb{P}^Y$. We use the abbreviation **i. i. d.** as shorthand for "independent and identically distributed".

Definition 2.12. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $X\colon \Omega \to \mathbb{R}$ be a random variable on this space. If X is integrable, we define the **expected value of** X, denoted by $\mathbb{E}(X)$, as

$$\mathbb{E}(X) := \int X \, \mathrm{d}\mathbb{P}.$$

If $X\colon \Omega \to \mathbb{R}^n$ is a random vector with components $X_i\colon \Omega \to \mathbb{R}$ for $i \in \{1, \dots, n\}$, then we say that X **is integrable** if all components are integrable. In that case, we define **the expected value of** X as the vector

$$\mathbb{E}(X) := (\mathbb{E}(X_1), \dots, \mathbb{E}(X_n))^\top.$$

The following three properties will be used multiple times throughout this text without explicit mention. They follow directly from proposition 2.6.

Proposition 2.13. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $X\colon \Omega \to \mathbb{R}^n$ and $Y\colon \Omega \to \mathbb{R}^n$ be integrable random variables. Then, the following hold:

(i) $\mathbb{E}(X + Y) = \mathbb{E}(X) + \mathbb{E}(Y)$.

(ii) $\mathbb{E}(cX) = c\,\mathbb{E}(X)$ for all $c \in \mathbb{R}$.

(iii) If $X \leq Y$, then $\mathbb{E}(X) \leq \mathbb{E}(Y)$. If, additionally, $X(\omega) < Y(\omega)$ for all ω in an event $A \in \mathcal{F}$ with $\mathbb{P}(A) > 0$, then $\mathbb{E}(X) < \mathbb{E}(Y)$.

Proposition 2.14 (Jensen's inequality)**.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $X\colon \Omega \to \mathbb{R}^n$ be a random variable. Then we have

$$\mathbb{E}(X^2) \geq \mathbb{E}(X)^2.$$

In particular: If X^2 is integrable, then X must also be integrable.

Definition 2.15. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $X\colon \Omega \to \mathbb{R}^n$ be a random variable. If X^2 is integrable, we define the **variance of** X, denoted by $\mathrm{Var}(X)$, as $\mathrm{Var}(X) := \mathbb{E}\|X - \mathbb{E}(X)\|^2$.

By proposition 2.7, for any integrable random variable $X : \Omega \to \mathbb{R}$, it holds that $\int X \, \mathrm{d}\mathbb{P} = \int I \, \mathrm{d}\mathbb{P}^X$, where $I\colon \mathbb{R} \to \mathbb{R}$ is the identity operator. It is now easy to see that if two random variables X and Y are identically distributed, then they have the same expected value and variance.

Proposition 2.16. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $X_1, \dots, X_N \colon \Omega \to \mathbb{R}^n$ be $N \in \mathbb{N}$ independent random variables, such that $\|X_i\|^2$ is integrable for all $i \in \{1, \dots, N\}$. Then $\mathrm{Var}(\sum_{i=1}^N a_i X_i) = \sum_{i=1}^N a_i^2 \mathrm{Var}(X_i)$, for any $a_1, \dots, a_N \in \mathbb{R}$. If, additionally, they are all identically distributed, it holds that $\mathrm{Var}(1/N \sum_{i=1}^N X_i) = \mathrm{Var}(X_1)/N$.

Proposition 2.17. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $X \colon \Omega \to \mathbb{R}^n$ be a random variable such that X^2 is integrable. Then, $\mathrm{Var}(X) = \mathbb{E}(X^2) - \mathbb{E}(X)^2$.

2.1.3. Densities

Definition 2.18 (Density)**.** Let $(E, \mathcal{E})$ be a measurable space, and let μ and ν be measures on $(E, \mathcal{E})$. A ν-integrable function $f \colon E \to [0, \infty)$ is called a **ν-density of** μ, if

$$\mu(A) = \int_A f \, \mathrm{d}\nu,$$

for all $A \in \mathcal{E}$. In the special case where $E = \mathbb{R}^n$ and ν is the Lebesgue-measure on $\mathbb{R}^n$, we refer to f as a **Lebesgue-density**, or simply **density**, of μ.

Definition 2.19 (Absolute continuity and equivalence)**.** Let $(\Omega, \mathcal{F})$ be a measurable space and let μ and ν be measures on $(\Omega, \mathcal{F})$. We say that μ is **absolutely continuous with respect to** ν, if

$$\nu(A) = 0 \quad \text{implies} \quad \mu(A) = 0,$$

for all $A \in \mathcal{F}$. If μ is absolutely continuous with respect to ν, then we use the notation $\mu \ll \nu$. We say that μ and ν are **equivalent**, if $\mu \ll \nu$ and $\nu \ll \mu$. If μ and ν are equivalent, then we use the notation $\mu \sim \nu$.

Proposition 2.20 (Radon-Nikodym)**.** Let $(\Omega, \mathcal{F})$ be a measurable space and let μ and ν be measures on $(\Omega, \mathcal{F})$. Then, $\mu \ll \nu$ if and only if there exists a ν-density of μ. If μ and ν are equivalent and f is a ν-density of μ, then $f > 0$ ν-almost everywhere and there exists a μ-density g of ν, such that $f \cdot g = 1$ ν-almost everywhere.

Proposition 2.21. Let $(E, \mathcal{E})$ be a measurable space and let μ, ν and τ be measures on $(E, \mathcal{E})$. Suppose that μ has a ν-density f and ν has a τ-density g. Then, the function $h := f \cdot g$ is a τ-density of μ.

If f is a ν-density of μ, then f is often referred to as a **Radon-Nikodym derivative of** μ **with respect to** ν. Commonly used notations are

$$f = \frac{\mathrm{d}\mu}{\mathrm{d}\nu}$$

and

$$\mathrm{d}\mu = f\,\mathrm{d}\nu.$$

Putting the two together, we formally have

$$\mathrm{d}\mu = \frac{\mathrm{d}\mu}{\mathrm{d}\nu}\mathrm{d}\nu.$$

This notation suggest that, if $\mu \ll \nu$ and $\nu \ll \tau$ for some other measure τ, then we have

$$\mathrm{d}\mu = \frac{\mathrm{d}\mu}{\mathrm{d}\nu}\mathrm{d}\nu = \frac{\mathrm{d}\mu}{\mathrm{d}\nu}\frac{\mathrm{d}\nu}{\mathrm{d}\tau}\mathrm{d}\tau,$$

and hence $\mu \ll \tau$ with density $\frac{\mathrm{d}\mu}{\mathrm{d}\nu}\frac{\mathrm{d}\nu}{\mathrm{d}\tau}$, which is consistent with the above proposition.

We now move to back to the setting of probability spaces, where the concept of densities are of great help to work with random variables.

Definition 2.22 (Density of a random variable)**.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $X\colon \Omega \to \mathbb{R}^n$ be a random variable. Further, let μ be a measure on $(\Omega, \mathcal{F})$. A function $f\colon \mathbb{R} \to [0, \infty)$ is called a **μ-density of** X, if f is a μ-density of $\mathbb{P}^X$. That is,

$$\mathbb{P}(X \in B) = \int_B f\,\mathrm{d}\mu,$$

for all $B \in \mathcal{B}(\mathbb{R}^n)$.

The following is an important special case of proposition 2.21.

Proposition 2.23. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $X\colon \Omega \to \mathbb{R}^n$ be a random variable. Further, let μ be a measure on $(\Omega, \mathcal{F})$ and suppose that X has a μ-density f such that $\int |x| f(x)\,\mathrm{d}x < \infty$. Then, X is integrable and

$$\mathbb{E}(X) = \int x f(x)\,\mu(dx).$$

In particular, if μ is the Lebesgue-measure on $\mathbb{R}^n$, it holds that

$$\mathbb{E}(X) = \int x f(x)\,\mathrm{d}x.$$

If μ is instead the counting measure on a discrete subset $S \subset \mathbb{R}^n$, then

$$f(x) = \mathbb{P}(X = x),$$

for all $x \in S$, and

$$\mathbb{E}(X) = \sum_{x \in S} x\, \mathbb{P}(X = x).$$

2.1.4. Conditioning

Definition 2.24 (Conditional expectation)**.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $X \colon \Omega \to \mathbb{R}^n$ be a random variable. Further, let $\mathcal{C} \subset \mathcal{F}$ be a σ-algebra. We call a random variable $Z \colon \Omega \to \mathbb{R}^n$ a **conditional expectation of** X **given** $\mathcal{C}$, if

1. Z is $\mathcal{C}$-measurable, and
2. for all $C \in \mathcal{C}$, it holds that

$$\int_C Z \, \mathrm{d}\mathbb{P} = \int_C X \, \mathrm{d}\mathbb{P}.$$

If Z is a conditional expectation of X given $\mathcal{C}$ then we use the notation $\mathbb{E}(X \mid \mathcal{C}) := Z$.

If $Y \colon \Omega \to \mathbb{R}^m$ is a random variable such that $\mathcal{C} = \sigma(Y)$, then we use the notation $\mathbb{E}(X \mid Y) := \mathbb{E}(X \mid \sigma(Y))$. In that case, we call $\mathbb{E}(X \mid Y)$ the **conditional expectation of** X **given** Y. Further, for $\omega \in \Omega$ with $Y(\omega) = y \in \mathbb{R}^m$, the **conditional expectation of** X **given** $Y = y$, denoted by $\mathbb{E}(X \mid Y = y)$, is defined as $\mathbb{E}(X \mid Y = y) := \mathbb{E}(X \mid Y)(\omega)$.

Note that $E(X \mid Y)$ is not unique. However, if Z_1 and Z_2 are both conditional expectations of X given Y, then we always have $Z_1 = Z_2$ almost surely. For simplicity, we will keep the "almost surely" implicit.

Remark 2.25. If X and Y are integrable random variables on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, then the conditional expectation of X given $Y = y$ can be thought of as the expected value of X on a different probabiliy space $(\Omega, \mathcal{F}, \mathbb{P}_{Y=y})$, where $\mathbb{P}_{Y=y}(A) := \mathbb{P}(A \mid Y = y)$. More precisely, $\mathbb{E}(X \mid Y = y) = \int X \, \mathrm{d}\mathbb{P}_{Y=y}$. It follows that $\mathbb{E}(X \mid Y = \cdot)$ inherits all basic properties of $\mathbb{E}(\cdot)$ from proposition 2.13.

Below, we state some further properties of conditional expectations.

Proposition 2.26 (Properties of $\mathbb{E}(X \mid Y)$)**.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $X \colon \Omega \to \mathbb{R}^n$, $Y \colon \Omega \to \mathbb{R}^m$ be integrable random variables. Further, let $F \colon \mathbb{R}^n \times \Omega \to \mathbb{R}$ such that $\omega \mapsto F(x, \omega)$ is $\mathcal{F}$-measurable for all $x \in \mathbb{R}^d$ and let $G \colon \mathbb{R}^n \to \mathbb{R}$ be Borel measurable. Then, the following hold:

(i) $\mathbb{E}(\mathbb{E}(X \mid Y)) = \mathbb{E}(X)$.

(ii) $\mathbb{E}(X \mid X) = X$.

(iii) If $F(x,\cdot) \leq G(x)$ a. s. for all $x \in \mathbb{R}^d$, it follows that $\mathbb{E}(F(X,\cdot) \mid X) \leq G(X)$.

Definition 2.27 (Conditional variance)**.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $X\colon \Omega \to \mathbb{R}^n$, $Y\colon \Omega \to \mathbb{R}^m$ be integrable random variables. The **conditional variance of X given Y**, denoted by $\mathrm{Var}(X \mid Y)$, is defined as

$$\mathrm{Var}(X \mid Y) := \mathbb{E}\big(||X - \mathbb{E}(X \mid Y)||^2 \mid Y\big).$$

For $y \in \mathbb{R}^d$, the **conditional variance of X given Y = y** is defined as

$$\mathrm{Var}(X \mid Y = y) := \mathbb{E}\big(||X - \mathbb{E}(X \mid Y = y)||^2 \mid Y = y\big).$$

Remark 2.28. Let X and Y be integrable random variables on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Similarly to the conditional expectation of X given $Y = y$, the conditional variance $\mathrm{Var}(X \mid Y = y)$ can be thought of as the variance of X on a different probability space $(\Omega, \mathcal{F}, \mathbb{P}_{Y=y})$ (see remark 2.25). Hence, all basic properties of $\mathrm{Var}(\cdot)$ are inherited by the conditional variance. In particular, if $X_1, \ldots, X_N$ are i. i. d. random variables, it holds that $\mathrm{Var}(X_1 + \cdots + X_N \mid Y) = \mathrm{Var}(X_1 \mid Y)/N$ – a fact that will be used later.

Proposition 2.29. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $X\colon \Omega \to \mathbb{R}^n$ be an integrable random variable. Further, let $F : \mathbb{R}^n \times \Omega \to \mathbb{R}$ such that $\omega \mapsto F(x, \omega)$ is $\mathcal{F}$-measurable for all $x \in \mathbb{R}^d$ and $\mathbb{E}(F(X,\cdot)^2) < \infty$, and let $G : \mathbb{R}^n \to \mathbb{R}$ be Borel measurable. Then, if $\mathrm{Var}(F(x,\cdot)) \leq G(x)$ for all $x \in \mathbb{R}^d$, it holds that

$$\mathrm{Var}(F(X,\cdot) \mid X) \leq G(X).$$

2.1.5. Martingales

Definition 2.30 (Filtration)**.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A sequence of σ-algebras $(\mathcal{F}_n)_{n\in\mathbb{N}_0}$ with $\mathcal{F}_n \subset \mathcal{F}\ \forall n \in \mathbb{N}_0$ is called a **filtration** if $\mathcal{F}_{n-1} \subset \mathcal{F}_n$ for all $n \in \mathbb{N}$. If $(\mathcal{F}_n)_{n\in\mathbb{N}_0}$ is a filtration on $(\Omega, \mathcal{F}, \mathbb{P})$, then we call $(\Omega, \mathcal{F}, \mathbb{P}, (\mathcal{F}_n)_{n\in\mathbb{N}_0})$ a **filtered probability space**.

Example 2.31 (Natural filtration)**.** Given a sequence of random variables $(X_n)_{n\in\mathbb{N}_0}$ on some common probabilty space $(\Omega, \mathcal{F}, \mathbb{P})$, one can define a filtration $(\mathcal{F}_n)_{n\in\mathbb{N}}$ via $\mathcal{F}_n := \sigma(X_0, \ldots, X_n)$ for all $n \in \mathbb{N}_0$. This filtration is referred to as the **natural filtration** of the sequence $(X_n)_{n\in\mathbb{N}_0}$.

Definition 2.32 (Stochastic process)**.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A sequence of random variables $(X_t)_{t\in T}$, where $T \subset \mathbb{R}$ is some set of indeces, is called a **stochastic**

process. Let $(\mathcal{F}_n)_{n\in\mathbb{N}_0}$ be a filtration on $(\Omega, \mathcal{F}, \mathbb{P})$. A stochastic process $X := (X_n)_{n\in\mathbb{N}_0}$ is called **adapted to** $(\mathcal{F}_n)_{n\in\mathbb{N}_0}$, if X_n is $\mathcal{F}_n$-measurable for all $n \in \mathbb{N}_0$.

Clearly, any stochastic process is adapted to their natural filtration.

Definition 2.33 (Martingale)**.** Let $X := (X_n)_{n\in\mathbb{N}}$ be a stochastic process on a filtered probability space $(\Omega, \mathcal{F}, \mathbb{P}, (\mathcal{F}_n)_{n\in\mathbb{N}_0})$. Then, we say that X is a **martingale**, if the following conditions are satisfied:

1. $\mathbb{E}|X_n| < \infty$ for all $n \in \mathbb{N}_0$.
2. X is adapted to $(\mathcal{F}_n)_{n\in\mathbb{N}_0}$, i. e. X_n is $\mathcal{F}_n$-measurable for all $n \in \mathbb{N}_0$.
3. $\mathbb{E}(X_{n+1} \mid X_n) = X_n$ for all $n \in \mathbb{N}_0$.

Proposition 2.34 (Azuma-Hoeffding inequality)**.** Let $(X_k)_{k\in\mathbb{N}_0}$ be a martingale on a filtered probability space $(\Omega, \mathcal{F}, \mathbb{P}, (\mathcal{F}_n)_{n\in\mathbb{N}_0})$ and let $(c_k)_{k\in\mathbb{N}}$ be a sequence such that

$$|X_k - X_{k-1}| \leq c_k$$

almost surely, for all $k \in \mathbb{N}$. Then, for all $K \in \mathbb{N}$ and all $t \in (0, \infty)$, it holds that

$$\mathbb{P}(X_K - X_0 \geq t) \leq \exp\left(\frac{-t^2}{2\sum_{k=1}^{K} c_k^2}\right).$$

2.2. Optimization theory

The contents of this section can be found in [**?**, **?**].

2.2.1. Lipschitz functions

Definition 2.35. Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be differentiable, and $L > 0$. We say that f is **Lipschitz-smooth with constant** L, or simply L**-smooth**, if its gradient is Lipschitz continuous, i. e. there exists a constant $L \in (0, \infty)$ such that

$$\|\nabla f(y) - \nabla f(x)\| \leq L\|y - x\|$$

for all $x, y \in \mathbb{R}^n$.

Proposition 2.36. Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be L-smooth. Then, it holds that

$$f(y) \leq f(x) + \langle \nabla f(x), y - x\rangle + \frac{L}{2}\|y - x\|^2$$

for all $x, y \in \mathbb{R}^n$.

2.2.2. Convex optimization

Definition 2.37 (Convex function)**.** A set $C \subset \mathbb{R}^n$ is called **convex** if, for all $x, y \in C$ and $t \in [0,1]$, it holds that $tx + (1-t)y \in C$. We say that a function $f\colon \mathbb{R}^n \to \mathbb{R}$ is **convex** if

$$f((1-t)x + ty) \leq (1-t)f(x) + tf(y)$$

for all $x, y \in \mathbb{R}^n$ and all $t \in (0,1)$. We say that f is **concave** if $-f$ is convex.

Proposition 2.38. Every convex function on $\mathbb{R}^n$ is continuous.

Definition 2.39 (Subgradient)**.** Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be convex. A vector $g \in \mathbb{R}^n$ is a **subgradient** of f at $x \in \mathbb{R}^n$ if

$$f(y) \geq f(x) + \langle g, y - x \rangle$$

for all $y \in \mathbb{R}^n$. The set of all subgradients of f at x is denoted by $\partial f(x)$ and we call this set the **subdifferential of** f **at** x. If $\partial f(x) \neq \emptyset$, then we call f **subdifferentiable at** x. If f is subdifferentiable for all $x \in \mathbb{R}^n$, we call f **subdifferentiable**. If f is subdifferentiable at $x \in \mathbb{R}^n$, then we use the notation

$$\tilde{\nabla} f(x)$$

to refer to an arbitrary subgradient of f at x.

Proposition 2.40. Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be differentiable. Then f is convex if and only if

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle$$

for all $x, y \in \mathbb{R}^n$. In particular, f is subdifferentiable. Moreover, the subgradients are unique, i. e. $\partial f(x) = \{\nabla f(x)\}$ for all $x \in \mathbb{R}^n$.

Proposition 2.41. Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be convex and subdifferentiable, and let $C \subset \mathbb{R}^n$ be a convex set. Then, a point $x^\star \in \mathbb{R}^n$ minimizes f on C if and only if there exists a subgradient $g \in \partial f(x^\star)$, such that

$$\langle g, x - x^\star \rangle \geq 0$$

for all $x \in C$. In particular, if $C = \mathbb{R}^n$, then $x^\star$ is a (global) minimizer of f if and only if

$$0 \in \partial f(x^\star).$$

Definition 2.42 (Strongly convex function)**.** Let $f : \mathbb{R}^n \to \mathbb{R}$ be subdifferentiable and let $\mu \in (0, \infty)$. We say that f is (μ-)**strongly convex** if

$$f(y) \geq f(x) + \langle \tilde{\nabla} f(x), y - x \rangle + \frac{\mu}{2} \left\| x - y \right\|^2$$

for all $x, y \in \mathbb{R}^n$.

Clearly, strongly convex functions are convex.

Proposition 2.43 (Convex functions have monotone gradients)**.** Let $f : \mathbb{R}^n \to \mathbb{R}$ be convex subdifferentiable. Then, it holds that

$$\langle \tilde{\nabla} f(y) - \tilde{\nabla} f(x), y - x \rangle \geq 0$$

for all $x, y \in \mathbb{R}^n$. If, additionally, $f : \mathbb{R}^n \to \mathbb{R}$ is μ-strongly convex for some $\mu \in (0, \infty)$, it holds that

$$\langle \tilde{\nabla} f(y) - \tilde{\nabla} f(x), y - x \rangle \geq \frac{\mu}{2} \left\| x - y \right\|^2$$

for all $x, y \in \mathbb{R}^n$.

Proposition 2.44 (Convex functions on convex sets have minimizers)**.** If $f : \mathbb{R}^n \to \mathbb{R}$ is convex and $C \subset \mathbb{R}^n$ is convex, then f admits a minimizer on C, i. e. there exists a point $x^\star \in C$ such that $f(x^\star) \leq f(x)$ for all $x \in C$. If, additionally, f is strongly convex, then the minimizer is unique.

Definition 2.45 (Projection operator)**.** Let $C \subset \mathbb{R}^n$ be a closed convex set. The **(euclidian) projection operator onto C** is the map $\Pi_C : \mathbb{R}^n \to C$, defined as

$$\Pi_C(x) := \inf_{y \in C} \|x - y\|$$

for $x \in \mathbb{R}^n$.

Proposition 2.46. For a closed convex set $C \subset \mathbb{R}^n$, the projection operator Π_C is 1-Lipschitz continuous, that is

$$\left\| \Pi_C(x) - \Pi_C(y) \right\| \leq \left\| x - y \right\|$$

for all $x, y \in \mathbb{R}^n$.

2.2.3. Stochastic optimization

Throughout this subsection, we fix an arbitrary probability space $(\Omega, \mathcal{F}, \mathbb{P})$. [**?**]

Algorithm 3 Stochastic Gradient Descent (SGD)

Require: Initial point $x_0 \in \mathbb{R}^n$, step sizes $(\eta_k)_{k\in\mathbb{N}_0}$

```
for k = 0 to K − 1 do
    Sample ξ_k from distribution of ξ
    g ← ∇̃F_{ξ_k}(x_k)
    x_{k+1} ← Π_X(x_k − η_k g)
end for
return x_K
```

Definition 2.47. Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be a convex function. A random vector $G\colon \Omega \to \mathbb{R}^n$ is called a **stochastic (sub-)gradient of** f **at** $x \in \mathbb{R}^n$, if

$$\mathbb{E}(G) \in \partial f(x).$$

Example 2.48 (Per-sample stochastic gradient)**.** Let $F\colon \mathbb{R}^n \times \Omega \to \mathbb{R}$ be continuously differentiable in its first argument and let $f(x) := \mathbb{E}(F(x,\cdot))$ for all $x \in \mathbb{R}^n$. Then, for any $x \in \mathbb{R}^n$, the random vector $G_x\colon \Omega \to \mathbb{R}^n$, defined by $G_x(\omega) := \nabla_x F(x,\omega)$, is a stochastic gradient of f at x.

A **convex stochastic optimization problem** has the form

$$\min_{x\in\mathcal{X}} \left\{ f(x) := \mathbb{E}(F_\xi(x)) \right\}, \tag{2.1}$$

where $\emptyset \neq \mathcal{X} \subset \mathbb{R}^n$, $\xi\colon \Omega \to \mathbb{R}^m$ is a random vector, and $F\colon \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}, (x,z) \mapsto F_z(x)$ is a function that satisfies

- $x \to F_{\xi(\omega)}(x)$ is convex for almost every $\omega \in \Omega$, and
- $\omega \to F_{\xi(\omega)}(x)$ is measurable for all $x \in \mathbb{R}^n$.

Under those assumptions, the problem is well-defined and f is convex. If, additionally, f is subdifferentiable, then a standard method to solve such a problem is **stochastic gradient descent**, which is described in algorithm 3. The convergence of the iterates $(x_k)_{k\in\mathbb{N}}$, generated by algorithm 3, to a solution of (2.1), depends heavily on the choice of step sizes $(\eta_k)_{k\in\mathbb{N}}$ and the variance of the stochastic gradients $\tilde{\nabla} F_\xi(x)$. A classical result by Robbins and Monro [2] states that if f is strongly convex and there exists a constant $\sigma^2 \in (0,\infty)$ such that $\sup_{x\in\mathbb{R}^n} \mathrm{Var}(\tilde{\nabla} F_\xi(x)) \leq \sigma^2$, then the condition

$$\sum_{k=0}^{\infty} \eta_k = \infty \quad \text{and} \quad \sum_{k=0}^{\infty} \eta_k^2 < \infty,$$

ensures that algorithm 3 converges to the minimum of f.

Algorithm 4 Stochastic Proximal Gradient Descent (SPGD)

Require: Initial point $x_0 \in \mathbb{R}^n$, step sizes $(\eta_k)_{k\in\mathbb{N}_0}$

1: **for** $k = 0$ to $K - 1$ **do**
2: Sample ξ_k from distribution of ξ
3: $g \leftarrow \tilde{\nabla} F_{\xi_k}(x_k)$
4: $x_{k+1} \leftarrow \operatorname{prox}_{\eta_k r}(x_k - \eta_k g)$
5: **end for**
6: **return** x_K

In many applications (such as the one we care about, see (P)), the objective function in (2.1) exhibits special structure, which can be exploited in algorithm design. This often leads to problems that can be formulated as

$$\min_{x\in\mathcal{X}} \left\{ f(x) := \mathbb{E}(F_\xi(x)) + r(x) \right\}, \tag{2.2}$$

where now the objective consists of a differentiable part $x \mapsto F_\xi(x)$, and a potentially less well-behaved part $r\colon \mathbb{R}^n \to \bar{\mathbb{R}}$, which is even allowed to take on the value $+\infty$. For example, the function r can be the ℓ_1-regularizer $x \mapsto \lambda \|x\|_1$ to promote sparsity, which is a non-differentiable operation at $x = 0$. Another popular choice is to use r to encode some constraints of the form $x \in \mathcal{S} \subset \mathbb{R}^n$ directly into f, by letting

$$r(x) := \begin{cases} 0, & \text{if } x \in \mathcal{S}, \\ \infty, & \text{else.} \end{cases}$$

To deal with such objectives effectively, we use the following concept.

Definition 2.49 (Proximity operator)**.** Let $r\colon \mathbb{R}^n \to \bar{\mathbb{R}}$ be proper and convex, and let $\eta \in (0, \infty)$. The **proximity operator** $\operatorname{prox}_{\eta r}\colon \mathbb{R}^n \to \mathbb{R}^n$ is defined as

$$\operatorname{prox}_{\eta r}(x) := \operatorname*{arg\,min}_{y\in\mathbb{R}^n} \left\{ r(x) + \frac{1}{2\eta} \left\|x - y\right\|^2 \right\}$$

for $x \in \mathbb{R}^n$.

Note that, under the stated assumptions, the minimization problem always has a unique solution, since the objective $x \mapsto r(x) + \frac{1}{2\eta} \left\|x - y\right\|^2$ is strongly convex (proposition 2.44).

We can now describe the **stochastic proximal gradient method** in algorithm 4.

2.2.4. Multifunctions and metric regularity

Definition 2.50 (Multifunction)**.** Let X and Y be Banach spaces. A function $\Psi\colon X \to 2^Y$ is called **multifunction**. The **domain** (denoted $\operatorname{dom}(\Psi)$) and the **range** (denoted $\operatorname{range}(\Psi)$) of a multifunction $\Psi\colon X \to 2^Y$ are defined as

$$\begin{aligned} \operatorname{dom}(\Psi) &:= \{\, x \in X \mid \Psi(x) \neq \emptyset \,\} \quad \text{and} \\ \operatorname{range}(\Psi) &:= \{\, y \in Y \mid y \in \Psi(x) \text{ for some } x \in X \,\}. \end{aligned}$$

The **graph** of Ψ is

$$\operatorname{graph}(\Psi) := \{\, (x, y) \in X \times Y \mid y \in \Psi(x), x \in X \,\}.$$

The **(graph) inverse** $\Psi^{-1}\colon Y \to 2^X$ of Ψ is defined as

$$\Psi^{-1}(y) := \{\, x \in X \mid y \in \Psi(x) \,\}.$$

The multifunction Ψ is said to be **closed at a point** $x \in X$ if $x_k \to x$, $y_k \in \Psi(x_k)$, and $y_k \to y$ imply that $y \in \Psi(x)$. If Ψ is closed at all points $x \in X$, then Ψ is said to be **closed**. Finally, it is said that Ψ is **convex** if, for any $x_1, x_2 \in X$ and w$t \in [0, 1]$,

$$t\Psi(x_1) + (1 - t)\Psi(x_2) \subset \Psi(tx_1 + (1 - t)x_2).$$

Definition 2.51 (Metric regularity)**.** We say that the multifunction $\Psi\colon X \to 2^Y$ is **metric regular** at a point $(x_0, y_0) \in \operatorname{graph}(\Psi)$ at rate $c \in \mathbb{R}_{\geq 0}$, if there exists a neighborhood $U \subset X \times Y$ containing (x_0, y_0), such that

$$\operatorname{dist}(x, \Psi^{-1}(y)) \leq c \operatorname{dist}(y, \Psi(x)),$$

for all $(x, y) \in U$.

Proposition 2.52 (Robinson-Ursescu stability theorem)**.** Let $\Psi\colon X \to 2^Y$ be a closed convex multifunction. Then Ψ is metric regular at $(x_0, y_0) \in \operatorname{graph}(\Psi)$ if and only if the regularity condition $y_0 \in \operatorname{int}(\operatorname{range} \Psi)$ holds.

3

Stochastic Inexact Penalty Method

In this chapter we will analyze the convergence properties of algorithm 2 applied to the constrained stochastic optimization problem

$$\begin{aligned} &\min_{x\in\mathbb{R}^n} \{\, f(x) := \mathbb{E}(F_\zeta(x)) + r(x) \,\} \\ &\text{s. t.} \quad A(\xi)x - b(\xi) \leq 0 \quad \text{a. s.}, \end{aligned} \tag{P}$$

where we implicitly assume the existence of a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ on which the random variables $\zeta\colon \Omega \to \mathbb{R}^q$, $\xi\colon \Omega \to \mathbb{R}^p$, as well as the expected value mapping $\mathbb{E}(\cdot)$, are defined. Further, we endow the probability space with a filtration $\mathcal{F} := (\mathcal{F}_k)_{k\in\mathbb{N}}$, defined by

$$\mathcal{F}_k := \sigma((\zeta_0, \xi_0), \ldots, (\zeta_{k-1}, \xi_{k-1})),$$

and denote the conditional expectation given $\mathcal{F}_k$ as

$$\mathbb{E}_k(X) := \mathbb{E}(X \mid \mathcal{F}_k),$$

for all $k \in \mathbb{N}$. Similarly, we write

$$\mathrm{Var}_k(X) := \mathbb{E}_k \|X - \mathbb{E}_k(X)\|^2$$

for the conditional variance given $\mathcal{F}_k$, for all $k \in \mathbb{N}$. Note that the sequence of iterates $(x_k)_{k\in\mathbb{N}}$ generated by algorithm 2 is adapted to $\mathcal{F}$ and thus $\mathbb{E}_k(x_k) = x_k$ for all $k \in \mathbb{N}$. We denote the feasible set for problem (P) as

$$\mathcal{X} := \{\, x \in \mathrm{dom}(f) \mid A(\xi)x - b(\xi) \leq 0 \ \text{ a. s.} \,\},$$

and the set of solutions as

$$\mathcal{X}^\star := \{\, x^\star \in \operatorname{dom}(f) \mid f(x^\star) \leq f(x)\, \forall x \in \mathcal{X} \,\}.$$

Algorithm 2 works with penalty functions $(\pi_k)_{k\in\mathbb{N}_0}$, which induce the sequence of unconstrained problems

$$\min_{x\in\mathbb{R}^n} \left\{ f_k(x) := f(x) + \gamma_k \pi_k(x) \right\}, \tag{P^{k}}$$

for $k \in \mathbb{N}_0$, where $(\gamma_k)_{k\in\mathbb{N}_0}$ is a sequence of positive real numbers. We assume that the penalty functions will always take the form

$$\pi_k(x) := \mathbb{E}(h_k(x; A(\xi), b(\xi))),$$

where, for all $k \in \mathbb{N}_0$, we let $(h_k)_{k\in\mathbb{N}_0}$ be any sequence of functions from $\mathbb{R}^n \times \mathbb{R}^{m\times n} \times \mathbb{R}^m$ to $\mathbb{R}$ with the following properties:

1. For all $k \in \mathbb{N}_0$, h_k is convex and differentiable.

2. $\pi_k(x) \geq \mathbb{E}\,\|(A(\xi)x - b(\xi))_+\|_1$ for all $x \in \mathbb{R}^n$ and $k \in \mathbb{N}_0$.

3. There exists a sequence $(\alpha_k)_{k\in\mathbb{N}_0}$ such that $\pi_k(\tilde{x}) \leq \alpha_k$ and $\sup_{k\in\mathbb{N}_0} \gamma_k \alpha_k < \infty$ for all $k \in \mathbb{N}_0$ and all feasible points $\tilde{x} \in \mathcal{X}$.

4. The gradients of $(h_k)_{k\in\mathbb{N}_0}$ have uniformly bounded gradient norm:

$$\sup_{k\in\mathbb{N}_0} \sup_{x\in\mathbb{R}^n} \|\nabla h_k(x; A(\xi), b(\xi))\| < \infty$$

   almost surely.

Further, for $k \in \mathbb{N}_0$, we define the solution set of problem (P^{k}) as

$$\mathcal{X}_k^\star := \{\, x_k^\star \in \operatorname{dom}(f) \mid f_k(x_k^\star) \leq f(x)\, \forall x \in \mathbb{R}^n \,\}.$$

Since proximal methods seperate the smooth part of the above objective, given by $x \mapsto \mathbb{E}(F_\zeta(x)) + \gamma_k \pi_k(x)$, from the possibly nonsmooth part, $x \mapsto r(x)$, it is useful to also define the functions

$$\psi_k(x) := \mathbb{E}(F_\zeta(x)) + \gamma_k \pi_k(x),$$

for all $k \in \mathbb{N}_0$.

We will denote the iterates of algorithm 2 by $(x_k)_{k\in\mathbb{N}}$. The parameters $(w_k)_{k\in\mathbb{N}_0}$ are used to compute the final output - a weighted average of the iterates of the first $K \in \mathbb{N}$ iterations - given by

$$\bar{x}_K := S_K^{-1} \sum_{k=1}^{K} w_k x_k,$$

where

$$S_K := \sum_{k=1}^{K} w_k.$$

Since this is a convex combination[1], it follows that $\bar{x}_K \in \mathrm{dom}(f)$ if f is convex and $(x_k)_{k\in\mathbb{N}} \subset \mathrm{dom}(f)$ (notice that the first iterate is excluded).

In the analysis of algorithm 2, we will need to control the variance of the error introduced by using stochastic gradients. We define the **stochastic error of the k-th stochastic gradient** of algorithm 2, g_k, as

$$z_k := \nabla\psi_k(x_k) - g_k,$$

for $k \in \mathbb{N}_0$. Note that $\mathbb{E}_k(z_k) = 0$, by definition of g_k and the aforementioned fact that x_k is $\mathcal{F}_k$-measurable.

To ensure that the the problems (P) and (P^{k}) are well-behaved enough to analyze convergence of algorithm 2, we will make the following basic assumptions.

Assumption 1. Problem (P) satisfies the following:

1. The function $x \mapsto F_\zeta(x)$ is almost surely L-smooth for some $L \in (0, \infty)$, and there exists a point $x \in \mathbb{R}^n$ such that $\mathbb{E}\,\|\nabla F_\zeta(x)\|^2 < \infty$. Further, the expectation $x \mapsto \mathbb{E}(F_\zeta(x))$ is μ-strongly convex for some $\mu \in [0, \infty)$. If $\mu = 0$, we additionally assume that f has bounded lower-level sets.

2. The function $r\colon \mathbb{R}^d \to \bar{\mathbb{R}}$ is proper, convex, and locally Lipschitz continuous on $\mathrm{dom}(r)$.

3. The matrix-valued map $A\colon \mathbb{R}^p \to \mathbb{R}^{m\times n}$, and the vector valued map $b\colon \mathbb{R}^p \to \mathbb{R}^m$, are both (Borel-)measurable.

4. The sequence $(\gamma_k)_{k\in\mathbb{N}_0}$ is nondecreasing and unbounded.

5. The iterate weights $(w_k)_{k\in\mathbb{N}}$ are nonnegative and $w_k > 0$ for at least one $k \in \{1, \ldots, K\}$, where $K \in \mathbb{N}$ is the number of iterations of algorithm 2.

[1] As long as $w_1, \ldots, w_K$ are nonnegative and at least one w_k is positive.

6. There exists at least one feasible point.

Notice that **we allow the strong convexity constant to take on the value zero**, because it allows for a unified treatment of both strongly convex and convex objectives. That is, the statement "f is μ-strongly convex with $\mu \in [0, \infty)$" means "f is either convex or μ-strongly convex with $\mu \in (0, \infty)$". For brevity, we will typically just say "f is μ-strongly convex", even if $\mu = 0$ is allowed. In cases where we need $\mu > 0$, we will make that clear. If $\mu = 0$, the additional assumption of bounded lower-level sets ensures that the family of sets $(\mathcal{X}_k^\star)_{k\in\mathbb{N}_0}$ is nonempty and uniformly bounded (see lemma 3.1). Simple ways to guarantee bounded lower-level sets in the (non-strongly) convex case are to choose r such that $\mathrm{dom}(r)$ is compact, or by letting $r(x) = \lambda \|x\|^2$ for some $\lambda > 0$.

Assumption 1 has multiple useful implications, which are captured by the following lemma.

Lemma 3.1. Assumption 1 implies the following:

1. The gradients of $(\pi_k)_{k\in\mathbb{N}_0}$ are uniformly bounded: There exists $G \in (0, \infty)$ such that
$$\sup_{k\in\mathbb{N}_0} \sup_{x\in\mathbb{R}^n} \|\nabla \pi_k(x)\| \leq G.$$
In particular, π_k is Lipschitz continuous for all $k \in \mathbb{N}$.

2. The objective f of (P) is μ-strongly convex with $\mu \in [0, \infty)$, locally Lipschitz continuous, and subdifferentiable on $\mathrm{dom}(f)$. Further, f has bounded lower-level sets.

3. The objectives $(f_k)_{k\in\mathbb{N}_0}$ of (P^k) are μ-strongly convex with $\mu \in [0, \infty)$, locally Lipschitz continuous, and subdifferentiable on $\mathrm{dom}(f_k)$. Further, f_k has bounded lower-level sets for all $k \in \mathbb{N}_0$.

4. The functions $(\psi_k)_{k\in\mathbb{N}_0}$ are differentiable and μ-strongly convex with $\mu \in [0, \infty)$.

5. Let $B \subset \mathbb{R}^n$ be a bounded subset. Then, for all $k \in \mathbb{N}_0$, the stochastic gradients defined by
$$g_k(x) := \nabla F_\zeta(x) + \gamma_k \nabla h_k(x; A(\xi), b(\xi)),$$
for $x \in \mathbb{R}^n$, satisfy $\sup_{x\in B} \left\|g_k(x)\right\| = \mathcal{O}(\gamma_k)$ almost surely.

6. The set $\mathcal{X}^\star$ is nonempty compact. For all $k \in \mathbb{N}_0$, the set $\mathcal{X}_k^\star$ is nonempty compact, and the family $(\mathcal{X}_k^\star)_{k\in\mathbb{N}_0}$ is uniformly bounded. In particular, the map $x \mapsto \Pi_\mathcal{X}(x)$ is bounded when restricted to the set $\cup_{k=0}^\infty \mathcal{X}_k^\star$.

7. The iterates $(x_k)_{k\in\mathbb{N}}$ of algorithm 2 (excluding x_0) are all included in $\mathrm{dom}(f)$. In particular, $\bar{x}_K \in \mathrm{dom}(f)$ for all $K \in \mathbb{N}$.

Proof. 1. Per definition, for all $k \in \mathbb{N}_0$,

$$\nabla\pi_k(x) = \nabla\mathbb{E}(h_k(x; A(\xi), b(\xi))) = \mathbb{E}(\nabla h_k(x; A(\xi), b(\xi))),$$

hence

$$\|\nabla\pi_k(x)\| \leq \mathbb{E}\,\|\nabla h_k(x; A(\xi), b(\xi))\|\,,$$

where the last step follows from Jensen's inequality. The claim follows by our assumption that

$$\sup_{k\in\mathbb{N}_0}\sup_{x\in\mathbb{R}^n}\|\nabla h_k(x; A(\xi), b(\xi))\| < \infty$$

almost surely. In particular, π_k is Lipschitz continuous for all $k \in \mathbb{N}_0$.

2. By L-smoothness of F_ζ, the expectation $\mathbb{E}(F_\zeta)$ is also L-smooth, thus locally Lipschitz. By local Lipschitz continuity of r, it follows that $f = \mathbb{E}(F_\zeta) + r$ is also locally Lipschitz. Similarly, as the sum of a μ-strongly convex function and a convex function, f is μ-strongly convex. Any proper, convex, and continuous function is subdifferentiable. Finally, f has bounded lower-level sets per assumption, if $\mu = 0$, and by properties of strongly convex functions, if $\mu > 0$ (see TODO: Ref a proposition).

3. By the above, f is μ-strongly convex, locally Lipschitz continuous, and subdifferentiable on $\mathrm{dom}(f)$. Combining this with the fact that π_k is convex, Lipschitz continuous, and differentiable, as well as $\gamma_k > 0$, it follows that $f_k = f + \gamma_k\pi_k$ is also μ-strongly convex, locally Lipschitz continuous, and subdifferentiable on $\mathrm{dom}(f)$, for all $k \in \mathbb{N}_0$. Bounded lower-level sets follow from the same argument as above. Finally, the claim follows from $\mathrm{dom}(f) = \mathrm{dom}(f_k)\ \ \forall k \in \mathbb{N}_0$.

4. For all $k \in \mathbb{N}_0$, ψ_k is the sum of two differentiable functions, one μ-strongly convex, the other convex. Hence, ψ_k is μ-strongly convex and differentiable for all $k \in \mathbb{N}_0$.

5. Let $B \subset \mathbb{R}^n$ be bounded. By the triangle inequality,

$$\left\|g_k(x)\right\| \leq \|\nabla F_\zeta(x)\| + \gamma_k\,\|\nabla h_k(x; A(\xi), b(\xi))\|\,.$$

for all $x \in B$. The claim now follows from local Lipschitz continuity of F_ζ (a. s.), and the assumption $\sup_{x\in B}\|\nabla h_k(x; A(\xi), b(\xi))\| < \infty$ (a. s.) $\forall k \in \mathbb{N}_0$.

6. As we have already shown, f is convex and has bounded lower-level sets. The set $\mathcal{X}$ is the intersection of the closed convex sets $\{ x \in \mathrm{dom}(f) \mid A(z)x - b(z) \leq 0 \}$ over z in the support of ξ. Thus, $\mathcal{X}$ is itself closed and convex, which implies that f attains a minimizer on $\mathcal{X}$ (TODO: Ref a proposition). For $k \in \mathbb{N}_0$, we have also shown that f_k is convex and has bounded lower-level sets, which implies that f_k has an unconstrained minimizer. Compactness of the solution sets follows from continuity and the bounded lower-level sets properties (TODO: Ref a proposition). To prove uniform boundedness of the family $(\mathcal{X}_k^\star)_{k\in\mathbb{N}_0}$, fix any feasible point $\hat{x} \in \mathcal{X}$ (which exists per assumption 1). By our assumptions on the penalty sequence $(\pi_k)_{k\in\mathbb{N}_0}$, there exists $c \in (0, \infty)$ such that, for all $k \in \mathbb{N}_0$ and $x_k^\star \in \mathcal{X}_k^\star$,

$$f(x_k^\star) \leq f_k(x_k^\star) \leq f_k(\hat{x}) = f(\hat{x}) + \gamma_k \pi_k(\hat{x}) \leq f(\hat{x}) + \gamma_k \alpha_k \leq f(\hat{x}) + c,$$

where we used optimality of $x_k^\star$ for f_k. Hence,

$$\mathcal{X}_k^\star \subset \{ x \in \mathbb{R}^n \mid f(x) \leq f(\hat{x}) + c \} =: f_{\leq f(\hat{x})+c}$$

for all $k \in \mathbb{N}_0$. Coupled with the fact that f has bounded lower-level sets, it follows that the family $(\mathcal{X}_k^\star)_{k\in\mathbb{N}_0}$ is uniformly bounded. In particular, since $f_{\leq f(\hat{x})+c}$ is also closed, by continuity of f, and thus compact, it follows that the image of the set $f_{\leq f(\hat{x})+c}$ under the continuous map $\Pi_{\mathcal{X}}$ is compact. Therefore, we obtain

$$\sup_{k\in\mathbb{N}} \sup_{x_k^\star \in \mathcal{X}_k^\star} \left\| \Pi_{\mathcal{X}}(x_k^\star) \right\| \leq \sup_{x \in f_{\leq f(\hat{x})+c}} \| \Pi_{\mathcal{X}}(x) \| < \infty,$$

as desired.

7. By definition of the proximal operator,

$$x_{k+1} = \arg\min_{x\in\mathbb{R}^d} \left\{ r(x) + \frac{1}{2\eta_k} \left\| x - (x_k - \eta_k g_k) \right\|^2 \right\},$$

for all $k \in \mathbb{N}_0$. Since r is proper, it must therefore hold that $x_k \in \mathrm{dom}(r) = \mathrm{dom}(f)$, for all $k \in \mathbb{N}$. By definition of $\bar{x}_K$ as a convex combination of the iterates $x_1, \ldots, x_K$, $K \in \mathbb{N}$, and convexity of f, it follows that $\bar{x}_K \in \mathrm{dom}(f)$.

This concludes the proof. □

A key challenge in the analysis of algorithm 2 is the fact that $x_k^\star$ must not be feasible, which prevents the use of standard arguments from the SGD literature to analyse convergence. Our proof methods combine approaches from recent works,

mainly the already mentioned [3], as well as [6]. The latter paper investiages *stochastic optimization problems under distributional drift*. While these kinds of problems are not exactly the same as the ones we are working with, the two settings do indeed exhibit striking resemblences. Namely, Cutler et al. [6] investigated a sequence of time-dependent composite problems of the form

$$\min_{x\in\mathbb{R}^n} g_t(x) + r_t(x),$$

where, for all $t \in \mathbb{N}$, g_t is smooth strongly convex, and r_t is convex. Comparing to our problem ($\mathrm{P^k}$), rewritten as

$$\min_{x\in\mathbb{R}^n} \left\{ f_k(x) = \psi_k(x) + r(x) \right\}, \tag{$\mathrm{P^k}$}$$

we see that the two settings almost match, except for smoothness properties. Namely, we do not need the family $(\psi_k)_{k\in\mathbb{N}}$ itself to be smooth, but only that it is composed of an L-smooth function and a differentiable one with uniformly bounded gradients. Even though it would be realistic to assume Lipschitz smoothness of the penalties π_k, this would lead to smoothness constants that blow up for $k \to \infty$, in order to satisfy the convergence assumption $\lim_{k\to\infty} \pi_k = \pi_{\ell_1}$. A naive application of the techniques in Cutler et al. [6] would then lead to a restriction on the step sizes $(\eta_k)_{k\in\mathbb{N}}$ of the form $\eta_k \in (0, 1/L_k)$ with L_k the smoothness constant of ψ_k. With some care however, we manage to circumvent this restriction, as well as the Lipschitz assumption on $\nabla\pi_k$, and only require $\eta_k \in (0, 1/L)$, which allows for a lot more freedom in the choice of step sizes. The resulting inequality can then be used to adapt the proof strategy from Nedić et al. [3] (where the authors did not use proximal maps in their algorithm) to our proximal method.

3.1. Almost sure convergence

In this section we will establish conditions under which we can guarantee almost sure convergence of the sequence of iterates $(x_k)_{k\in\mathbb{N}_0}$. The proof will also yield convergence in expectation of $(x_k)_{k\in\mathbb{N}_0}$ to $x^\star$ along a subsequence. The main use of the almost sure convergence result will be that we will have conditions on the stepsizes $(\eta_k)_{k\in\mathbb{N}_0}$ and the penalty parameters $(\gamma_k)_{k\in\mathbb{N}_0}$ to ensure that $(x_k)_{k\in\mathbb{N}_0}$ is bounded with probability one. This, together with some of the results we prove along the way, will come in very handy in the subsequent analysis of the quantitative convergence rates of our methods.

The proof for almost sure convergence hinges on two technical lemmata. The first

is the well-known Robbins-Siegmund lemma, which provides a general sufficient condition to guarantee almost sure convergence of so-called "almost supermartingales".

Lemma 3.2 (Robbins-Siegmund)**.** Let $(\mathcal{F}_k)_{k\in\mathbb{N}}$ be an increasing sequence of σ-algebras and v_k, a_k, b_k, c_k be nonnegative $\mathcal{F}_k$-measurable random variables. If, for all $k \in \mathbb{N}$,

$$\mathbb{E}(v_{k+1} \mid \mathcal{F}_k) \leq v_k(1 + a_k) + b_k - c_k, \tag{3.1}$$

and $\sum_{k=1}^{\infty} a_k < \infty$, $\sum_{k=1}^{\infty} b_k < \infty$ a. s., then with probability one, $(v_k)_{k\in\mathbb{N}}$ is convergent and it holds that $\sum_{k=1}^{\infty} c_k < \infty$.

Proof. See [7]. □

Our goal for the rest of this section is to derive a recursive inequality for the sequence $\|x_k - x^\star\|^2$, which resembles (3.1). As a first step, we will analyze the convergence of the sequence $(x_k^\star)_{k\in\mathbb{N}_0}$. In particular, we will establish that $\operatorname{dist}(x_k^\star, \mathcal{X})$ converges to zero for $k \to \infty$, and that this convergence is independent of the sequence $(\gamma_k)_{k\in\mathbb{N}_0}$, as long as one insures that γ_k is eventually large enough.

To do this, we will first show that one can locally bound the distance $\operatorname{dist}(x, \mathcal{X})$ by (a term proportional to) the penalty $\pi_k(x)$. We will rely on an extension of a classic result by Hoffman [8], who analyzed the distance of points $x \in \mathbb{R}^n$ to the set of solutions of linear systems of inequalities $Ax \leq b$ with $A \in \mathbb{R}^{m\times n}$, $b \in \mathbb{R}^m$. Crucially, there always exists a constant $\tau \in (0, \infty)$, such that

$$\tau \operatorname{dist}(x, S) \leq \|(Ax - b)_+\|_\infty ,$$

where $S := \{\, y \in \mathbb{R}^n \mid Ay \leq b \,\}$. Since we are essentially dealing with infnitely inequality systems, we cannot directly apply Hoffman's lemma. Instead, we will use the theory of metric regularity, which will lead us to a local version of the above bound. Luckily, a local bound is all that is needed for our purposes and only requires the following natural assumptions.

Assumption 2. There exists a compact set $\Xi \in \mathbb{R}^p$ such that ξ is supported on Ξ and $\mathbb{P}^\xi$ has a Lebesgue-density q such that

1. q is continuous on Ξ and
2. $q(x) > 0$ for all $x \in \Xi$.

Further, we assume that

$$\int_\Xi \|A(z)x - b(z)\|_1 \, \mathrm{d}z < \infty,$$

for all $x \in \mathbb{R}^n$.

A simple sufficient condition for the last part of the assumption is continuity of the maps A and b on Ξ. One can always satisfy assumption 2 if the set Ξ (or a superset thereof) is known, by sampling uniformly. In fact, lemma 3.3 will demonstrate that uniform sampling is optimal[2], in the sense that it minimizes constants that depend on the distribution of ξ in the derived upper bound. This is very intuitive: Since we are working in the setting of almost sure constraints, no realization of ξ is more important than any other. However, it should be noted that this principle only applies if the set Ξ is fixed and cannot be somehow reduced to a smaller set, as ideally one would only need to sample from the set of maximizers of $z \mapsto A(z)x - b(z)$ on Ξ. Nevertheless, in the case where such a reduction is possible, the optimal way to sample would still be to sample uniformly, albeit from the reduced set.

The next assumption is a common one in the optimization literature.

Assumption 3. There exists a *Slater point*, i. e. a point $x \in \mathcal{X}$ such that $A(\xi)x < b(\xi)$ almost surely.

We will now formulate and prove the lemma.

Lemma 3.3 (Subfeasibility bound via penalty functions)**.** (TODO: Not all of assumption 1 is needed here, should probably split the assumption up into two parts.) Assume assumptions 1 to 3 hold and let $C \subset \mathbb{R}^n$ be a compact subset. Then there exists a constant $\tau \in (0, \infty)$ such that

$$\tau \operatorname{dist}(x, \mathcal{X}) \leq \pi_k(x),$$

for all $x \in C$ and $k \in \mathbb{N}_0$.

Proof. We consider the Banach space $Y := L_1(\Xi, \mathbb{R}^m)$ equipped with the norm

$$\|y\|_Y := \int_\Xi \|y(z)\|_1 \, dz.$$

The distance map on Y, denoted dist_Y, is then given by

$$\operatorname{dist}_Y(y, S) := \inf_{y' \in S} \|y - y'\|_Y$$

for $y \in Y$ and $S \subset Y$. We define the multifunction $\Psi \colon \mathbb{R}^n \to 2^Y$ as

$$\Psi(x) := \{\, y \in Y \mid A(\xi)x - b(\xi) \leq y(\xi) \text{ a. s.} \,\}.$$

[2]Of course, one may not have control over the sampling distribution.

This multifunction is closed and convex. Indeed, let $(x_k)_{k\in\mathbb{N}} \in \mathbb{R}^n$ be a sequence that converges to some $x \in \mathbb{R}^n$, and assume that there exists a sequence $(y_k)_{k\in\mathbb{N}}$ with $y_k \in \Psi(x_k)$ for all $k \in \mathbb{N}$, such that $\lim_{k\to\infty} y_k = y \in Y$. Then, with probability one,

$$A(\xi)x - b(\xi) = \lim_{k\to\infty} A(\xi)x_k - b(\xi) \leq \lim_{k\to\infty} y_k(\xi) = y(\xi).$$

Hence $y \in \Psi(x)$, which proves closedness. For convexity, let $x_1, x_2 \in \mathbb{R}^n$ and $t \in [0,1]$. Then

$$\begin{aligned} t\Psi(x_1) + (1-t)\Psi(x_2) = \{\, y \mid y = ty_1 + (1-t)y_2 \text{ for } y_1, y_2 \in Y \\ \text{such that } A(\xi)x_1 - b(\xi) \leq y_1(\xi) \text{ a.s.} \\ \text{and } A(\xi)x_2 - b(\xi) \leq y_2(\xi) \text{ a.s.}\,\}. \end{aligned}$$

Let $x := tx_1 + (1-t)x_2$. Then, for any $y \in t\Psi(x_1) + (1-t)\Psi(x_2)$, there exist $y_1, y_2 \in Y$, such that

$$A(\xi)x - b(\xi) = t(A(\xi)x_1 - b(\xi)) + (1-t)(A(\xi)x_2 - b(\xi)) \leq ty_1(\xi) + (1-t)y_2(\xi) = y(\xi).$$

Hence,

$$t\Psi(x_1) + (1-t)\Psi(x_2) \subset \Psi(tx_1 + (1-t)x_2),$$

proving convexity. Now let $x_0 \in C$ be some arbitrary point. Per definition of the inverse Ψ^{-1}, it holds that

$$\Psi^{-1}(0) = \{\, x \in \mathbb{R}^n \mid 0 \in \Psi(x)\,\} = \{\, x \in \mathbb{R}^n \mid A(\xi)x - b(\xi) \leq 0 \text{ a.s.}\,\} = \mathcal{X}.$$

Hence we can write $\operatorname{dist}(x, \mathcal{X}) = \operatorname{dist}(x, \Psi^{-1}(0))$. Note that assumption 3 implies $0 \in \operatorname{int}(\operatorname{range}\Psi)$. Therefore we can apply proposition 2.52, which guarantees the existence of a constant $c \in \mathbb{R}_{\geq 0}$ such that

$$\operatorname{dist}(x, \Psi^{-1}(y)) \leq c \operatorname{dist}_Y(y, \Psi(x)),$$

for all (x, y) in some open neighborhood $U \subset \mathbb{R}^n \times Y$ containing $(x_0, 0)$. In particular,

$$\operatorname{dist}(x, \mathcal{X}) \leq c \operatorname{dist}_Y(0, \Psi(x)) \tag{3.2}$$

for all x in the open set $U \cap (\mathbb{R}^n \times \{0\})$. Since x_0 is arbitrary, we can derive a similar bound that holds around an open neighborhood $U_x \subset \mathbb{R}^n$ of a point $x \in C$, for any $x \in C$, yielding corresponding constants $(c_x)_{x\in C}$. By compactness of C, the open

covering

$$C \subset \bigcup_{x \in C} U_x$$

has a finite subcovering

$$C \subset \bigcup_{i=1}^{\ell} U_{x_i}$$

with $(x_i)_{i \in \{1,\dots,\ell\}} \subset C$. The corresponding constants $(c_{x_i})_{i \in \{1,\dots,\ell\}}$ have a maximum $c := \max_{i \in \{1,\dots,\ell\}} c_{x_i}$. Thus, we have shown that there exists $c \in \mathbb{R}_{\geq 0}$ such that

$$\operatorname{dist}(x, \mathcal{X}) \leq c \operatorname{dist}_Y(0, \Psi(x))$$

for all $x \in C$. Next, we will show that

$$\operatorname{dist}_Y(0, \Psi(x)) = \int_\Xi \|(A(z)x - b(z))_+\|_1 \, \mathrm{d}z = \|(Ax - b)_+\|_Y \tag{3.3}$$

for any $x \in C$. Fix $x \in C$ and let $y \in \Psi(x)$. By definition of $\Psi(x)$ and positivity of q on Ξ, it holds that $y \in \Psi(x) \iff A(z)x - b(z) \leq y(z)$ for all $z \in \Xi$. Set

$$\phi(z) := (A(z)x - b(z))_+$$

for $z \in \Xi$. Clearly, $\phi \in \Psi(x)$. We will show that $\|\phi(z)\| \leq \|y(z)\|$ for all $z \in \Xi$. We denote by $\phi_i(z), a_i(z), b_i(z), y_i$, the i-th row of $\phi(z), A(z), b(z), y$. It holds that

$$|\phi_i(z)| = \begin{cases} 0, & a_i(z)x - b_i(z) \leq 0 \\ a_i(z)x - b_i(z), & \text{else} \end{cases}$$

and thus

$$|y_i| \geq \begin{cases} a_i(z)x - b_i(z), & \text{if } a_i(z)x - b_i(z) \geq 0, \\ 0, & \text{else.} \end{cases} = |\phi_i(z)|,$$

for all $i \in \{1, \dots, m\}$. It follows that

$$\begin{aligned} \|\phi(z)\|_1 &= \sum_{i=1}^m |\phi_i(z)| \\ &\leq \sum_{i=1}^m |y_i| \\ &= \|y\|_1 , \end{aligned}$$

and therefore,

$$\inf_{y\in\Psi(x)} \int_{\Xi} \|y\|_1 \, \mathrm{d}z \geq \int_{\Xi} \|\phi(z)\| \, \mathrm{d}z,$$

proving (3.3). To finish the proof, we need to establish a relationship between (3.3) and $\pi_k(x)$. By compactness of Ξ and continuity of q, the image $q(\Xi)$ must be compact. In particular, since $q(x) > 0$ for all $x \in \Xi$, there must exist some uniform positive lower bound $c_q \in (0, \infty)$ such that $q(x) \geq c_q$ for all $x \in \Xi$. If we denote the Lebesgue-measure by λ, we see that

$$\mathbb{P}^{\xi}(A) = \int_A q(z) \, \mathrm{d}z \geq c_q \lambda(A),$$

for all measurable $A \subset \Xi$. Therefore, $\mathbb{P}^{\xi}$ and λ are equivalent on Ξ, hence proposition 2.20 implies that q^{-1} is a $\mathbb{P}^{\xi}$-density of λ. Using proposition 2.21 and proposition 2.7, we can conclude

$$\begin{aligned}
\|y\|_Y &= \int_{z\in\Xi} \|y(z)\|_1 \, \mathrm{d}z \\
&= \int_{z\in\Xi} \|y(z)\|_1 \, q^{-1}(z) \, \mathbb{P}^{\xi}(dz) \\
&\leq c_q^{-1} \int_{z\in\Xi} \|y(z)\|_1 \, \mathbb{P}^{\xi}(dz) \\
&= c_q^{-1} \int \|y(\xi)\|_1 \, \mathrm{d}\mathbb{P} \\
&= c_q^{-1} \, \mathbb{E} \|y(\xi)\|_1 .
\end{aligned}$$

Combining with (3.2) and (3.3), we obtain

$$\operatorname{dist}(x, \mathcal{X}) \leq c c_q^{-1} \mathbb{E} \|(A(\xi)x - b(\xi))_+\|_1 .$$

The claim follows after setting $\tau := c^{-1} c_q$ and applying one of the defining properties of $(\pi_k)_{k\in\mathbb{N}_0}$. □

Theorem 3.4 (Convergence of $x_k^\star$)**.** Assume assumptions 1 to 3 hold and let $x^\star \in \mathcal{X}^\star$. Then, there exist constants $M, \tau \in (0, \infty)$, such that

$$\frac{\mu}{2}\|x^\star - x_k^\star\|^2 + \frac{\mu}{2}\|x^\star - \Pi_{\mathcal{X}}(x_k^\star)\|^2 + (\tau\gamma_k - M) \operatorname{dist}(x_k^\star, \mathcal{X}) \leq \gamma_k \alpha_k,$$

for all $k \in \mathbb{N}_0$ and $x_k^\star \in \mathcal{X}_k^\star$. In particular,

$$\operatorname{dist}(x_k^\star, \mathcal{X}) = \mathcal{O}(\alpha_k)$$

and, if $\mu > 0$, we further have

$$\|x^\star - x_k^\star\|^2 = O(\gamma_k \alpha_k).$$

Proof. Let $k \in \mathbb{N}_0$ and $x_k^\star \in \mathcal{X}_k^\star$. By optimality of $x_k^\star$ for f_k, there exists $0 \in \partial f_k(x_k^\star)$. Hence, by strong convexity, we obtain

$$\frac{\mu}{2}\|x^\star - x_k^\star\|^2 \leq f_k(x^\star) - f_k(x_k^\star) = f(x^\star) - f(x_k^\star) + \gamma_k \pi_k(x^\star) - \gamma_k \pi_k(x_k^\star). \tag{3.4}$$

We can write

$$\begin{aligned} f(x^\star) - f(x_k^\star) &= f(x^\star) - f(\Pi_{\mathcal{X}}(x_k^\star)) + f(\Pi_{\mathcal{X}}(x_k^\star)) - f(x_k^\star) \\ &\leq -\frac{\mu}{2}\|x^\star - \Pi_{\mathcal{X}}(x_k^\star)\|^2 - \langle \tilde{\nabla} f(x^\star), \Pi_{\mathcal{X}}(x_k^\star) - x^\star \rangle + f(\Pi_{\mathcal{X}}(x_k^\star)) - f(x_k^\star), \end{aligned}$$

where we again used strong convexity in the second step. Since $\Pi_{\mathcal{X}}(x_k^\star) \in \mathcal{X}$ and $x^\star$ is optimal for f on $\mathcal{X}$, it holds that

$$\langle \tilde{\nabla} f(x^\star), \Pi_{\mathcal{X}}(x_k^\star) - x^\star \rangle \geq 0,$$

and thus

$$f(x^\star) - f(x_k^\star) \leq -\frac{\mu}{2}\|x^\star - \Pi_{\mathcal{X}}(x_k^\star)\|^2 + f(\Pi_{\mathcal{X}}(x_k^\star)) - f(x_k^\star).$$

By lemma 3.1, we know that the family $(\mathcal{X}_k^\star)_{k \in \mathbb{N}_0}$ is uniformly bounded and

$$\sup_{k \in \mathbb{N}_0} \sup_{x_k^\star \in \mathcal{X}_k^\star} \left\|\Pi_{\mathcal{X}}(x_k^\star)\right\| < \infty.$$

Hence, by local Lipschitz continuity of f (lemma 3.1) and Cauchy-Schwarz, there exists a constant $M \in (0, \infty)$, such that

$$f(\Pi_{\mathcal{X}}(x_k^\star)) - f(x_k^\star) \leq M \operatorname{dist}(x_k^\star, \mathcal{X}),$$

for all $k \in \mathbb{N}_0$ and $x_k^\star \in \mathcal{X}_k^\star$. We now obain

$$f(x^\star) - f(x_k^\star) \leq -\frac{\mu}{2}\|x^\star - \Pi_{\mathcal{X}}(x_k^\star)\|^2 + M \operatorname{dist}(x_k^\star, \mathcal{X}).$$

Plugging this into (3.4), we obtain

$$\frac{\mu}{2}\|x^\star - x_k^\star\|^2 \leq -\frac{\mu}{2}\|x^\star - \Pi_{\mathcal{X}}(x_k^\star)\|^2 + M \operatorname{dist}(x_k^\star, \mathcal{X}) + \gamma_k \pi_k(x^\star) - \gamma_k \pi_k(x_k^\star).$$

Now, using our lower bound on $\pi_k(x_k^\star)$ from lemma 3.3, and combining terms, we arrive at the inequality

$$\frac{\mu}{2}\|x^\star - x_k^\star\|^2 \leq -\frac{\mu}{2}\|x^\star - \Pi_{\mathcal{X}}(x_k^\star)\|^2 + (M - \gamma_k\tau)\operatorname{dist}(x_k^\star, \mathcal{X}) + \gamma_k\pi_k(x^\star).$$

Using $\pi_k(x^\star) \leq \alpha_k$ and rearranging, we obtain

$$\frac{\mu}{2}\|x^\star - x_k^\star\|^2 + \frac{\mu}{2}\|x^\star - \Pi_{\mathcal{X}}(x_k^\star)\|^2 + (\gamma_k\tau - M)\operatorname{dist}(x_k^\star, \mathcal{X}) \leq \gamma_k\alpha_k.$$

The asymptotic rate for $\|x^\star - x_k^\star\|^2$ now follows in the case $\mu > 0$. For the bound on $\operatorname{dist}(x_k^\star, \mathcal{X})$, we let K be large enough such that $\gamma_k\tau > M$ for all $k \geq K$. Dividing by γ_k on both sides and using the nonnegativity of the other terms on the left-hand side, we get

$$c \cdot \operatorname{dist}(x_k^\star, \mathcal{X}) \leq \frac{\gamma_k\tau - M}{\gamma_k}\operatorname{dist}(x_k^\star, \mathcal{X}) \leq \alpha_k,$$

for all $k \geq K$ and some constant $c \in (0, 1)$, as desired. □

Having established the convergence of the sequences $(x_k^\star)_{k\in\mathbb{N}_0}$ to the feasible set $\mathcal{X}$, we will now shift our attention to the iterates $(x_k)_{k\in\mathbb{N}_0}$ of algorithm 2. We begin with the following fundamental recursive inequality.

Lemma 3.5 (One-step improvement)**.** Let assumption 1 hold and let $\rho \in (0, 1)$ and $\eta_k \in (0, \rho L^{-1}]$ for all $k \in \mathbb{N}$. Then the iterates $(x_k)_{k\in\mathbb{N}_0}$ generated by algorithm 2 with step size schedule $(\eta_k)_{k\in\mathbb{N}_0}$ satisfy

$$\begin{aligned} 2\eta_k(f_k(x_{k+1}) - f_k(x)) \leq (1 - \mu\eta_k)\|x - x_k\|^2 + 2\eta_k\langle z_k, x_k - x\rangle - \|x - x_{k+1}\|^2 \\ + \frac{8G^2\eta_k^2\gamma_k^2}{1-\rho} + \frac{2\eta_k^2}{1-\rho}\|z_k\|^2 \end{aligned} \tag{3.5}$$

almost surely, for all $x \in \mathbb{R}^n$ and $k \in \mathbb{N}_0$.

Proof. For $k \in \mathbb{N}_0$ we denote by g_k the stochastic gradient of ψ_k at x_k that is used in iteration k of algorithm 2. We also let $\psi(x) := \mathbb{E}(F_\zeta(x))$ for $x \in \mathbb{R}^n$, so that $\psi_k = \psi + \gamma_k\pi_k$ and $\nabla\psi_k = \nabla\psi + \gamma_k\nabla\pi_k$. By L-smoothness of ψ (assumption 1), we

have

$$\begin{aligned}
f_k(x_{k+1}) &= \psi_k(x_{k+1}) + r(x_{k+1}) \\
&= \psi(x_{k+1}) + \gamma_k \pi_k(x_{k+1}) + r(x_{k+1}) \\
&\leq \psi(x_k) + \langle \nabla\psi(x_k), x_{k+1} - x_k \rangle + \frac{L}{2} \|x_{k+1} - x_k\|^2 + \gamma_k \pi_k(x_{k+1}) + r(x_{k+1}) \\
&= \psi(x_k) + \langle \nabla\psi_k(x_k), x_{k+1} - x_k \rangle + \frac{L}{2} \|x_{k+1} - x_k\|^2 + \gamma_k \pi_k(x_{k+1}) + r(x_{k+1}) \\
&\quad + \gamma_k \langle \nabla\pi_k(x_k), x_k - x_{k+1} \rangle.
\end{aligned}$$

By convexity of π_k, we further have

$$\pi_k(x_{k+1}) \leq \pi_k(x_k) + \langle \nabla\pi_k(x_{k+1}), x_{k+1} - x_k \rangle,$$

and an application of Cauchy-Schwarz and Young's inequality yields

$$\pi_k(x_{k+1}) \leq \pi_k(x_k) + \frac{\epsilon_k^{-1}}{2} \|\nabla\pi_k(x_{k+1})\|^2 + \frac{\epsilon_k}{2} \|x_{k+1} - x_k\|^2,$$

for any $\epsilon_k \in (0, \infty)$. Note that the gradients $(\nabla\pi_k)_{k\in\mathbb{N}_0}$ are bounded uniformly (lemma 3.1), hence there exists $G \in (0, \infty)$ such that

$$\pi_k(x_{k+1}) \leq \pi_k(x_k) + \frac{\epsilon_k}{2} G^2 + \frac{\epsilon_k^{-1}}{2} \|x_{k+1} - x_k\|^2,$$

for any $\epsilon_k \in (0, \infty)$. With this, we can further bound $f_k(x_{k+1})$ by

$$\begin{aligned}
f_k(x_{k+1}) &\leq \psi(x_k) + \langle \nabla\psi_k(x_k), x_{k+1} - x_k \rangle + \frac{L}{2} \|x_{k+1} - x_k\|^2 + r(x_{k+1}) \\
&\quad + \gamma_k \left(\pi_k(x_k) + \frac{\epsilon_k}{2} G^2 + \frac{\epsilon_k^{-1}}{2} \|x_{k+1} - x_k\|^2 \right) + \gamma_k \langle \nabla\pi_k(x_k), x_k - x_{k+1} \rangle. \\
&= \psi_k(x_k) + \langle \nabla\psi_k(x_k), x_{k+1} - x_k \rangle + \frac{L + \gamma_k \epsilon_k^{-1}}{2} \|x_{k+1} - x_k\|^2 + r(x_{k+1}) \\
&\quad + \frac{\gamma_k \epsilon_k G^2}{2} + \gamma_k \langle \nabla\pi_k(x_k), x_k - x_{k+1} \rangle.
\end{aligned}$$

By another application of Cauchy-Schwarz and Young's inequality, we have for any $\epsilon_k \in (0, \infty)$

$$\langle \nabla\pi_k(x_k), x_k - x_{k+1} \rangle \leq \frac{\epsilon_k}{2} G^2 + \frac{\epsilon_k^{-1}}{2} \|x_{k+1} - x_k\|^2,$$

where again used the bound $\sup_{k\in\mathbb{N}_0}\sup_{x\in\mathbb{R}^n}\|\nabla\pi_k(x)\| \leq G$, and therefore we obtain

$$f_k(x_{k+1}) \leq \psi_k(x_k) + \langle \nabla\psi_k(x_k), x_{k+1} - x_k\rangle + \frac{L + 2\gamma_k\epsilon_k^{-1}}{2}\|x_{k+1} - x_k\|^2 + r(x_{k+1}) + \gamma_k\epsilon_k G^2$$

for any $\epsilon_k \in (0,\infty)$. The rest of the proof follows the strategy of Cutler et al. [6], lemma 2. Recall the error in the k-th stochastic gradient, $z_k = \nabla\psi_k(x_k) - g_k$. By adding and subtracting $\langle z_k, x_{k+1} - x_k\rangle$ in the above inequality, we get

$$f_k(x_{k+1}) \leq \psi_k(x_k) + \langle g_k, x_{k+1} - x_k\rangle + \frac{L + 2\gamma_k\epsilon_k^{-1}}{2}\|x_{k+1} - x_k\|^2 + r(x_{k+1}) + \gamma_k\epsilon_k G^2 + \langle z_k, x_{k+1} - x_k\rangle.$$

By yet another application of Cauchy-Schwarz and Young's inequality, we have

$$\langle z_k, x_{k+1} - x_k\rangle \leq \frac{\delta_k}{2}\|z_k\|^2 + \frac{\delta_k^{-1}}{2}\|x_{k+1} - x_k\|^2,$$

for all $\delta_k \in (0,\infty)$, and therefore

$$\begin{aligned} f_k(x_{k+1}) &\leq \psi_k(x_k) + \langle g_k, x_{k+1} - x_k\rangle + \frac{L + 2\gamma_k\epsilon_k^{-1} + \delta_k^{-1}}{2}\|x_{k+1} - x_k\|^2 + r(x_{k+1}) \\ &\quad + \gamma_k\epsilon_k G^2 + \frac{\delta_k}{2}\|z_k\|^2, \\ &= \psi_k(x_k) + r(x_{k+1}) + \langle g_k, x_{k+1} - x_k\rangle + \frac{1}{2\eta_k}\|x_{k+1} - x_k\|^2 \\ &\quad + \frac{L + 2\gamma_k\epsilon_k^{-1} + \delta_k^{-1} - \eta_k^{-1}}{2}\|x_{k+1} - x_k\|^2 + \gamma_k\epsilon_k G^2 + \frac{\delta_k}{2}\|z_k\|^2, \end{aligned}$$

where in the last step we added and subtracted $(2\eta_k)^{-1}\|x_{k+1} - x_k\|^2$ and moved $r(x_{k+1})$ further forward. From the definition of the proximal operator, it follows that

$$\begin{aligned} x_{k+1} &= \operatorname{prox}_{\eta_k r}(x_k - \eta_k g_k) \\ &= \arg\min_{x\in\mathbb{R}^n}\left\{ r(x) + \frac{1}{2\eta_k}\left\|x - (x_k - \eta_k g_k)\right\|^2 \right\} \\ &= \arg\min_{x\in\mathbb{R}^n}\left\{ r(x) + \langle g_k, x - x_k\rangle + \frac{1}{2\eta_k}\|x - x_k\|^2 \right\}, \end{aligned}$$

where the last step follows from expanding the square and dropping the constant term $\eta_k^2\left\|g_k\right\|^2$ from the minimization. The function $x \mapsto r(x) + \langle g_k, x - x_k\rangle + (2\eta_k)^{-1}\|x - x_k\|^2$ is $(2\eta_k)^{-1}$-strongly convex and minimized by x_{k+1}. Thus, com-

paring with our previous bound on $f_k(x_{k+1})$, we can conclude

$$\begin{aligned} f_k(x_{k+1}) \leq \psi_k(x_k) + r(x) + \langle g_k, x - x_k \rangle + \frac{1}{2\eta_k} \|x - x_k\|^2 - \frac{1}{2\eta_k} \|x - x_{k+1}\|^2 \\ + \frac{L + 2\gamma_k \epsilon_k^{-1} + \delta_k^{-1} - \eta_k^{-1}}{2} \|x_{k+1} - x_k\|^2 + \gamma_k \epsilon_k G^2 + \frac{\delta_k}{2} \|z_k\|^2, \end{aligned}$$

for all $x \in \mathbb{R}^n$. By μ-strong convexity of ψ_k, we have

$$\begin{aligned} \psi_k(x_k) + r(x) + \langle g_k, x - x_k \rangle &= \psi_k(x_k) + r(x) + \langle \nabla \psi_k(x_k), x - x_k \rangle + \langle z_k, x_k - x \rangle \\ &\leq \psi_k(x) - \frac{\mu}{2} \|x - x_k\|^2 + r(x) + \langle z_k, x_k - x \rangle \\ &= f_k(x) - \frac{\mu}{2} \|x - x_k\|^2 + \langle z_k, x_k - x \rangle, \end{aligned}$$

for all $x \in \mathbb{R}^n$. Hence,

$$\begin{aligned} f_k(x_{k+1}) \leq f_k(x) + \left(\frac{1}{2\eta_k} - \frac{\mu}{2}\right) \|x - x_k\|^2 + \langle z_k, x_k - x \rangle - \frac{1}{2\eta_k} \|x - x_{k+1}\|^2 \\ + \frac{L + 2\gamma_k \epsilon_k^{-1} + \delta_k^{-1} - \eta_k^{-1}}{2} \|x_{k+1} - x_k\|^2 + \gamma_k \epsilon_k G^2 + \frac{\delta_k}{2} \|z_k\|^2. \end{aligned} \tag{3.6}$$

Fix $\alpha \in (0, 1 - \rho)$ and define $\epsilon_k := 2\alpha^{-1}\eta_k\gamma_k$. Then

$$(L + 2\gamma_k \epsilon_k^{-1})\eta_k = \left(L + 2\gamma_k \frac{\alpha}{2\eta_k\gamma_k}\right)\eta_k = L\eta_k + \alpha \leq \rho + \alpha < 1,$$

where we used that $\eta_k \leq \rho L^{-1}$, which holds per assumption. Choosing

$$\delta_k := \frac{\eta_k}{1 - (L\eta_k + \alpha)} \in (0, \infty)$$

therefore yields

$$L + 2\gamma_k \epsilon_k^{-1} + \delta_k^{-1} - \eta_k^{-1} = L + \frac{\alpha}{\eta_k} - \frac{1 - (L\eta_k + \alpha)}{\eta_k} - \frac{1}{\eta_k} = 0.$$

Hence, we can drop the $\|x_{k+1} - x_k\|^2$ term from (3.6) and get

$$\begin{aligned} f_k(x_{k+1}) \leq f_k(x) + \left(\frac{1}{2\eta_k} - \frac{\mu}{2}\right) \|x - x_k\|^2 + \langle z_k, x_k - x \rangle - \frac{1}{2\eta_k} \|x - x_{k+1}\|^2 \\ + 2\alpha^{-1}\eta_k\gamma_k^2 G^2 + \frac{\eta_k}{2(1 - (L\eta_k + \alpha))} \|z_k\|^2. \end{aligned}$$

Subtracting by $f_k(x)$ and multiplying both sides by $2\eta_k$ yields

$$2\eta_k(f_k(x_{k+1}) - f_k(x)) \leq (1 - \mu\eta_k)\|x - x_k\|^2 + 2\eta_k\langle z_k, x_k - x\rangle - \|x - x_{k+1}\|^2 + 4\alpha^{-1}\eta_k^2\gamma_k^2 G^2 + \frac{\eta_k^2}{1 - (L\eta_k + \alpha)}\|z_k\|^2 .$$

For the specific choice $\alpha := (1-\rho)/2$, it holds that $L\eta_k + \alpha \leq \rho + (1-\rho)/2 = (1+\rho)/2$. Hence, we arrive at

$$2\eta_k(f_k(x_{k+1}) - f_k(x)) \leq (1 - \mu\eta_k)\|x - x_k\|^2 + 2\eta_k\langle z_k, x_k - x\rangle - \|x - x_{k+1}\|^2 + \frac{8G^2\eta_k^2\gamma_k^2}{1-\rho} + \frac{2\eta_k^2}{1-\rho}\|z_k\|^2 ,$$

as desired. □

After rearranging the inequality from lemma 3.5, setting $x = x^\star$, and dropping the $-\mu\eta_k$ term, we get

$$\left\|x^\star - x_{k+1}\right\|^2 \leq \left\|x^\star - x_k\right\|^2 + 2\eta_k\langle z_k, x_k - x^\star\rangle + 16G^2\eta_k^2\gamma_k^2 + 4\eta_k^2\|z_k\|^2 - 2\eta_k(f_k(x_{k+1}) - f_k(x^\star)),$$

provided $\eta_k \leq 1/(2L)$. By properties of conditional expectation and the definition of stochastic gradients, we know that $\mathbb{E}_k\langle z_k, x_k - x\rangle = \langle \mathbb{E}_k(z_k), x_k - x\rangle = 0$. Further note that $\mathbb{E}_k\|z_k\|^2 = \mathbb{E}_k\left\|g_k - \nabla\psi_k(x_k)\right\|^2$ and $\mathbb{E}_k(g_k) = \nabla\psi_k(x_k)$ per definition of g_k, hence $\mathbb{E}_k\|z_k\|^2 = \mathrm{Var}_k(g_k)$. Therefore, after applying conditional expectations to both sides of the above inequality, we obtain

$$\mathbb{E}_k\left\|x^\star - x_{k+1}\right\|^2 \leq \left\|x^\star - x_k\right\|^2 + 16G^2\eta_k^2\gamma_k^2 + 4\eta_k^2\,\mathrm{Var}_k(g_k) - 2\eta_k\mathbb{E}_k(f_k(x_{k+1}) - f_k(x^\star)).$$

This is not yet quite in the form needed to apply lemma 3.2. For one, we need to bound the noise term $\mathrm{Var}_k(g_k)$. Second, we cannot in general guarantee that $\mathbb{E}_k(f_k(x_{k+1}) - f_k(x^\star)) \geq 0$. However, as we show in the next lemma, we can find a lower bound that involves a nonnegative term plus a small negative term.

Lemma 3.6. Assume that assumptions 1 to 3 hold, and let $\eta_k \in (0, \rho L^{-1}]$ for all $k \in \mathbb{N}_0$ and some $\rho \in (0,1)$. Then the iterates $(x_k)_{k\in\mathbb{N}_0}$ generated by algorithm 2 with step

size schedule $(\eta_k)_{k\in\mathbb{N}_0}$ satisfy

$$\begin{aligned}\left\|x^\star - x_{k+1}\right\|^2 \leq (1-\mu\eta_k)\left\|x^\star - x_k\right\|^2 + \frac{2\eta_k^2}{1-\rho}\|z_k\|^2 + \frac{8G^2\eta_k^2\gamma_k^2}{1-\rho} \\ + 2\eta_k\langle z_k, x_k - x^\star\rangle + \mathcal{O}(\eta_k\gamma_k\alpha_k) - 2\eta_k(f_k(x_{k+1}) - f_k(x_k^\star))\end{aligned}$$

almost surely, for all $k \in \mathbb{N}_0$.

Proof. Let $k \in \mathbb{N}_0$. From lemma 3.5, we have

$$\begin{aligned}2\eta_k(f_k(x_{k+1}) - f_k(x^\star)) \leq (1-\mu\eta_k)\left\|x^\star - x_k\right\|^2 + 2\eta_k\langle z_k, x_k - x^\star\rangle - \left\|x^\star - x_{k+1}\right\|^2 \\ + \frac{8G^2\eta_k^2\gamma_k^2}{1-\rho} + \frac{2\eta_k^2}{1-\rho}\|z_k\|^2\end{aligned}$$

almost surely. Write

$$\begin{aligned}f_k(x_{k+1}) - f_k(x^\star) = (f_k(x_{k+1}) - f_k(x_k^\star)) + (f_k(x_k^\star) - f_k(\Pi_{\mathcal{X}}(x_k^\star))) \\ + (f_k(\Pi_{\mathcal{X}}(x_k^\star)) - f_k(x^\star)).\end{aligned} \tag{3.7}$$

By definition, it holds that

$$\begin{aligned}f_k(\Pi_{\mathcal{X}}(x_k^\star)) - f_k(x^\star) &= f(\Pi_{\mathcal{X}}(x_k^\star)) + \gamma_k\pi_k(\Pi_{\mathcal{X}}(x_k^\star)) - f(x^\star) - \gamma_k\pi_k(x^\star) \\ &\geq f(\Pi_{\mathcal{X}}(x_k^\star)) - f(x^\star) - \gamma_k\pi_k(x^\star) \\ &\geq -\gamma_k\pi_k(x^\star),\end{aligned}$$

where the last step follows from the fact that $x^\star$ minimizes f on $\mathcal{X}$ and $\Pi_{\mathcal{X}}(x_k^\star) \in \mathcal{X}$. Further, feasibility implies $\pi_k(x^\star) \leq \alpha_k$. Hence, combining with (3.7), we have

$$f_k(x_{k+1}) - f_k(x^\star) \geq (f_k(x_{k+1}) - f_k(x_k^\star)) + f_k(x_k^\star) - f_k(\Pi_{\mathcal{X}}(x_k^\star)) - \gamma_k\alpha_k.$$

For the next steps, we let $\tilde{\nabla} f(x)$ denote the minimum-norm subgradient in $\partial f(x)$, for all $x \in \operatorname{dom}(f)$. To analyze $f_k(x_k^\star) - f_k(\Pi_{\mathcal{X}}(x_k^\star))$, we first use convexity and Cauchy-Schwarz to get

$$f_k(x_k^\star) - f_k(\Pi_{\mathcal{X}}(x_k^\star)) \geq \langle \tilde{\nabla} f_k(\Pi_{\mathcal{X}}(x_k^\star)), x_k^\star - \Pi_{\mathcal{X}}(x_k^\star)\rangle \geq -\left\|\tilde{\nabla} f_k(\Pi_{\mathcal{X}}(x_k^\star))\right\| \operatorname{dist}(x_k^\star, \mathcal{X}).$$

The sequence $(x_k^\star)_{k\in\mathbb{N}}$ converges to $x^\star$ (TODO: this is wrong, but need only boundedness anyway.), which implies, by continuity of the projection map, $\lim_{k\to\infty}\Pi_{\mathcal{X}}(x_k^\star) = \Pi_{\mathcal{X}}(x^\star) = x^\star$. In particular, $(\Pi_{\mathcal{X}}(x_k^\star))_{k\in\mathbb{N}}$ is bounded, so lemma 3.1 implies that there exists $K_1 \in \mathbb{N}$ such that $\sup_{k\in\mathbb{N}}\left\|\tilde{\nabla} f_k(\Pi_{\mathcal{X}}(x_k^\star))\right\| \leq c_1\gamma_k$ for some $c_1 \in (0,\infty)$ and all

$k \geq K_1$. Hence,

$$f_k(x_k^\star) - f_k(\Pi_{\mathcal{X}}(x_k^\star)) \geq -c_1\gamma_k \operatorname{dist}(x_k^\star, \mathcal{X}),$$

for all $k \geq K_1$. By theorem 3.4, it holds that there exist $K_2 \in \mathbb{N}$ and $c_2 \in (0, \infty)$ such that, for all $k \geq K_2$, $\operatorname{dist}(x_k^\star, \mathcal{X}) \leq c_2\,\alpha_k$. Setting $K := \max(K_1, K_2)$, it therefore holds that

$$f_k(x_k^\star) - f_k(\Pi_{\mathcal{X}}(x_k^\star)) \geq -c_1c_2 \cdot \gamma_k\alpha_k$$

and

$$\begin{aligned} f_k(x_{k+1}) - f_k(x^\star) &\geq f_k(x_{k+1}) - f_k(x_k^\star) - c_1c_2 \cdot \gamma_k\alpha_k - \gamma_k\alpha_k \\ &= f_k(x_{k+1}) - f_k(x_k^\star) - (1 + c_1c_2)\,\gamma_k\alpha_k, \end{aligned}$$

for all $k \geq K$. Plugging this into our original estimate, we have

$$\begin{aligned} 2\eta_k\big(f_k(x_{k+1}) - f_k(x_k^\star) - O(1)\gamma_k\alpha_k\big) \leq (1 - \mu\eta_k)\left\|x^\star - x_k\right\|^2 - \left\|x^\star - x_{k+1}\right\|^2 \\ + \frac{8G^2\eta_k^2\gamma_k^2}{1-\rho} + \frac{2\eta_k^2}{1-\rho}\left\|z_k\right\|^2 + 2\eta_k\langle z_k, x_k - x^\star\rangle \end{aligned}$$

almost surely. Hence

$$\begin{aligned} 2\eta_k(f_k(x_{k+1}) - f_k(x_k^\star)) \leq (1 - \mu\eta_k)\left\|x^\star - x_k\right\|^2 - \left\|x^\star - x_{k+1}\right\|^2 + \frac{8G^2\eta_k^2\gamma_k^2}{1-\rho} \\ + \frac{2\eta_k^2}{1-\rho}\left\|z_k\right\|^2 + 2\eta_k\langle z_k, x_k - x^\star\rangle + O(1)\eta_k\gamma_k\alpha_k \end{aligned}$$

almost surely, for all $k \geq K$. The claim follows. □

From the above lemma, we can derive

$$\begin{aligned} \mathbb{E}_k\left\|x^\star - x_{k+1}\right\|^2 \leq \left\|x^\star - x_k\right\|^2 + \frac{2\eta_k^2}{1-\rho}\operatorname{Var}_k(g_k) + \frac{8G^2\eta_k^2\gamma_k^2}{1-\rho} + O(\eta_k\gamma_k\alpha_k) \\ - 2\eta_k\mathbb{E}_k(f_k(x_{k+1}) - f_k(x_k^\star)). \end{aligned}$$

The term $2\eta_k\mathbb{E}_k(f_k(x_{k+1}) - f_k(x_k^\star))$ is indeed nonnegative by optimality of $x_k^\star$ for f_k. Assuming that $\sum_{k=0}^\infty \eta_k\gamma_k\alpha_k < \infty$ and $\sum_{k=0}^\infty \eta_k^2\gamma_k^2 < \infty$, we are almost ready to apply lemma 3.2. The following lemma will give a bound on the gradient noise $\operatorname{Var}_k(g_k)$.

Lemma 3.7 (Bound on gradient noise)**.** Let assumption 1 hold. Then, in the situation of algorithm 2, there exists a constant $\sigma^2 \in (0, \infty)$ such that

$$\operatorname{Var}_k(g_k) \leq 4L^2\left\|x_k - x^\star\right\|^2 + (1 + \gamma_k^2)\sigma^2,$$

for all $k \in \mathbb{N}_0$.

Proof. We will use the inequality $(a+b)^2 \leq 2a^2 + 2b^2 \;\; \forall a, b \in \mathbb{R}$ multiple times. For $k \in \mathbb{N}_0$, we have

$$\mathrm{Var}_k(g_k) \leq \mathbb{E}_k \left\|g_k\right\|^2 \leq 2\mathbb{E}_k \left\|\nabla F_{\zeta_k}(x_k)\right\|^2 + 2\gamma_k^2 \mathbb{E}_k \left\|\nabla h_k(x_k; A(\xi_k), b(\xi_k))\right\|^2.$$

Using the (a. s.) L-smoothness of $x \mapsto F_\zeta(x)$, we have

$$\begin{aligned}
\mathbb{E}_k \left\|\nabla F_{\zeta_k}(x_k)\right\|^2 &= \mathbb{E}_k \left\|\nabla F_{\zeta_k}(x_k) - \nabla F_{\zeta_k}(x^\star) + \nabla F_{\zeta_k}(x^\star)\right\|^2 \\
&\leq 2\Big(\mathbb{E}_k \left\|\nabla F_{\zeta_k}(x_k) - \nabla F_{\zeta_k}(x^\star)\right\|^2 + \mathbb{E}_k \left\|\nabla F_{\zeta_k}(x^\star)\right\|^2\Big) \\
&\leq 2L^2 \left\|x_k - x^\star\right\|^2 + 2\,\mathbb{E}_k \left\|\nabla F_{\zeta_k}(x^\star)\right\|^2,
\end{aligned}$$

where in the last step we also used that x_k is $\mathcal{F}_k$-measurable. By one of our assumptions, we can find a point $x \in \mathbb{R}^n$ such that $\mathbb{E}\left\|F_\zeta(x)\right\|^2 < \infty$. By properties of conditional expectation, it then also follows that

$$\mathbb{E}\Big(\mathbb{E}_k \left\|\nabla F_{\zeta_k}(x)\right\|^2\Big) = \mathbb{E}\left\|\nabla F_\zeta(x)\right\|^2 < \infty,$$

and therefore

$$\mathbb{E}_k \left\|\nabla F_{\zeta_k}(x)\right\|^2 < \infty$$

almost surely. Hence, using a. s. L-smoothness of $x \mapsto F_\zeta(x)$ again, there exists a constant $M^2 \in (0, \infty)$ such that

$$\begin{aligned}
\mathbb{E}_k \left\|\nabla F_{\zeta_k}(x^\star)\right\|^2 &= \mathbb{E}_k \left\|\nabla F_{\zeta_k}(x^\star) - \nabla F_{\zeta_k}(x) + \nabla F_{\zeta_k}(x)\right\|^2 \\
&\leq 2\,\mathbb{E}_k \left\|\nabla F_{\zeta_k}(x^\star) - \nabla F_{\zeta_k}(x)\right\|^2 + 2\,\mathbb{E}_k \left\|\nabla F_{\zeta_k}(x)\right\|^2 \\
&\leq 2L^2 \left\|x^\star - x\right\|^2 + 2\,\mathbb{E}_k \left\|\nabla F_{\zeta_k}(x)\right\|^2 \\
&\leq \frac{1}{4}M^2.
\end{aligned}$$

Therefore, combining with the previous inequality, we get the bound

$$\mathbb{E}_k \left\|\nabla F_{\zeta_k}(x_k)\right\|^2 \leq 2L^2 \left\|x_k - x^\star\right\|^2 + \frac{1}{2}M^2.$$

Since, per assumption 1, it holds that the family $(\nabla h_k)_{k\in\mathbb{N}}$ is uniformly bounded, there exists a constant $\tilde{M}^2 \in (0, \infty)$ such that

$$\mathbb{E}_k \left\|\nabla h_k(x_k; A(\xi_k), b(\xi_k))\right\|^2 \leq \frac{1}{2}\tilde{M}^2.$$

Putting everything together, we obtain

$$\mathrm{Var}_k(g_k) \leq 4L^2 \left\|x_k - x^\star\right\|^2 + M^2 + \gamma_k^2 \tilde{M}^2 = 4L^2 \left\|x_k - x^\star\right\|^2 + M^2 + \gamma_k^2 \tilde{M}^2.$$

Setting $\sigma^2 := \max(M^2, \tilde{M}^2)$ yields the desired result. □

We are now ready to prove the main theorem of this section.

Theorem 3.8 (Almost sure convergence)**.** Assume assumptions 1 to 3 hold. Let $(x_k)_{k\in\mathbb{N}_0}$ be a sequence generated by algorithm 2 with parameters $(\eta_k)_{k\in\mathbb{N}_0}$, $(\gamma_k)_{k\in\mathbb{N}_0}$, $(h_k)_{k\in\mathbb{N}_0}$ that satisfiy

1. $\sum_{k=0}^{\infty} \eta_k = \infty$ and there exists $\rho \in (0,1)$ such that $\eta_k \leq \rho L^{-1}$ for all $k \in \mathbb{N}$.

2. $\sum_{k=0}^{\infty} \eta_k \gamma_k \alpha_k < \infty$.

3. $\sum_{k=0}^{\infty} \eta_k^2 \gamma_k^2 < \infty$.

Then, $\|x_k - x^\star\|$ converges almost surely and, in particular, $(x_k)_{k\in\mathbb{N}}$ is bounded almost surely. In the case $\mu > 0$, it further holds that $\liminf_{k\to\infty} \mathbb{E}\,\|x_k - x^\star\|^2 = 0$.

Proof. By lemma 3.6 (dropping the $-\mu\eta_k$ term), we have

$$\begin{aligned}\mathbb{E}_k \left\|x^\star - x_{k+1}\right\|^2 \leq \left\|x^\star - x_k\right\|^2 + \frac{2\eta_k^2}{1-\rho}\,\mathrm{Var}_k(g_k) + \frac{8G^2\eta_k^2\gamma_k^2}{1-\rho} \\ + O(\eta_k\gamma_k\alpha_k) - 2\eta_k\mathbb{E}_k(f_k(x_{k+1}) - f_k(x_k^\star)),\end{aligned}$$

Lemma 3.7 lets us bound the variance by

$$\mathrm{Var}_k(g_k) \leq 4L^2 \left\|x_k - x^\star\right\|^2 + (1+\gamma_k^2)\sigma^2$$

for a constant $\sigma^2 \in (0,\infty)$ and all $k \in \mathbb{N}_0$. Thus, we have

$$\begin{aligned}\mathbb{E}_k \left\|x^\star - x_{k+1}\right\|^2 &\leq \left\|x^\star - x_k\right\|^2 + \frac{2\eta_k^2}{1-\rho}\left(4L^2 \left\|x_k - x^\star\right\|^2 + (1+\gamma_k^2)\sigma^2\right) \\ &\quad + O(\eta_k\gamma_k\alpha_k) - 2\eta_k\mathbb{E}_k(f_k(x_{k+1}) - f_k(x_k^\star)) \\ &= \left(1 + 8L^2(1-\rho)^{-1}\eta_k^2\right)\left\|x^\star - x_k\right\|^2 + 2\eta_k^2(1-\rho)^{-1}(1+\gamma_k^2)\sigma^2 \\ &\quad + O(\eta_k\gamma_k\alpha_k) - 2\eta_k\mathbb{E}_k(f_k(x_{k+1}) - f_k(x_k^\star)),\end{aligned}$$

for all $k \in \mathbb{N}_0$. Define the nonnegative sequences $(a_k)_{k\in\mathbb{N}_0}, (b_k)_{k\in\mathbb{N}_0}, (c_k)_{k\in\mathbb{N}_0}$ by

$$\begin{aligned} a_k &:= 8L^2(1-\rho)^{-1}\eta_k^2 \\ b_k &:= 2\eta_k^2(1-\rho)^{-1}(1+\gamma_k^2)\sigma^2 + O(\eta_k\gamma_k\alpha_k) \\ c_k &:= 2\eta_k\mathbb{E}_k(f_k(x_{k+1}) - f_k(x_k^\star)). \end{aligned}$$

Note that c_k is indeed nonnegative, since $x_k^\star$ minimizes f_k. The above inequality now takes the form

$$\mathbb{E}_k \left\|x^\star - x_{k+1}\right\|^2 \leq (1+a_k)\left\|x^\star - x_k\right\|^2 + b_k - c_k.$$

Our assumptions imply that $\sum_{k=0}^\infty a_k < \infty$ and $\sum_{k=0}^\infty b_k < \infty$, so we can apply lemma 3.2, which implies that with probability one the sequence $\|x^\star - x_k\|^2$ converges and $\sum_{k=0}^\infty \eta_k\mathbb{E}_k(f_k(x_{k+1}) - f_k(x_k^\star)) < \infty$. By the bounded convergence theorem (TODO: add ref), it further holds that

$$\begin{aligned} \infty > \mathbb{E}\Big(\sum_{k=0}^\infty \eta_k\mathbb{E}_k(f_k(x_{k+1}) - f_k(x_k^\star))\Big) &= \sum_{k=0}^\infty \eta_k\mathbb{E}(\mathbb{E}_k(f_k(x_{k+1}) - f_k(x_k^\star))) \\ &= \sum_{k=0}^\infty \eta_k\mathbb{E}(f_k(x_{k+1}) - f_k(x_k^\star)). \end{aligned}$$

Since $\sum_{k=0}^\infty \eta_k = \infty$, it must therefore hold that

$$\liminf_{k\to\infty} \mathbb{E}(f_k(x_{k+1}) - f_k(x_k^\star)) = 0.$$

Strong convexity of f_k and optimaliy of $x_k^\star$ for f_k imply

$$f_k(x_{k+1}) - f_k(x_k^\star) \geq \frac{\mu}{2}\left\|x_{k+1} - x_k^\star\right\|^2.$$

If $\mu > 0$, we therefore have $\liminf_{k\to\infty} \mathbb{E}\left\|x_{k+1} - x_k^\star\right\|^2 = 0$, which implies that

$$\liminf_{k\to\infty} \mathbb{E}\left\|x_k - x^\star\right\|^2 = 0,$$

since we also know that $x_k^\star$ converges to $x^\star$ in the case $\mu > 0$, by theorem 3.4. □

3.2. Convergence rates in expectation

In the previous section, we established that the iterates $(x_k)_{k\in\mathbb{N}_0}$ of algorithm 2 are bounded with probability one, provided that the parameters $(h_k)_{k\in\mathbb{N}_0}$, $(\eta_k)_{k\in\mathbb{N}_0}$, $(\gamma_k)_{k\in\mathbb{N}_0}$ satisfy certain conditions, which are captured in the following assumption.

Assumption 4. The parameters $(h_k)_{k\in\mathbb{N}_0}$, $(\eta_k)_{k\in\mathbb{N}_0}$, $(\gamma_k)_{k\in\mathbb{N}_0}$ of algorithm 2 satisfiy

1. $\sum_{k=0}^{\infty}\eta_k = \infty$ and there exist constants $\rho \in (0,1)$ and $K \in \mathbb{N}$ such that $\eta_k \leq \rho L^{-1}$ for all $k \in \mathbb{N}$ with $k \geq K$.

2. $\sum_{k=0}^{\infty}\eta_k\gamma_k\alpha_k < \infty$.

3. $\sum_{k=0}^{\infty}\eta_k^2\gamma_k^2 < \infty$.

Lemma 3.9. Let assumptions 1 to 4 hold. Then, for all $k \in \mathbb{N}$, it holds that

$$\begin{aligned}2\eta_k(f_k(x_k) - f_k(x)) \leq (1-\mu\eta_k)\|x - x_k\|^2 + 2\eta_k\langle z_k, x_k - x\rangle - \|x - x_{k+1}\|^2 \\ + \frac{8G^2\eta_k^2\gamma_k^2}{1-\rho} + \frac{2\eta_k^2}{1-\rho}\|z_k\|^2 + \eta_k^2 O(\gamma_k^2)\end{aligned}$$

almost surely.

Proof. By lemma 3.5,

$$\begin{aligned}2\eta_k(f_k(x_{k+1}) - f_k(x)) \leq (1-\mu\eta_k)\|x - x_k\|^2 + 2\eta_k\langle z_k, x_k - x\rangle - \|x - x_{k+1}\|^2 \\ + \frac{8G^2\eta_k^2\gamma_k^2}{1-\rho} + \frac{2\eta_k^2}{1-\rho}\|z_k\|^2\end{aligned} \tag{3.8}$$

almost surely, for all $k \in \mathbb{N}_0$ and $x \in \mathbb{R}^n$. Let $\tilde{\nabla} f_k(x_k)$ denote a subgradient of f_k at x_k. By convexity of f_k and Cauchy-Schwarz, we have

$$f_k(x_{k+1}) \geq f_k(x_k) + \langle \tilde{\nabla} f_k(x_k), x_{k+1} - x_k\rangle \geq f_k(x_k) - \left\|\tilde{\nabla} f_k(x_k)\right\| \|x_{k+1} - x_k\| .$$

Therefore,

$$\begin{aligned}2\eta_k(f_k(x_k) - f_k(x)) \leq (1-\mu\eta_k)\|x - x_k\|^2 + 2\eta_k\langle z_k, x_k - x\rangle - \|x - x_{k+1}\|^2 \\ + \frac{8G^2\eta_k^2\gamma_k^2}{1-\rho} + \frac{2\eta_k^2}{1-\rho}\|z_k\|^2 + 2\eta_k\left\|\tilde{\nabla} f_k(x_k)\right\| \|x_{k+1} - x_k\|\end{aligned} \tag{3.9}$$

almost surely, for all $x \in \mathbb{R}^n$. The sequence $(x_k)_{k\in\mathbb{N}_0}$ is bounded almost surely

(theorem 3.8), hence lemma 3.1 implies

$$\left\|\tilde{\nabla} f_k(x_k)\right\| = O(\gamma_k)$$

almost surely. We will now analyze $\|x_{k+1} - x_k\|$. Let $y_k := \operatorname{prox}_{\eta_k r}(x_k)$. Then, using the triangle inequality and the nonexpansiveness property of the proximal operator (TODO), we have

$$\begin{aligned}
\|x_{k+1} - x_k\| &= \left\|x_{k+1} - y_k + y_k - x_k\right\| \\
&\leq \left\|\operatorname{prox}_{\eta_k r}(x_k - \eta_k g_k) - \operatorname{prox}_{\eta_k r}(x_k)\right\| + \mathbb{E}\left\|y_k - x_k\right\| \\
&\leq \eta_k \left\|g_k\right\| + \left\|y_k - x_k\right\|.
\end{aligned}$$

For the second term, we use the definition of y_k as the solution to

$$\min_{x \in \mathbb{R}^n} r(x) + \frac{1}{2\eta_k} \|x - x_k\|^2.$$

By the first-order optimality condition, there exists a subgradient $\tilde{\nabla} r(y_k) \in \partial r(y_k)$ such that

$$\tilde{\nabla} r(y_k) + \frac{y_k - x_k}{\eta_k} = 0 \iff \frac{x_k - y_k}{\eta_k} \in \partial r(y_k).$$

Since x_k is in the domain of r for all $k \in \mathbb{N}$ (TODO) (not necessarily for $k = 0$), we can now use the above, together with convexity, to get

$$r(x_k) - r(y_k) \geq \langle \eta_k^{-1}(x_k - y_k), x_k - y_k \rangle = \eta_k^{-1} \left\|x_k - y_k\right\|^2,$$

for all $k \in \mathbb{N}$. Finally, local Lipschitz continuity of r (assumption 1) yields (a. s.)

$$L_r \left\|x_k - y_k\right\| \geq r(x_k) - r(y_k) \geq \eta_k^{-1} \left\|x_k - y_k\right\|^2 \iff \left\|x_k - y_k\right\| \leq \eta_k L_r,$$

for some $L_r \in (0, \infty)$, where we again used that $(x_k)_{k \in \mathbb{N}_0}$ is bounded almost surely (theorem 3.8) and the prox operator is nonexpansive (TODO). Combining with (3.9), we therefore obtain

$$\begin{aligned}
2\eta_k(f_k(x_k) - f_k(x)) \leq (1 - \mu\eta_k)\|x - x_k\|^2 + 2\eta_k\langle z_k, x_k - x\rangle - \|x - x_{k+1}\|^2 \\
+ \frac{8G^2\eta_k^2\gamma_k^2}{1-\rho} + \frac{2\eta_k^2}{1-\rho}\|z_k\|^2 + \eta_k^2 O(\gamma_k)\left(\left\|g_k\right\| + L_r\right),
\end{aligned}$$

almost surely, for all $k \in \mathbb{N}$. Finally, we can use lemma 3.1, which tells us that $\left\|g_k\right\| = O(\gamma_k)$ almost surely (again using boundedness of $(x_k)_{k \in \mathbb{N}_0}$), hence we arrive

at

$$2\eta_k(f_k(x_k) - f_k(x)) \leq (1 - \mu\eta_k)\|x - x_k\|^2 + 2\eta_k\langle z_k, x_k - x\rangle - \|x - x_{k+1}\|^2 + \frac{8G^2\eta_k^2\gamma_k^2}{1-\rho} + \frac{2\eta_k^2}{1-\rho}\|z_k\|^2 + \eta_k^2 O(\gamma_k^2),$$

as desired. □

Lemma 3.10. Let assumptions 1 and 4 hold. For all $k \in \mathbb{N}_0$, it holds that

$$\|z_k\|^2 \leq 8L_{\text{loc}}^2 + 8\gamma_k^2 G^2,$$

with probability one, where $L_{\text{loc}} \in (0, \infty)$.

Proof. We have

$$\begin{aligned}\|g_k\| &= \|\nabla F_{\zeta_k}(x_k) + \gamma_k h_k(x_k; A(\xi)_k, b(\xi_k))\| \\ &\leq \|\nabla F_{\zeta_k}(x_k)\| + \gamma_k \|h_k(x_k; A(\xi_k), b(\xi_k))\|.\end{aligned}$$

By almost-sure boundedness of $(x_k)_{k\in\mathbb{N}_0}$ (theorem 3.8) and local Lipschitz smoothness of $x \mapsto F_\zeta(x)$ (a. s.), it holds that

$$\|F_{\zeta_k}(x_k)\| \leq L_{\text{loc}} \text{ (a. s.)}.$$

By assumption 1, we have

$$\sup_{k\in\mathbb{N}_0}\sup_{x\in\mathbb{R}^n}\|\nabla h_k(x; A(\xi_k), b(\xi_k))\| \leq G.$$

Using the inequality $(a+b)^2 \leq 2(a^2+b^2)\ \forall a, b \in \mathbb{R}$, we have

$$\|g_k\|^2 \leq 2L_{\text{loc}}^2 + 2\gamma_k^2 G^2$$

with probability one. The claim follows from $\nabla\psi_k(x_k) = \mathbb{E}_k(g_k)$ and

$$\|g_k - \nabla\psi_k(x_k)\|^2 \leq 2\|g_k\|^2 + 2\|\nabla\psi_k(x_k)\|^2 \leq 2\|g_k\|^2 + 2\mathbb{E}_k\|g_k\|^2,$$

where the last step uses Jensen's inequality. □

Lemma 3.11. Let the parameters $(\eta_k)_{k\in\mathbb{N}_0}, (\gamma_k)_{k\in\mathbb{N}_0}, (h_k)_{k\in\mathbb{N}_0}$ of algorithm 2 satisfy

$$\eta_k = O\left(\frac{1}{k^c \log^{(1+3e)/2}(k+1)}\right),\ \gamma_k = O(\log^e(k+1)),\ \alpha_k = O\left(\frac{1}{k^d}\right)$$

with $c \in [1/2, 1)$, $d \in (1/2, \infty)$, and $e \in (0, \infty)$. Then, assumption 4 holds.

Proof. See the proof of lemma 9 in [3]. □

Theorem 3.12. Let assumptions 1 to 3 hold. Let the parameters $(\eta_k)_{k\in\mathbb{N}_0}, (\gamma_k)_{k\in\mathbb{N}_0}$, $(h_k)_{k\in\mathbb{N}_0}$ be as in lemma 3.11. Further, let the iterate weights $(w_k)_{k\in\mathbb{N}}$ be defined as $w_k := \eta_k$, for all $k \in \mathbb{N}$. Then there exists a constant $k_0 \in \mathbb{N}$ such that the averaged iterate $\bar{x}_K$, generated by algorithm 2 with parameters $(\eta_k)_{k\in\mathbb{N}_0}$, $(\gamma_k)_{k\in\mathbb{N}_0}$, $(h_k)_{k\in\mathbb{N}_0}$, $(w_k)_{k\in\mathbb{N}}$, satisfies

$$f(\bar{x}_K) - f(x^\star) \leq \frac{S_{1,k_0-1}}{S_K}(f(\bar{x}_{1,k_0-1}) - f(x^\star)) + \frac{\left\|x^\star - x_{k_0}\right\|^2}{2S_K} + \frac{\sum_{k=k_0}^K \eta_k^2 O(\gamma_k^2)}{2S_K} + \frac{\sum_{k=k_0}^K \eta_k \gamma_k \alpha_k}{S_K} + \frac{\sum_{k=k_0}^K \eta_k \langle z_k, x_k - x^\star\rangle}{S_K}$$

almost surely, for all $K \in \mathbb{N}$ with $K \geq k_0$, where $S_{t,k} := \sum_{i=t}^k w_i$, and $\bar{x}_{t,k} := S_{t,k}^{-1} \sum_{i=t}^k w_i x_i \ \forall t, k \in \mathbb{N}$. Furthermore, there exist constants $k_1 \in \mathbb{N}$ and $\tau \in (0, \infty)$ such that

$$\operatorname{dist}(\bar{x}_K, \mathcal{X}) \leq \frac{S_{1,k_1-1}}{S_K} \operatorname{dist}(\bar{x}_{1,k_1-1}, \mathcal{X}) + \frac{\gamma_{k_1}^{-1} \left\|x^\star - x_{k_1}\right\|^2}{\tau S_K} + \frac{\sum_{k=k_1}^K \eta_k^2 O(\gamma_k)}{\tau S_K} + 2\frac{\sum_{k=k_1}^K \eta_k \alpha_k}{\tau S_K} + 2\frac{\sum_{k=k_1}^K \gamma_k^{-1} \eta_k \langle z_k, x_k - \Pi_{\mathcal{X}}(x_k)\rangle}{\tau S_K}$$

almost surely, for all $K \in \mathbb{N}$ with $K \geq k_1$.

Proof. Let $k_0 \in \mathbb{N}$ be large enough such that $\eta_{k_0} \leq 1/(2L)$ and fix some $k \in \mathbb{N}$ with $k \geq k_0$. Lemma 3.11 implies that the parameters $(\eta_k)_{k\in\mathbb{N}_0}, (\gamma_k)_{k\in\mathbb{N}_0}, (h_k)_{k\in\mathbb{N}_0}$ satisfy assumption 4. Thus, we can apply lemma 3.9, which implies

$$2\eta_k(f_k(x_k) - f_k(x)) \leq (1 - \mu\eta_k)\left\|x - x_k\right\|^2 + 2\eta_k\langle z_k, x_k - x\rangle - \left\|x - x_{k+1}\right\|^2 + 16G^2\eta_k^2\gamma_k^2 + 4\eta_k^2\left\|z_k\right\|^2 + \eta_k^2 O(\gamma_k^2),$$

almost surely, for all $x \in \mathbb{R}^n$. Combining with lemma 3.10, we find that

$$2\eta_k(f_k(x_k) - f_k(x)) \leq (1 - \mu\eta_k)\left\|x - x_k\right\|^2 + 2\eta_k\langle z_k, x_k - x\rangle - \left\|x - x_{k+1}\right\|^2 + \eta_k^2 O(\gamma_k^2)$$

almost surely, for all $x \in \mathbb{R}^n$. Assume now that $x \in \mathcal{X}$. By a property of π_k, the fact that $(x_k)_{k\in\mathbb{N}}$ is bounded almost surely (theorem 3.8), and lemma 3.3, we can deduce

that there exists $\tau \in (0, \infty)$ such that

$$\begin{aligned} f_k(x_k) - f_k(x) &= f(x_k) - f(x) + \gamma_k(\pi_k(x_k) - \pi_k(x)) \\ &\geq f(x_k) - f(x) + \gamma_k(\tau \operatorname{dist}(x_k, \mathcal{X}) - \alpha_k) \end{aligned}$$

with probability one. Hence, we have

$$\begin{aligned} 2\eta_k(f(x_k) - f(x)) \leq (1 - \mu\eta_k)\|x - x_k\|^2 - \|x - x_{k+1}\|^2 + 2\eta_k\langle z_k, x_k - x\rangle + \eta_k^2 O(\gamma_k^2) \\ + 2\eta_k\gamma_k\alpha_k - 2\tau\eta_k\gamma_k \operatorname{dist}(x_k, \mathcal{X}), \end{aligned} \tag{3.10}$$

for all $x \in \mathcal{X}$. We can drop the $-\mu\eta_k$ and $-\operatorname{dist}(x_k, \mathcal{X})$ terms. Then, summing both sides from $k = k_0$ to $K \in \mathbb{N}$ with $K \geq k_0$, we obtain

$$\begin{aligned} 2\sum_{k=k_0}^{K} \eta_k(f(x_k) - f(x)) &\leq \left\|x - x_{k_0}\right\|^2 - \|x - x_{K+1}\|^2 + 2\sum_{k=k_0}^{K} \eta_k\langle z_k, x_k - x\rangle \\ &\quad + 2\sum_{k=k_0}^{K} \eta_k\gamma_k\alpha_k + \sum_{k=k_0}^{K} \eta_k^2 O(\gamma_k^2) \\ &\leq \left\|x - x_{k_0}\right\|^2 + 2\sum_{k=k_0}^{K} \eta_k\langle z_k, x_k - x\rangle \\ &\quad + 2\sum_{k=k_0}^{K} \eta_k\gamma_k\alpha_k + \sum_{k=k_0}^{K} \eta_k^2 O(\gamma_k^2) \end{aligned}$$

almost surely, for all $x \in \mathcal{X}$. We define $S_{t,k} := \sum_{i=t}^{k} \eta_i$, $S_k := S_{1,k}$, and $\bar{x}_{t,k} := S_{t,k}^{-1} \sum_{i=t}^{k} \eta_i x_i$ for $t, k \in \mathbb{N}$. Setting $x := x^\star \in \mathcal{X}^\star$, dividing both sides by $2\sum_{k=k_0}^{K} \eta_k$ and using convexity of f, we obtain

$$\begin{aligned} f(\bar{x}_{k_0,K}) - f(x^\star) \leq \frac{\left\|x^\star - x_{k_0}\right\|^2}{2S_{k_0,K}} + \frac{\sum_{k=k_0}^{K} \eta_k\langle z_k, x_k - x^\star\rangle}{S_{k_0,K}} \\ + \frac{\sum_{k=k_0}^{K} \eta_k\gamma_k\alpha_k}{S_{k_0,K}} + \frac{\sum_{k=k_0}^{K} \eta_k^2 O(\gamma_k^2)}{2S_{k_0,K}}. \end{aligned}$$

Note that

$$\bar{x}_K = \frac{S_{1,k_0-1}}{S_K}\bar{x}_{1,k_0-1} + \frac{S_{k_0,K}}{S_K}\bar{x}_{k_0,K}$$

and

$$\frac{S_{1,k_0-1}}{S_K} + \frac{S_{k_0,K}}{S_K} = 1,$$

hence, using convexity of f again, we have

$$f(\bar{x}_K) - f(x^\star) \leq \frac{S_{1,k_0-1}}{S_K}(f(\bar{x}_{1,k_0-1}) - f(x^\star)) + \frac{S_{k_0,K}}{S_K}(f(\bar{x}_{k_0,K}) - f(x^\star)).$$

Combining with the latest bound on $(f(\bar{x}_{k_0,K}) - f(x^\star))$, we arrive at

$$\begin{aligned} f(\bar{x}_K) - f(x^\star) \leq \frac{S_{1,k_0-1}}{S_K}(f(\bar{x}_{1,k_0-1}) - f(x^\star)) + \frac{\left\|x^\star - x_{k_0}\right\|^2}{2S_K} + \frac{\sum_{k=k_0}^K \eta_k \langle z_k, x_k - x^\star\rangle}{S_K} \\ + \frac{\sum_{k=k_0}^K \eta_k \gamma_k \alpha_k}{S_K} + \frac{\sum_{k=k_0}^K \eta_k^2 \mathcal{O}(\gamma_k^2)}{2S_K} \end{aligned}$$

almost surely. Next, we will derive the desired bound for $\operatorname{dist}(\bar{x}_K, \mathcal{X})$. Again fix $k \geq k_0$. For the choice $x := \Pi_{\mathcal{X}}(x_k)$, inequality (3.10) gives us

$$\begin{aligned} 2\eta_k(f(x_k) - f(\Pi_{\mathcal{X}}(x_k))) \leq (1 - \mu\eta_k)\operatorname{dist}(x_k, \mathcal{X})^2 - \|\Pi_{\mathcal{X}}(x_k) - x_{k+1}\|^2 + \eta_k^2 \mathcal{O}(\gamma_k^2) \\ + 2\eta_k\langle z_k, x_k - \Pi_{\mathcal{X}}(x_k)\rangle + 2\eta_k\gamma_k\alpha_k - 2\tau\eta_k\gamma_k \operatorname{dist}(x_k, \mathcal{X}). \end{aligned}$$

Note that $\|\Pi_{\mathcal{X}}(x_k) - x_{k+1}\| \geq \operatorname{dist}(x_{k+1}, \mathcal{X})$, hence, after additionally rearranging, we obtain

$$\begin{aligned} \operatorname{dist}(x_{k+1}, \mathcal{X})^2 \leq (1 - \mu\eta_k)\operatorname{dist}(x_k, \mathcal{X})^2 + \eta_k^2 \mathcal{O}(\gamma_k^2) + 2\eta_k\langle z_k, x_k - \Pi_{\mathcal{X}}(x_k)\rangle \\ + 2\eta_k\gamma_k\alpha_k - 2\tau\eta_k\gamma_k \operatorname{dist}(x_k, \mathcal{X}) + 2\eta_k(f(\Pi_{\mathcal{X}}(x_k)) - f(x_k)). \end{aligned}$$

Almost sure boundedness of $(x_k)_{k\in\mathbb{N}}$ (theorem 3.8) implies that $(\Pi_{\mathcal{X}}(x_k))_{k\in\mathbb{N}_0}$ is also bounded almost surely, since continuous functions map compacta to compacta. Therefore, by local Lipschitz continuity of f (lemma 3.1), there exists a constant $M \in [0, \infty)$ such that

$$f(\Pi_{\mathcal{X}}(x_k)) - f(x_k) \leq M \operatorname{dist}(x_k, \mathcal{X}) \ \text{(a. s.)}.$$

Combining with the previous inequality and gathering terms involving $\operatorname{dist}(x_k, \mathcal{X})$, we arrive at

$$\begin{aligned} \operatorname{dist}(x_{k+1}, \mathcal{X})^2 \leq (1 - \mu\eta_k)\operatorname{dist}(x_k, \mathcal{X})^2 + \eta_k^2 \mathcal{O}(\gamma_k^2) + 2\eta_k\langle z_k, x_k - \Pi_{\mathcal{X}}(x_k)\rangle \\ + 2\eta_k\gamma_k\alpha_k - 2\eta_k(\tau\gamma_k - M)\operatorname{dist}(x_k, \mathcal{X}). \end{aligned}$$

Since $\gamma_k \uparrow \infty$, there exists $k_1 \in \mathbb{N}$ such that $\tau_k\gamma_k - M \geq \tau\gamma_k/2$ for all natural numbers

$k \geq k_1$. Thus,

$$\begin{aligned}\operatorname{dist}(x_{k+1}, \mathcal{X})^2 \leq (1 - \mu\eta_k) \operatorname{dist}(x_k, \mathcal{X})^2 + \eta_k^2 O(\gamma_k^2) + 2\eta_k \langle z_k, x_k - \Pi_{\mathcal{X}}(x_k)\rangle \\ + 2\eta_k\gamma_k\alpha_k - \tau\eta_k\gamma_k \operatorname{dist}(x_k, \mathcal{X}),\end{aligned} \tag{3.11}$$

for all $k \geq k_1$. Dropping the $-\mu\eta_k$ term and multiplying both sides by γ_k^{-1}, we get

$$\begin{aligned}\gamma_k^{-1} \operatorname{dist}(x_{k+1}, \mathcal{X})^2 \leq \gamma_k^{-1} \operatorname{dist}(x_k, \mathcal{X})^2 + \eta_k^2 O(\gamma_k) \\ + 2\gamma_k^{-1}\eta_k \langle z_k, x_k - \Pi_{\mathcal{X}}(x_k)\rangle + 2\eta_k\alpha_k - \tau\eta_k \operatorname{dist}(x_k, \mathcal{X}),\end{aligned}$$

for all $k \geq k_1$. Since γ_k is nondecreasing, we have $\gamma_k^{-1} \geq \gamma_{k+1}^{-1}$ for all $k \in \mathbb{N}$. Setting $d_j := \gamma_j^{-1} \operatorname{dist}(x_j, \mathcal{X})^2 \ \ \forall j \in \mathbb{N}$, we therefore have

$$d_{k+1} \leq d_k + \eta_k^2 O(\gamma_k) + 2\gamma_k^{-1}\eta_k \langle z_k, x_k - \Pi_{\mathcal{X}}(x_k)\rangle + 2\eta_k\alpha_k - \tau\eta_k \operatorname{dist}(x_k, \mathcal{X}),$$

for all $k \geq k_1$. Rearranging again and summing from $k = k_1$ to $k = K$ for $K \in \mathbb{N}$ with $K \geq k_1$, we have

$$\begin{aligned}\tau \sum_{k=k_1}^{K} \eta_k \operatorname{dist}(x_k, \mathcal{X}) &\leq d_{k_1} - d_{K+1} + \sum_{k=k_1}^{K} \eta_k^2 O(\gamma_k) + 2\sum_{k=k_1}^{K} \eta_k\alpha_k \\ &\quad + 2\sum_{k=k_1}^{K} \gamma_k^{-1}\eta_k \langle z_k, x_k - \Pi_{\mathcal{X}}(x_k)\rangle \\ &\leq d_{k_1} + \sum_{k=k_1}^{K} \eta_k^2 O(\gamma_k) + 2\sum_{k=k_1}^{K} \eta_k\alpha_k \\ &\quad + 2\sum_{k=k_1}^{K} \gamma_k^{-1}\eta_k \langle z_k, x_k - \Pi_{\mathcal{X}}(x_k)\rangle,\end{aligned}$$

where we used $d_{K+1} \geq 0$ in the last step. The distance functional $x \mapsto \operatorname{dist}(x, \mathcal{X})$ is convex (TODO: Show this in an example.), so if we divide both sides of the above inequality by $\tau \cdot S_{k_1,K}$, we obtain

$$\begin{aligned}\operatorname{dist}(\bar{x}_{k_1,K}, \mathcal{X}) \leq \frac{d_{k_1}}{\tau S_{k_1,K}} + \frac{\sum_{k=k_1}^{K} \eta_k^2 O(\gamma_k)}{\tau S_{k_1,K}} + 2\frac{\sum_{k=k_1}^{K} \eta_k\alpha_k}{\tau S_{k_1,K}} \\ + 2\frac{\sum_{k=k_1}^{K} \gamma_k^{-1}\eta_k \langle z_k, x_k - \Pi_{\mathcal{X}}(x_k)\rangle}{\tau S_{k_1,K}}.\end{aligned}$$

To derive a bound for $\operatorname{dist}(\bar{x}_K, \mathcal{X})$, we proceed similarly as before. We have

$$\bar{x}_K = \frac{S_{1,k_1-1}}{S_K}\bar{x}_{1,k_1-1} + \frac{S_{k_1,K}}{S_K}\bar{x}_{k_1,K}$$

and

$$\frac{S_{1,k_1-1}}{S_K} + \frac{S_{k_1,K}}{S_K} = 1.$$

Combining with the latest bound on $\operatorname{dist}(\bar{x}_{k_1,K}, \mathcal{X})$, we obtain the desired bound

$$\begin{aligned}\operatorname{dist}(\bar{x}_K, \mathcal{X}) &\leq \frac{S_{1,k_1-1}}{S_K}\operatorname{dist}(\bar{x}_{1,k_1-1}, \mathcal{X}) + \frac{S_{k_1,K}}{S_K}\operatorname{dist}(\bar{x}_{k_1,K}, \mathcal{X}) \\ &\leq \frac{S_{1,k_1-1}}{S_K}\operatorname{dist}(\bar{x}_{1,k_1-1}, \mathcal{X}) + \frac{d_{k_1}}{\tau S_K} + \frac{\sum_{k=k_1}^K \eta_k^2 \mathcal{O}(\gamma_k)}{\tau S_K} + 2\frac{\sum_{k=k_1}^K \eta_k \alpha_k}{\tau S_K} \\ &\quad + 2\frac{\sum_{k=k_1}^K \gamma_k^{-1}\eta_k \langle z_k, x_k - \Pi_{\mathcal{X}}(x_k)\rangle}{\tau S_K}\end{aligned}$$

almost surely, or all $K \geq k_1$. This concludes the proof. □

Corollary 3.13 (Convergence rates in expectation - convex case)**.** Let assumptions 1 to 3 hold. Let the parameters $(\eta_k)_{k\in\mathbb{N}_0}$, $(\gamma_k)_{k\in\mathbb{N}_0}$, $(h_k)_{k\in\mathbb{N}_0}$ be the same as in lemma 3.11. Further, let the iterate weights $(w_k)_{k\in\mathbb{N}}$ be defined as $w_k := \eta_k$, for all $k \in \mathbb{N}$. Then, the averaged iterate $\bar{x}_K$, generated by algorithm 2 with parameters $(\eta_k)_{k\in\mathbb{N}_0}$, $(\gamma_k)_{k\in\mathbb{N}_0}$, $(h_k)_{k\in\mathbb{N}_0}$, $(w_k)_{k\in\mathbb{N}}$, satisfies

$$\mathbb{E}|f(\bar{x}_K) - f(x^\star)| = \max(1, \tau^{-1})\,\mathcal{O}\Big(\frac{\log^{(1+3e)/2}(K)}{K^{1-c}}\Big).$$

Further, it also holds that

$$\mathbb{E}(\operatorname{dist}(\bar{x}_K, \mathcal{X})) = \tau^{-1}\mathcal{O}\Big(\frac{\log^{(1+3e)/2}(K)}{K^{1-c}}\Big).$$

Proof. Theorem 3.12 gives us an upper bound for $f(\bar{x}_K) - f(x^\star)$,

$$\begin{aligned}f(\bar{x}_K) - f(x^\star) \leq \frac{S_{1,k_0-1}}{S_K}(f(\bar{x}_{1,k_0-1}) - f(x^\star)) + \frac{\left\|x^\star - x_{k_0}\right\|^2}{2S_K} + \frac{\sum_{k=k_0}^K \eta_k^2 \mathcal{O}(\gamma_k^2)}{2S_K} \\ + \frac{\sum_{k=k_0}^K \eta_k \gamma_k \alpha_k}{S_K} + \frac{\sum_{k=k_0}^K \eta_k \langle z_k, x_k - x^\star\rangle}{S_K}\end{aligned}$$

almost surely, for all $K \in \mathbb{N}$ with $K \geq k_0$, where $S_K := \sum_{k=1}^K \eta_k$. Note that

$$\mathbb{E}(\langle z_k, x_k - x^\star\rangle) = \mathbb{E}(\mathbb{E}_k(\langle z_k, x_k - x^\star\rangle)) = \mathbb{E}(\langle \mathbb{E}_k(z_k), x_k - x^\star\rangle) = 0,$$

for all $k \in \mathbb{N}$. Hence, taking expectations on both sides of the above inequality, we get

$$\mathbb{E}(f(\bar{x}_K) - f(x^\star)) \leq \frac{S_{1,k_0-1}}{S_K}\mathbb{E}(f(\bar{x}_{1,k_0-1}) - f(x^\star)) + \frac{\mathbb{E}\left\|x^\star - x_{k_0}\right\|^2}{2S_K} + \frac{\sum_{k=k_0}^K \eta_k^2 \mathcal{O}(\gamma_k^2)}{2S_K} + \frac{\sum_{k=k_0}^K \eta_k \gamma_k \alpha_k}{S_K}. \tag{3.12}$$

By convexity, we have for all $k \in \mathbb{N}$

$$\begin{aligned} f(\bar{x}_k) - f(x^\star) &\geq \langle \tilde{\nabla} f(x^\star), \bar{x}_k - x^\star \rangle \\ &= \langle \tilde{\nabla} f(x^\star), \bar{x}_k - \Pi_{\mathcal{X}}(\bar{x}_k) \rangle + \langle \tilde{\nabla} f(x^\star), \Pi_{\mathcal{X}}(\bar{x}_k) - x^\star \rangle, \end{aligned}$$

for any subgradient $\tilde{\nabla} f(x^\star) \in \partial f(x^\star)$. Optimality of $x^\star$ for f on $\mathcal{X}$ implies that there exists a subgradient $\tilde{\nabla} f(x^\star) \in \partial f(x^\star)$ such that $\langle \tilde{\nabla} f(x^\star), \Pi_{\mathcal{X}}(\bar{x}_k) - x^\star \rangle \geq 0$. Combining these facts with the above inequality and applying Cauchy-Schwarz, we obtain

$$f(\bar{x}_k) - f(x^\star) \geq -\left\|\tilde{\nabla} f(x^\star)\right\| \operatorname{dist}(\bar{x}_k, \mathcal{X}), \tag{3.13}$$

for all $k \in \mathbb{N}$. Taking expectations in the bound for $\operatorname{dist}(\bar{x}_K, \mathcal{X})$ from theorem 3.15, we know that there exists $k_1 \in \mathbb{N}$ such that for all $K \in \mathbb{N}$ with $K \geq k_1$, it holds that

$$\mathbb{E}(\operatorname{dist}(\bar{x}_K, \mathcal{X})) \leq \frac{S_{1,k_1-1}}{S_K}\mathbb{E}(\operatorname{dist}(\bar{x}_{1,k_1-1}, \mathcal{X})) + \frac{\gamma_{k_1}^{-1}\left\|x^\star - x_{k_1}\right\|^2}{\tau S_K} + \frac{\sum_{k=k_1}^K \eta_k^2 \mathcal{O}(\gamma_k)}{\tau S_K} + 2\frac{\sum_{k=k_1}^K \eta_k \alpha_k}{\tau S_K}.$$

The first two terms are constants. Since $\eta_k = \mathcal{O}\left(\left(k^c \log^{(1+3e)/2}(k+1)\right)^{-1}\right)$, it follows that (TODO: Show this with integral test.)

$$S_K = \mathcal{O}\left(\frac{K^{1-c}}{\log^{(1+3e)/2}(K)}\right) \tag{3.14}$$

By lemma 3.11, it holds that $\sum_{k=1}^K \gamma_k \eta_k \alpha_k = \mathcal{O}(1)$ and $\sum_{k=1}^K \eta_k^2 \gamma_k^2 = \mathcal{O}(1)$. Therefore, we obtain

$$\mathbb{E}(\operatorname{dist}(\bar{x}_K, \mathcal{X})) = \mathcal{O}\left(\frac{\log^{(1+3e)/2}(K)}{\tau K^{1-c}}\right).$$

Combining with (3.13), we obtain

$$\mathbb{E}(f(\bar{x}_k) - f(x^\star)) \geq -\left\|\nabla f(x^\star)\right\| \cdot \mathcal{O}\left(\frac{\log^{(1+3e)/2}(K)}{\tau K^{1-c}}\right).$$

For the upper bound in (3.12), we similarly find that

$$\mathbb{E}(f(\bar{x}_K) - f(x^\star)) \leq \mathcal{O}\Big(\frac{\log^{(1+3e)/2}(K)}{K^{1-c}}\Big).$$

Putting the two bounds together, we obtain

$$\mathbb{E}|f(\bar{x}_K) - f(x^\star)| = \max(1, \tau^{-1})\,\mathcal{O}\Big(\frac{\log^{(1+3e)/2}(K)}{K^{1-c}}\Big),$$

as desired. □

Lemma 3.14. In the setting $\mu > 0$, let $\eta_k := 2/(\mu k)$ for all $k \in \mathbb{N}$ and $\eta_0 := 2/\mu$. Then, for any $e \in (0, \infty)$, and any sequences $(\gamma_k)_{k\in\mathbb{N}_0}$, $(h_k)_{k\in\mathbb{N}_0}$ with $\gamma_k = \mathcal{O}(\log^e(k+1))$ and $\alpha_k = \mathcal{O}(1/k)$, assumption 4 holds.

Proof. Fix $e \in (0, \infty)$ and let $\gamma_k = \mathcal{O}(\log^e(k))$, $\alpha_k = \mathcal{O}(1/k)$. Clearly, there exist $\rho \in (0, 1)$ and $K \in \mathbb{N}$ large enough such that $\eta_k \leq \rho L^{-1}$ for all $k \geq K$. It also holds that

1. $\sum_{k=0}^{\infty} \eta_k = \infty$,
2. $\sum_{k=0}^{\infty} \gamma_k^2 \eta_k^2 = \sum_{k=1}^{\infty} \mathcal{O}(\log^{2e}(k)/k^2) < \infty$,
3. $\sum_{k=0}^{\infty} \gamma_k \eta_k \alpha_k = \sum_{k=1}^{\infty} \mathcal{O}(\log^e(k)/k^2) < \infty$.

This proves the claim. □

Theorem 3.15. Let assumptions 1 to 3 hold, and assume that we are in the setting $\mu > 0$. Let the parameters $(\eta_k)_{k\in\mathbb{N}_0}$, $(\gamma_k)_{k\in\mathbb{N}_0}$, $(h_k)_{k\in\mathbb{N}_0}$ be as in lemma 3.14. Further, let the iterate weights $(w_k)_{k\in\mathbb{N}}$ be defined as $w_k := \eta_k^{-1}$, for all $k \in \mathbb{N}$. Then there exists a constant $k_0 \in \mathbb{N}$ such that the averaged iterate x_K, generated by algorithm 2 with parameters $(\eta_k)_{k\in\mathbb{N}_0}$, $(\gamma_k)_{k\in\mathbb{N}_0}$, $(h_k)_{k\in\mathbb{N}_0}$, $(w_k)_{k\in\mathbb{N}}$, satisfies

$$\begin{aligned} f(\bar{x}_K) - f(x^\star) \leq \frac{S_{1,k_0-1}}{S_K}(f(\bar{x}_{1,k_0-1}) - f(x^\star)) + \frac{e_{k_0}}{2S_K} + \frac{\sum_{k=k_0}^{K} \eta_k^{-1}\langle z_k, x_k - x^\star\rangle}{S_K} \\ + \frac{\sum_{k=k_0}^{K} \mathcal{O}(\gamma_k^2)}{S_K} + \frac{\sum_{k=k_0}^{K} \eta_k^{-1}\gamma_k\alpha_k}{S_K}. \end{aligned}$$

almost surely, for all $K \in \mathbb{N}$ with $K \geq k_0$, where $e_{k_0} := \eta_{k_0-1}^{-2} \left\|x^\star - x_{k_0}\right\|^2$ $S_{t,k} := \sum_{i=t}^{k} w_i$, and $\bar{x}_{t,k} := S_{t,k}^{-1} \sum_{i=t}^{k} w_i x_i$ $\forall t, k \in \mathbb{N}$. Furthermore, there exist constants $k_1 \in \mathbb{N}$ and

$\tau \in (0, \infty)$ such that

$$\operatorname{dist}(\bar{x}_K, \mathcal{X}) \leq \frac{S_{1,k_1-1}}{S_K} \operatorname{dist}(\bar{x}_{1,k_1-1}, \mathcal{X}) + \frac{d_{k_1}}{\tau S_K} + \frac{\sum_{k=k_1}^K O(\gamma_k)}{\tau S_K} + 2\frac{\sum_{k=k_1}^K \eta_k^{-1}\alpha_k}{\tau S_K} + 2\frac{\sum_{k=k_1}^K \gamma_k^{-1}\eta_k^{-1}\langle z_k, x_k - \Pi_{\mathcal{X}}(x_k)\rangle}{\tau S_K}$$

almost surely, for all $K \in \mathbb{N}$ with $K \geq k_1$, where $d_{k_1} := \gamma_{k_1}^{-1}\eta_{k_1-1}^{-2} \operatorname{dist}(x_{k_1}, \mathcal{X})^2$.

Proof. Let $k_0 \in \mathbb{N}$ be large enough such that $\eta_{k_0} \leq 1/(2L)$ and fix some $k \in \mathbb{N}$ with $k \geq k_0$. Lemma 3.14 implies that the parameters $(\eta_k)_{k\in\mathbb{N}_0}, (\gamma_k)_{k\in\mathbb{N}_0}, (h_k)_{k\in\mathbb{N}_0}$ satisfy assumption 4. We can apply the exact same steps as in the proof of theorem 3.12, until we arrive at inequality (3.10). From this starting point, we can closely follow the proof stategy of lemma 13 in [3] to arrive at our desired result. First, we will prove the claim for $f(\bar{x}_k) - f(x^\star)$. Dropping the $-\operatorname{dist}(x_k, \mathcal{X})$ term from (3.10), setting $x := x^\star \in \mathcal{X}^\star$, and multiplying both sides by η_k^{-2}, we get

$$2\eta_k^{-1}(f(x_k) - f(x^\star)) \leq \eta_k^{-2}(1 - \mu\eta_k)\left\|x^\star - x_k\right\|^2 - \eta_k^{-2}\left\|x^\star - x_{k+1}\right\|^2 + 2\eta_k^{-1}\langle z_k, x_k - x^\star\rangle + O(\gamma_k^2) + 2\eta_k^{-1}\gamma_k\alpha_k.$$

For the choice of step size $\eta_k = 2/(\mu k)$, it holds that

$$\frac{1 - \mu\eta_k}{\eta_k^2} = \frac{\mu^2 k^2(1 - 2/k)}{4} = \frac{\mu^2(k^2 - 2k)}{4} = \frac{\mu^2((k-1)^2 - 1)}{4} \leq \frac{\mu^2(k-1)^2}{4} = \eta_{k-1}^{-2}.$$

Setting $e_j := \eta_{j-1}^{-2}\left\|x^\star - x_j\right\|^2$ for all $j \in \mathbb{N}$, we thus have

$$2\eta_k^{-1}(f(x_k) - f(x^\star)) \leq e_k - e_{k+1} + 2\eta_k^{-1}\langle z_k, x_k - x^\star\rangle + O(\gamma_k^2) + 2\eta_k^{-1}\gamma_k\alpha_k$$

almost surely, for all $k \in \mathbb{N}$. Summing both sides over $k = k_0, \dots, K$ for $K \in \mathbb{N}$ yields

$$\begin{aligned}
2\sum_{k=k_0}^{K} \eta_k^{-1}(f(x_k) - f(x^\star)) &\le e_{k_0} - e_K + 2\sum_{k=k_0}^{K} \eta_k^{-1}\langle z_k, x_k - x^\star\rangle + 2\sum_{k=k_0}^{K} O(\gamma_k^2) \\
&\quad + 2\sum_{k=k_0}^{K} \eta_k^{-1}\gamma_k\alpha_k \\
&\le e_{k_0} + 2\sum_{k=k_0}^{K} \eta_k^{-1}\langle z_k, x_k - x^\star\rangle + 2\sum_{k=k_0}^{K} O(\gamma_k^2) \\
&\quad + 2\sum_{k=k_0}^{K} \eta_k^{-1}\gamma_k\alpha_k,
\end{aligned}$$

where we used $e_K \ge 0$ in the second step. We define $S_{t,k} := \sum_{i=t}^{k} \eta_i^{-1}$, $S_k := S_{1,k}$, and $\bar{x}_{t,k} := S_{t,k}^{-1}\sum_{i=t}^{k} \eta_i^{-1} x_i$ for $t, k \in \mathbb{N}$. Using convexity of f, we get

$$\begin{aligned}
f(\bar{x}_{k_0,K}) - f(x^\star) &\le S_K^{-1}\sum_{k=k_0}^{K} \eta_k^{-1} f(x_k) - f(x^\star) \\
&\le \frac{e_{k_0}}{2S_{k_0,K}} + \frac{\sum_{k=k_0}^{K} \eta_k^{-1}\langle z_k, x_k - x^\star\rangle}{S_{k_0,K}} + \frac{\sum_{k=k_0}^{K} O(\gamma_k^2)}{S_{k_0,K}} \\
&\quad + \frac{\sum_{k=k_0}^{K} \eta_k^{-1}\gamma_k\alpha_k}{S_{k_0,K}},
\end{aligned}$$

for all $K \in \mathbb{N}$, as desired. Note that

$$\bar{x}_K = \frac{S_{1,k_0-1}}{S_K}\bar{x}_{1,k_0-1} + \frac{S_{k_0,K}}{S_K}\bar{x}_{k_0,K}$$

and

$$\frac{S_{1,k_0-1}}{S_K} + \frac{S_{k_0,K}}{S_K} = 1,$$

hence, using convexity of f again, we have

$$f(\bar{x}_K) - f(x^\star) \le \frac{S_{1,k_0-1}}{S_K}(f(\bar{x}_{1,k_0-1}) - f(x^\star)) + \frac{S_{k_0,K}}{S_K}(f(\bar{x}_{k_0,K}) - f(x^\star)).$$

Combining with the latest bound on $(f(\bar{x}_{k_0,K}) - f(x^\star))$, we arrive at

$$f(\bar{x}_K) - f(x^\star) \leq \frac{S_{1,k_0-1}}{S_K}(f(\bar{x}_{1,k_0-1}) - f(x^\star)) + \frac{e_{k_0}}{2S_K} + \frac{\sum_{k=k_0}^K \eta_k^{-1}\langle z_k, x_k - x^\star\rangle}{S_K} + \frac{\sum_{k=k_0}^K \mathcal{O}(\gamma_k^2)}{S_K} + \frac{\sum_{k=k_0}^K \eta_k^{-1}\gamma_k\alpha_k}{S_K}.$$

Next, we will derive the desired bound for $\operatorname{dist}(\bar{x}_K, \mathcal{X})$. Our starting point is inequality (3.11) from the proof of theorem 3.12. Multiplying both sides by $\gamma_k^{-1}\eta_k^{-2}$ yields

$$\gamma_k^{-1}\eta_k^{-2}\operatorname{dist}(x_{k+1},\mathcal{X})^2 \leq \gamma_k^{-1}\eta_k^{-2}(1-\mu\eta_k)\operatorname{dist}(x_k,\mathcal{X})^2 + \mathcal{O}(\gamma_k) + 2\gamma_k^{-1}\eta_k^{-1}\langle z_k, x_k - \Pi_{\mathcal{X}}(x_k)\rangle + 2\eta_k^{-1}\alpha_k - \tau\eta_k^{-1}\operatorname{dist}(x_k,\mathcal{X}),$$

for all $k \geq k_1$. We have already shown that $\eta_k^{-2}(1-\mu\eta_k) \leq \eta_{k-1}^2$ for all $k \in \mathbb{N}$. Also, since γ_k is nondecreasing, we have $\gamma_k^{-1} \geq \gamma_{k+1}^{-1}$ for all $k \in \mathbb{N}$. Combining these two facts, we get

$$\gamma_{k+1}^{-1}\eta_k^{-2}\operatorname{dist}(x_{k+1},\mathcal{X})^2 \leq \gamma_k^{-1}\eta_{k-1}^2\operatorname{dist}(x_k,\mathcal{X})^2 + \mathcal{O}(\gamma_k) + 2\gamma_k^{-1}\eta_k^{-1}\langle z_k, x_k - \Pi_{\mathcal{X}}(x_k)\rangle + 2\eta_k^{-1}\alpha_k - \tau\eta_k^{-1}\operatorname{dist}(x_k,\mathcal{X}),$$

for all $k \geq k_1$. Setting $d_j := \gamma_j^{-1}\eta_{j-1}^{-2}\operatorname{dist}(x_j,\mathcal{X})^2 \ \forall j \in \mathbb{N}$ and rearranging again, we obtain

$$\tau\eta_k^{-1}\operatorname{dist}(x_k,\mathcal{X}) \leq d_k - d_{k+1} + \mathcal{O}(\gamma_k) + 2\eta_k^{-1}\alpha_k + 2\gamma_k^{-1}\eta_k^{-1}\langle z_k, x_k - \Pi_{\mathcal{X}}(x_k)\rangle,$$

for all $k \geq k_1$. Summing over $k = k_1, \ldots, K$ for $K \in \mathbb{N}$, we have

$$\begin{aligned}\tau\sum_{k=k_1}^K \eta_k^{-1}\operatorname{dist}(x_k,\mathcal{X}) &\leq d_{k_1} - d_{K+1} + \sum_{k=k_1}^K \mathcal{O}(\gamma_k) + 2\sum_{k=k_1}^K \eta_k^{-1}\alpha_k \\ &\quad + 2\sum_{k=k_1}^K \gamma_k^{-1}\eta_k^{-1}\langle z_k, x_k - \Pi_{\mathcal{X}}(x_k)\rangle \\ &\leq d_{k_1} + \sum_{k=k_1}^K \mathcal{O}(\gamma_k) + 2\sum_{k=k_1}^K \eta_k^{-1}\alpha_k \\ &\quad + 2\sum_{k=k_1}^K \gamma_k^{-1}\eta_k^{-1}\langle z_k, x_k - \Pi_{\mathcal{X}}(x_k)\rangle,\end{aligned}$$

where we used $d_{K+1} \geq 0$ in the last step. The distance functional $x \mapsto \operatorname{dist}(x,\mathcal{X})$ is

convex (TODO: Show this in an example.), so if we divide both sides of the above inequality by $\tau \cdot S_{k_1,K}$, we obtain

$$\begin{aligned}\operatorname{dist}(\bar{x}_{k_1,K}, \mathcal{X}) \leq \frac{d_{k_1}}{\tau S_{k_1,K}} + \frac{\sum_{k=k_1}^{K} \mathcal{O}(\gamma_k)}{\tau S_{k_1,K}} + 2\frac{\sum_{k=k_1}^{K} \eta_k^{-1}\alpha_k}{\tau S_{k_1,K}} \\ + 2\frac{\sum_{k=k_1}^{K} \gamma_k^{-1}\eta_k^{-1}\langle z_k, x_k - \Pi_{\mathcal{X}}(x_k)\rangle}{\tau S_{k_1,K}}.\end{aligned}$$

To derive a bound for $\operatorname{dist}(\bar{x}_K, \mathcal{X})$, we proceed similarly as before. We have

$$\bar{x}_K = \frac{S_{1,k_1-1}}{S_K}\bar{x}_{1,k_1-1} + \frac{S_{k_1,K}}{S_K}\bar{x}_{k_1,K}$$

and

$$\frac{S_{1,k_1-1}}{S_K} + \frac{S_{k_1,K}}{S_K} = 1.$$

Combining with the latest bound on $\operatorname{dist}(\bar{x}_{k_1,K}, \mathcal{X})$, we obtain the desired bound

$$\begin{aligned}\operatorname{dist}(\bar{x}_K, \mathcal{X}) &\leq \frac{S_{1,k_1-1}}{S_K}\operatorname{dist}(\bar{x}_{1,k_1-1}, \mathcal{X}) + \frac{S_{k_1,K}}{S_K}\operatorname{dist}(\bar{x}_{k_1,K}, \mathcal{X}) \\ &\leq \frac{S_{1,k_1-1}}{S_K}\operatorname{dist}(\bar{x}_{1,k_1-1}, \mathcal{X}) + \frac{d_{k_1}}{\tau S_K} + \frac{\sum_{k=k_1}^{K} \mathcal{O}(\gamma_k)}{\tau S_K} + 2\frac{\sum_{k=k_1}^{K} \eta_k^{-1}\alpha_k}{\tau S_K} \\ &\quad + 2\frac{\sum_{k=k_1}^{K} \gamma_k^{-1}\eta_k^{-1}\langle z_k, x_k - \Pi_{\mathcal{X}}(x_k)\rangle}{\tau S_K}\end{aligned}$$

almost surely, or all $K \geq k_1$. This concludes the proof. □

Corollary 3.16 (Convergence rates in expectation - strongly convex case)**.** Let assumptions 1 to 3 hold and assume that we are in the setting $\mu > 0$. Let the parameters $(\eta_k)_{k\in\mathbb{N}_0}, (\gamma_k)_{k\in\mathbb{N}_0}$, $(h_k)_{k\in\mathbb{N}_0}$ be the same as in lemma 3.14. Further, let the iterate weights $(w_k)_{k\in\mathbb{N}}$ be defined as $w_k := \eta_k^{-1}$, for all $k \in \mathbb{N}$. Then, the averaged iterate $\bar{x}_K$, generated by algorithm 2 with parameters $(\eta_k)_{k\in\mathbb{N}_0}$, $(\gamma_k)_{k\in\mathbb{N}_0}$, $(h_k)_{k\in\mathbb{N}_0}$, $(w_k)_{k\in\mathbb{N}}$, satisfies

$$\mathbb{E}|f(\bar{x}_K) - f(x^\star)| = \max(1, \tau^{-1})\,\mathcal{O}\Big(\frac{\log^{2e}(K)}{K}\Big).$$

Further, it also holds that

$$\mathbb{E}(\operatorname{dist}(\bar{x}_K, \mathcal{X})) = \tau^{-1}\,\mathcal{O}\Big(\frac{\log^{e}(K)}{K}\Big).$$

Proof. Theorem 3.15 gives us an upper bound for $f(\bar{x}_K) - f(x^\star)$,

$$f(\bar{x}_K) - f(x^\star) \leq \frac{S_{1,k_0-1}}{S_K}(f(\bar{x}_{1,k_0-1}) - f(x^\star)) + \frac{e_{k_0}}{2S_K} + \frac{\sum_{k=k_0}^K \eta_k^{-1}\langle z_k, x_k - x^\star\rangle}{S_K} + \frac{\sum_{k=k_0}^K \mathcal{O}(\gamma_k^2)}{S_K} + \frac{\sum_{k=k_0}^K \eta_k^{-1}\gamma_k\alpha_k}{S_K}.$$

almost surely, for all $K \in \mathbb{N}$ with $K \geq k_0$. Note that

$$\mathbb{E}(\langle z_k, x_k - x^\star\rangle) = \mathbb{E}(\mathbb{E}_k(\langle z_k, x_k - x^\star\rangle)) = \mathbb{E}(\langle \mathbb{E}_k(z_k), x_k - x^\star\rangle) = 0,$$

for all $k \in \mathbb{N}$. Hence, taking expectations on both sides of the above inequality, we get

$$\mathbb{E}(f(\bar{x}_K) - f(x^\star)) \leq \frac{S_{1,k_0-1}}{S_K}\mathbb{E}(f(\bar{x}_{1,k_0-1}) - f(x^\star)) + \frac{\mathbb{E}(e_{k_0})}{2S_K} + \frac{\sum_{k=k_0}^K \mathcal{O}(\gamma_k^2)}{S_K} + \frac{\sum_{k=k_0}^K \eta_k^{-1}\gamma_k\alpha_k}{S_K}. \tag{3.15}$$

By convexity, we have for all $k \in \mathbb{N}$

$$\begin{aligned} f(\bar{x}_k) - f(x^\star) &\geq \langle \tilde{\nabla} f(x^\star), \bar{x}_k - x^\star\rangle \\ &= \langle \tilde{\nabla} f(x^\star), \bar{x}_k - \Pi_{\mathcal{X}}(\bar{x}_k)\rangle + \langle \tilde{\nabla} f(x^\star), \Pi_{\mathcal{X}}(\bar{x}_k) - x^\star\rangle, \end{aligned}$$

for any subgradient $\tilde{\nabla} f(x^\star) \in \partial f(x^\star)$. Optimality of $x^\star$ for f on $\mathcal{X}$ implies that there exists a subgradient $\tilde{\nabla} f(x^\star) \in \partial f(x^\star)$ such that $\langle \tilde{\nabla} f(x^\star), \Pi_{\mathcal{X}}(\bar{x}_k) - x^\star\rangle \geq 0$. Combining these facts with the above inequality, and applying Cauchy-Schwarz, we obtain

$$f(\bar{x}_k) - f(x^\star) \geq -\left\|\tilde{\nabla} f(x^\star)\right\| \operatorname{dist}(\bar{x}_k, \mathcal{X}), \tag{3.16}$$

for all $k \in \mathbb{N}$. Taking expectations in the bound for $\operatorname{dist}(\bar{x}_K, \mathcal{X})$ from theorem 3.15, we know that there exists $k_1 \in \mathbb{N}$ such that for all $K \in \mathbb{N}$ with $K \geq k_1$, it holds that

$$\mathbb{E}(\operatorname{dist}(\bar{x}_K, \mathcal{X})) \leq \frac{S_{1,k_1-1}}{S_K}\mathbb{E}(\operatorname{dist}(\bar{x}_{1,k_1-1}, \mathcal{X})) + \frac{\mathbb{E}(d_{k_1})}{\tau S_K} + \frac{\sum_{k=k_1}^K \mathcal{O}(\gamma_k)}{\tau S_K} + 2\frac{\sum_{k=k_1}^K \eta_k^{-1}\alpha_k}{\tau S_K}.$$

The first two terms are constants. Since $\eta_k^{-1} = (\mu k)/2$, it follows that $S_K = \mathcal{O}(K^2)$.

Also, since $\alpha_k = \mathcal{O}(1/k)$, we have $\eta_k^{-1}\alpha_k = \mathcal{O}(1)$. Therefore,

$$\begin{aligned}\mathbb{E}(\operatorname{dist}(\bar{x}_K, \mathcal{X})) &= \mathcal{O}(K^{-2}) + \mathcal{O}\Big(K^{-2}\sum_{k=1}^{K}\log^e(k)\Big) + \mathcal{O}(K^{-1}) \\ &= \mathcal{O}(\tau^{-1}K^{-2}) + \mathcal{O}(\tau^{-1}\log^e(K)/K) + \mathcal{O}(\tau^{-1}K^{-1}) \\ &= \tau^{-1}\mathcal{O}\Big(\frac{\log^e(K)}{K}\Big).\end{aligned}$$

Combining with (3.16), we obtain

$$\mathbb{E}(f(\bar{x}_k) - f(x^\star)) \geq -\big\|\nabla f(x^\star)\big\| \cdot \mathcal{O}\Big(\frac{\log^{2e}(K)}{\tau K}\Big).$$

Note that $\mathcal{O}(\gamma_k^2) = \mathcal{O}(\log^{2e}(k))$ and $\eta_k^{-1}\gamma_k\alpha_k = \mathcal{O}(\log^e(k))$, so for the upper bound in (3.15), we find that

$$\begin{aligned}\mathbb{E}(f(\bar{x}_K) - f(x^\star)) &\leq \mathcal{O}(K^{-2}) + \mathcal{O}\Big(K^{-2}\sum_{k=1}^{K}\log^{2e}(k)\Big) + \mathcal{O}\Big(K^{-2}\sum_{k=1}^{K}\log^e(k)\Big) \\ &= \mathcal{O}\Big(\frac{\log^{2e}(K)}{K}\Big).\end{aligned}$$

Putting the two bounds together, we obtain

$$\mathbb{E}|f(\bar{x}_K) - f(x^\star)| = \max(1, \tau^{-1})\,\mathcal{O}\Big(\frac{\log^{2e}(K)}{K}\Big),$$

as desired. □

3.3. High-probability guarantees

In the previous section, we proved that the expected values $\mathbb{E}|f(\bar{x}_k) - f(x^\star)|$ and $\operatorname{dist}(\bar{x}_k, \mathcal{X})$ converge to zero as $k \to \infty$, provided the parameters $(\eta_k)_{k\in\mathbb{N}_0}$, $(\gamma_k)_{k\in\mathbb{N}_0}$, $(h_k)_{k\in\mathbb{N}_0}$, and $(w_k)_{k\in\mathbb{N}}$ are suitably chosen. We will now turn our attention towards establishing *high-probability guarantees* for convergence, in contrast to guarantees that hold only in expectation. In particular, we will investigate under which conditions we can guarantee that an iterate $\bar{x}_k$ is both a) close to the feasible set and b) has function value close to the optimal value $f(x^\star)$ with high probability. This notion of "closeness with high probability" is formalized in the following definition.

Definition 3.17 ((ϵ, δ)-solution)**.** Let $\epsilon \in (0, \infty)$ and $\delta \in (0, 1)$. We call a random

variable $x\colon \Omega \to \mathbb{R}^n$ an (ϵ, δ)**-solution** of (P), if

$$\mathbb{P}\Big(\max\big(|f(x) - f(x^\star)|, \operatorname{dist}(x, \mathcal{X})\big) \geq \epsilon\Big) \leq \delta.$$

In other words, an iterate x_k of algorithm 2 is an (ϵ, δ)-solution, if we can guarantee that $|f(x_k) - f(x^\star)| < \epsilon$ and $\operatorname{dist}(x_k, \mathcal{X}) < \epsilon$ with probability greater than $1 - \delta$. We can now state the central question we aim to answer in this section as follows:

For any given (ϵ, δ), how many iterations are needed to guarantee that the sequence $(\bar{x}_k)_{k\in\mathbb{N}}$ generated by algorithm 2 reaches an (ϵ, δ)-solution of (P)*?*

Our approach to answer this question is standard. We rely on the Azuma-Hoeffding inequality (proposition 2.34) and the almost-sure bounds we derived in the previous sections.

Lemma 3.18. Let $(\nu_k)_{k\in\mathbb{N}}$ be an arbitrary sequence of real numbers and let $(Y_k)_{k\in\mathbb{N}}$ be a bounded sequence of random variables such that Y_k is a function of x_k for all $k \in \mathbb{N}$. Then the stochastic process $(X_k)_{k\in\mathbb{N}_0}$, where $X_0 := 0$ and

$$X_k := \sum_{i=1}^{k} \nu_i \langle z_i, Y_i \rangle$$

for $k \in \mathbb{N}$, is a martingale with respect to the filtration $(\mathcal{G}_k)_{k\in\mathbb{N}_0}$ defined by $\mathcal{G}_k := \sigma(z_0, \ldots, z_k)\, \forall k \in \mathbb{N}_0$.

Proof. First, note that Y_k is a function of $z_0, z_1 \ldots, z_{k-1}$ for all $k \in \mathbb{N}$. Indeed, for $k = 1$, we have

$$x_1 = x_0 + \eta_0 g_0 = x_0 + \eta_0(z_0 + \nabla\psi_0(x_0)),$$

and $x_0, \nabla\psi_0(x_0), \eta_0$ are deterministic quantities. Assuming that x_k is a function of $z_0, z_1, \ldots, z_{k-1}$, it follows inductively that

$$x_{k+1} = x_k + \eta_k g_k = x_k + \eta_k(z_k + \nabla\psi_k(x_k))$$

is a function of $z_1, \ldots, z_k$. It follows that Y_k, as a function of x_k, is also a function of $z_0, z_1, \ldots, z_{k-1}$, hence Y_k is $\sigma(z_0, \ldots, z_{k-1})$-measurable. Thus, the process $(X_k)_{k\in\mathbb{N}_0}$ is adapted to the filtration $\mathcal{G}$. With probability one we also have, for all $k \in \mathbb{N}_0$,

$$|X_k| \leq \sum_{i=1}^{k} |\nu_i \langle z_i, Y_i \rangle| \leq \sum_{i=1}^{k} \nu_i \, \|z_i\| \, \|Y_i\| = \sum_{i=1}^{k} \nu_i \mathcal{O}(\gamma_i)\mathcal{O}(1) \leq \nu_k \mathcal{O}(k\gamma_k),$$

where the third step follows from lemma 3.10 and the assumption that $(Y_k)_{k\in\mathbb{N}}$ is bounded. In particular,

$$\mathbb{E}|X_k| < \infty$$

for all $k \in \mathbb{N}_0$. Finally, we have

$$\begin{aligned}
\mathbb{E}(X_k \mid \mathcal{G}_{k-1}) &= \mathbb{E}\big(\nu_k\langle z_k, Y_k\rangle + X_{k-1} \mid \mathcal{G}_{k-1}\big) \\
&= \nu_k\mathbb{E}\big(\langle z_k, Y_k\rangle \mid \mathcal{G}_{k-1}\big) + X_{k-1} \\
&= \nu_k\big\langle \mathbb{E}(z_k \mid \mathcal{G}_{k-1}), Y_k\big\rangle + X_{k-1} \\
&= \nu_k\big\langle \mathbb{E}(z_k \mid x_k), Y_k\big\rangle + X_{k-1} \\
&= X_{k-1},
\end{aligned}$$

where we used that x_k is a function of $z_0, \dots, z_{k-1}$ and, per definition of stochastic gradients, $\mathbb{E}(z_k \mid x_k) = 0$. Therefore, $(X_k)_{k\in\mathbb{N}_0}$ is a martingale. □

Theorem 3.19 (Convergence with high probability - convex case)**.** Let assumptions 1 to 3 hold. Let the parameters $(\eta_k)_{k\in\mathbb{N}_0}, (\gamma_k)_{k\in\mathbb{N}_0}, (h_k)_{k\in\mathbb{N}_0}$ be the same as in lemma 3.11. Further, let the iterate weights $(w_k)_{k\in\mathbb{N}}$ be defined as $w_k := \eta_k$, for all $k \in \mathbb{N}$. Then, for arbitrary $\delta \in (0, \infty)$ and $K \in \mathbb{N}$, the averaged iterate $\bar{x}_K$, generated by algorithm 2 with parameters $(\eta_k)_{k\in\mathbb{N}_0}$, $(\gamma_k)_{k\in\mathbb{N}_0}$, $(h_k)_{k\in\mathbb{N}_0}$, $(w_k)_{k\in\mathbb{N}}$, satisfies the bounds (TODO: Define M and R somewhere, τ too.)

$$|f(\bar{x}_K) - f(x^\star)| \le \delta \max(1, \tau^{-1})\, \mathcal{O}\Big(\frac{MR\log^{(2+3e)/2}(K)}{\sqrt{K}}\Big)$$

and

$$\operatorname{dist}(\bar{x}_K, \mathcal{X}) \le \delta\tau^{-1}\, \mathcal{O}\Big(\frac{MR\log^{(2+3e)/2}(K)}{\sqrt{K}}\Big)$$

with probability at least $1 - \exp(-\delta^2/2)$.

Proof. Let $K \in \mathbb{N}$. From theorem 3.12, we have the almost-sure bound

$$\begin{aligned}
f(\bar{x}_K) - f(x^\star) \le \frac{S_{1,k_0-1}}{S_K}(f(\bar{x}_{1,k_0-1}) - f(x^\star)) + \frac{\|x^\star - x_{k_0}\|^2}{2S_K} + \frac{\sum_{k=k_0}^K \eta_k^2 \mathcal{O}(\gamma_k^2)}{2S_K} \\
+ \frac{\sum_{k=k_0}^K \eta_k\gamma_k\alpha_k}{S_K} + \frac{\sum_{k=k_0}^K \eta_k\langle z_k, x_k - x^\star\rangle}{S_K}.
\end{aligned}$$

By theorem 3.8, the numerators in the first two terms are bounded almost surely, whereas the sum of the last two terms is on the order $\mathcal{O}\big(\log^{(1+3e)/2}(K)/\sqrt{K}\big)$, as argued

in the proof of corollary 3.13. Thus,

$$f(\bar{x}_K) - f(x^\star) \leq \mathcal{O}\Big(\frac{\log^{(1+3e)/2}(K)}{\sqrt{K}}\Big) + \frac{\sum_{k=k_0}^K \eta_k \langle z_k, x_k - x^\star\rangle}{S_K}$$

almost surely. To prove the claim, we will first bound the probability that the random variable

$$\sum_{k=k_0}^K \nu_k \langle z_k, x_k - x^\star\rangle$$

is large, where

$$\nu_k := \frac{\eta_k}{S_K}.$$

The sequence $(x_k)_{k\in\mathbb{N}_0}$ is bounded a. s. by theorem 3.8. Therefore, lemma 3.18 implies that $(X_k)_{k\in\mathbb{N}_0}$ is a martingale, and we also have

$$|X_k - X_{k-1}| = \nu_k |\langle z_k, x_k - x^\star\rangle| \leq \nu_k \|z_k\| \left\|x_k - x^\star\right\| =: c_k$$

for all $k \in \mathbb{N}$. We can thus apply the Azuma-Hoeffding inequality (proposition 2.34) to obtain

$$\mathbb{P}\Big(\sum_{k=1}^K \nu_k \langle z_k, x_k - x^\star\rangle \geq t\Big) \leq \exp\Big(\frac{-t^2}{2\sum_{k=1}^K c_k^2}\Big),$$

for all $t \in (0,\infty)$. By lemma 3.10, it holds that $\|z_k\| \leq M \log^e(k)\, \forall k \in \mathbb{N}_0$, where $M := 4\max(1, \min_{k\in\mathbb{N}_0} \gamma_k)\max(L_{\text{loc}}, G)$. Further, using (3.14), we have

$$\nu_k = \frac{\eta_k}{S_K} = \mathcal{O}\Big(\frac{1}{\log^{(1+3e)/2}(k)\sqrt{k}S_K}\Big) = \mathcal{O}\Big(\frac{\log^{(1+3e)/2}(K)}{\sqrt{k}\sqrt{K}}\Big).$$

for all $k \in \{1, \dots K\}$. Let $R \in (0,\infty)$ be large enough such that $\sup_{k\in\mathbb{N}} \|x_k - x^\star\| \leq R$ almost surely. We obtain

$$c_k = \nu_k \|z_k\| \left\|x_k - x^\star\right\| \leq \mathcal{O}\Big(\frac{MR\log^{(1+3e)/2}(K)}{\sqrt{k}\sqrt{K}}\Big),$$

which implies

$$\sum_{k=1}^K c_k^2 \leq \mathcal{O}\Big(\frac{M^2R^2\log^{1+3e}(K)}{K}\Big)\sum_{k=1}^K \frac{1}{k} = \mathcal{O}\Big(\frac{M^2R^2\log^{2+3e}(K)}{K}\Big)$$

almost surely, where the last step follows from

$$\sum_{k=1}^{K} \frac{1}{k} = \mathcal{O}(\log(K)).$$

Hence, it holds that

$$\mathbb{P}\Big(\sum_{k=1}^{K} \nu_k \langle z_k, x_k - x^\star \rangle \geq t\Big) \leq \exp\left(-\frac{1}{2} t^2 \mathcal{O}\Big(\frac{K}{M^2 R^2 \log^{2+3e}(K)}\Big)\right),$$

for all $t \in (0, \infty)$. Substituting $t := \delta\, \mathcal{O}(MR \log^{(2+3e)/2}(K)/\sqrt{K})$ for arbitrary $\delta \in (0,1)$, we obtain

$$\mathbb{P}\left(\sum_{k=1}^{K} \nu_k \langle z_k, x_k - x^\star \rangle \geq \delta\, \mathcal{O}\Big(\frac{MR \log^{(2+3e)/2}(K)}{\sqrt{K}}\Big)\right) \leq \exp\Big(-\frac{1}{2}\delta^2\Big).$$

Therefore,

$$f(\bar{x}_K) - f(x^\star) \leq \mathcal{O}\Big(\frac{\log^{2e}(K)}{K}\Big) + \delta\, \mathcal{O}\Big(\frac{MR \log^{(2+3e)/2}(K)}{\sqrt{K}}\Big) \tag{3.17}$$

with probability at least $1 - \exp(-\delta^2/2)$. To derive a lower bound, we will first prove the second claim, which follows from essentially the same argument. For $k \in \mathbb{N}$, define

$$\tilde{\nu}_k := \frac{\gamma_k^{-1} \eta_k}{\tau S_K}$$

and

$$\tilde{X}_k := \sum_{k=1}^{K} \tilde{\nu}_k \langle z_k, x_k - \Pi_{\mathcal{X}}(x_k) \rangle,$$

as well as $\tilde{X}_0 := 0$. The process $(x_k - \Pi_{\mathcal{X}}(x_k))_{k \in \mathbb{N}}$ is a function of x_k, and bounded by theorem 3.8. Hence, applying lemma 3.18, we find that $(\tilde{X}_k)_{k \in \mathbb{N}_0}$ is a martingale. Further, we have

$$|\tilde{X}_k - \tilde{X}_{k-1}| = \tilde{\nu}_k |\langle z_k, x_k - \Pi_{\mathcal{X}}(x_k) \rangle| \leq \tilde{\nu}_k \|z_k\| \operatorname{dist}(x_k, \mathcal{X}) =: d_k,$$

for all $k \in \mathbb{N}$. It holds that

$$\tilde{\nu}_k = \frac{\gamma_k^{-1} \eta_k}{\tau S_K} \leq (\tau \gamma_0)^{-1} \frac{\eta_k}{S_K} = (\tau \gamma_0)^{-1} \nu_k \leq (\tau \gamma_0)^{-1} \mathcal{O}\Big(\frac{\log^{(1+3e)/2}(K)}{k^{1/2} K^{1/2}}\Big),$$

for all $k \in \mathbb{N}$. Letting $\tilde{R} \in (0, \infty)$ be large enough such that $\sup_{k\in\mathbb{N}} \operatorname{dist}(x_k, \mathcal{X}) \leq \tilde{R}$ a. s., we thus have

$$d_k \leq (\tau\gamma_0)^{-2}\, \mathcal{O}\Big(\frac{M\tilde{R}\log^{(1+3e)/2}(K)}{k^{1/2}K^{1/2}}\Big),$$

for all $k \in \mathbb{N}$. It follows that

$$\sum_{k=1}^{K} d_k^2 \leq (\tau\gamma_0)^{-2}\mathcal{O}\Big(\frac{M^2\tilde{R}^2\log^{1+3e}(K)}{K}\Big)\sum_{k=1}^{K}\frac{1}{k} = (\tau\gamma_0)^{-2}\mathcal{O}\Big(\frac{M^2\tilde{R}^2\log^{2+3e}(K)}{K}\Big).$$

Applying the Azuma-Hoeffding inequality (proposition 2.34), we obtain

$$\begin{aligned}\mathbb{P}\Big(\sum_{k=1}^{K}\tilde{v}_k\langle z_k, x_k - \Pi_{\mathcal{X}}(x_k)\rangle \geq t\Big) &\leq \exp\Big(\frac{-t^2}{2\sum_{k=1}^{K} d_k^2}\Big)\\ &\leq \exp\Bigg(-\frac{1}{2}t^2\mathcal{O}\Big(\frac{\tau_k^2\gamma_0^2K}{2M^2\tilde{R}^2\log^{2+3e}(K)}\Big)\Bigg),\end{aligned}$$

for all $t \in (0, \infty)$ and $K \in \mathbb{N}$. Substituting $t := \delta\,\mathcal{O}(M\tilde{R}\log^{(2+3e)/2}(K)/\tau\gamma_0\sqrt{K})$ for arbitrary $\delta \in (0, \infty)$, we obtain

$$\mathbb{P}\Bigg(\sum_{k=1}^{K}\tilde{v}_k\langle z_k, x_k - \Pi_{\mathcal{X}}(x_k)\rangle \geq \delta(\tau\gamma_0)^{-1}\mathcal{O}\Big(\frac{M\tilde{R}\log^{(2+3e)/2}(K)}{\sqrt{K}}\Big)\Bigg) \leq \exp\Big(-\frac{1}{2}\delta^2\Big).$$

for all $K \in \mathbb{N}$. We therefore have, for any $\delta \in (0, \infty)$ and $K \in \mathbb{N}$,

$$\begin{aligned}\operatorname{dist}(\bar{x}_K, \mathcal{X}) &\leq \tau^{-1}\,\mathcal{O}\Big(\frac{\log^{(1+3e)/2}(K)}{\sqrt{K}}\Big) + 2\delta(\tau\gamma_0)^{-1}\mathcal{O}\Big(\frac{M\tilde{R}\log^{(2+3e)/2}(K)}{\sqrt{K}}\Big)\\ &= \delta\tau^{-1}\,\mathcal{O}\Big(\frac{M\tilde{R}\log^{(2+3e)/2}(K)}{\sqrt{K}}\Big)\end{aligned} \tag{3.18}$$

with probability at least $1 - \exp(-\delta^2/2)$. Note that for any valid choice of R, we can choose $\tilde{R}$ such that $\tilde{R} \leq R$, which brings us to the desired bound. We can derive a lower bound on $f(\bar{x}_K) - f(x^\star)$ via

$$\begin{aligned}f(\bar{x}_K) - f(x^\star) &\geq \langle\tilde{\nabla}f(x^\star), \bar{x}_K - x^\star\rangle\\ &= \langle\tilde{\nabla}f(x^\star), \bar{x}_K - \Pi_{\mathcal{X}}(\bar{x}_K)\rangle + \langle\tilde{\nabla}f(x^\star), \Pi_{\mathcal{X}}(\bar{x}_K) - x^\star\rangle\\ &\geq \langle\tilde{\nabla}f(x^\star), \bar{x}_K - \Pi_{\mathcal{X}}(\bar{x}_K)\rangle,\end{aligned}$$

where the first step follows from convexity, and the second from optimality of $x^\star$ on

$\mathcal{X}$. Cauchy-Schwarz now implies

$$f(\bar{x}_K) - f(x^\star) \geq -\left\|\tilde{\nabla} f(x^\star)\right\| \operatorname{dist}(\bar{x}_K, \mathcal{X}).$$

Combining with the high-probability bound (3.18), we have, for any $\delta \in (0, \infty)$,

$$f(\bar{x}_K) - f(x^\star) \geq -\tau^{-1} \left\|\tilde{\nabla} f(x^\star)\right\| \delta \tau^{-1} \mathcal{O}\Big(\frac{M\tilde{R}\log^{(2+3e)/2}(K)}{\sqrt{K}}\Big)$$

with probability at least $1 - \exp(-\delta^2/2)$. Note that we can choose $\tilde{R} \leq R$ without loss of generality. Hence, by combining with the upper bound (3.17), we have for any $\delta \in (0, \infty)$ and all $K \in \mathbb{N}$,

$$|f(\bar{x}_K) - f(x^\star)| \leq \max(1, \tau^{-1}) \mathcal{O}\Big(\frac{MR\log^{(2+3e)/2}(K)}{\sqrt{K}}\Big)$$

with probability at least $1 - \exp(-\delta^2/2)$, as desired. □

Corollary 3.20. Let assumptions 1 to 3 hold. Let the parameters $(\eta_k)_{k\in\mathbb{N}_0}$, $(\gamma_k)_{k\in\mathbb{N}_0}$, $(h_k)_{k\in\mathbb{N}_0}$ be the same as in lemma 3.11. Further, let the iterate weights $(w_k)_{k\in\mathbb{N}}$ be defined as $w_k := \eta_k$, for all $k \in \mathbb{N}$. Then, there exists a constant $c \in (0, \infty)$ such that, for any $\epsilon \in (0, \infty)$, $\delta \in (0, 1)$, and all $K \in \mathbb{N}$ with

$$\frac{K}{\log^{2+3e}(K)} \geq c\frac{2\log(\delta^{-1})}{\epsilon^2\tau^2},$$

the output $\bar{x}_K$ of algorithm 2 is an (ϵ, δ)-solution of (P).

Theorem 3.21 (Convergence with high probability - strongly convex case)**.** Let assumptions 1 to 3 hold and assume that we are in the setting $\mu > 0$. Let the parameters $(\eta_k)_{k\in\mathbb{N}_0}$, $(\gamma_k)_{k\in\mathbb{N}_0}$, $(h_k)_{k\in\mathbb{N}_0}$ be the same as in lemma 3.14. Further, let the iterate weights $(w_k)_{k\in\mathbb{N}}$ be defined as $w_k := \eta_k^{-1}$, for all $k \in \mathbb{N}$. Additionally, we define $a_{\tau^{-1}} := \max\left(2\left\|\tilde{\nabla} f(x^\star)\right\|(\tau\gamma_0)^{-1}, 1\right)$. Then, for arbitrary $\delta \in (0, \infty)$ and $K \in \mathbb{N}$, the averaged iterate $\bar{x}_K$, generated by algorithm 2 with parameters $(\eta_k)_{k\in\mathbb{N}_0}$, $(\gamma_k)_{k\in\mathbb{N}_0}$, $(h_k)_{k\in\mathbb{N}_0}$, $(w_k)_{k\in\mathbb{N}}$, satisfies the bounds

$$|f(\bar{x}_K) - f(x^\star)| \leq \tau^{-1}\mathcal{O}\Big(\frac{\log^{2e}(K)}{K}\Big) + \delta a_{\tau^{-1}} \frac{MR\log^e(K)}{\sqrt{K}}$$

and

$$\operatorname{dist}(\bar{x}_K, \mathcal{X}) \leq \tau^{-1}\mathcal{O}\Big(\frac{\log^e(K)}{K}\Big) + 2\delta\tau^{-1}\gamma_0^{-1}\frac{MR\log^e(K)}{\sqrt{K}}$$

with probability at least $1 - \exp(-\delta^2/2)$.

Proof. Let $K \in \mathbb{N}$. From theorem 3.15, we have the almost-sure bound

$$f(\bar{x}_K) - f(x^\star) \leq \frac{S_{1,k_0-1}}{S_K}(f(\bar{x}_{1,k_0-1}) - f(x^\star)) + \frac{e_{k_0}}{2S_K} + \frac{\sum_{k=k_0}^K \eta_k^{-1}\langle z_k, x_k - x^\star\rangle}{S_K} + \frac{\sum_{k=k_0}^K O(\gamma_k^2)}{S_K} + \frac{\sum_{k=k_0}^K \eta_k^{-1}\gamma_k\alpha_k}{S_K}.$$

By theorem 3.8, the numerators in the first two terms are bounded almost surely, whereas the sum of the last two terms is on the order $O(\log^{2e}(K)/K)$, as argued in the proof of corollary 3.16. Thus,

$$f(\bar{x}_K) - f(x^\star) \leq O\Big(\frac{\log^{2e}(K)}{K}\Big) + \frac{\sum_{k=k_0}^K \eta_k^{-1}\langle z_k, x_k - x^\star\rangle}{S_K}$$

almost surely. To prove the claim, we will first bound the probability that the random variable

$$\sum_{k=k_0}^K \nu_k\langle z_k, x_k - x^\star\rangle$$

is large, where

$$\nu_k := \frac{\eta_k^{-1}}{S_K}.$$

The sequence $(x_k)_{k\in\mathbb{N}_0}$ is bounded a. s. by theorem 3.8. Therefore, lemma 3.18 implies that $(X_k)_{k\in\mathbb{N}_0}$ is a martingale, and we also have

$$|X_k - X_{k-1}| = \nu_k|\langle z_k, x_k - x^\star\rangle| \leq \nu_k \|z_k\| \left\|x_k - x^\star\right\| =: c_k$$

for all $k \in \mathbb{N}$. We can thus apply the Azuma-Hoeffding inequality (proposition 2.34) to obtain

$$\mathbb{P}\Big(\sum_{k=1}^K \nu_k\langle z_k, x_k - x^\star\rangle \geq t\Big) \leq \exp\Big(\frac{-t^2}{2\sum_{k=1}^K c_k^2}\Big),$$

for all $t \in (0,\infty)$. By lemma 3.10, it holds that $\|z_k\| \leq M\log^e(k)\,\forall k \in \mathbb{N}_0$, where $M := 4\max(1, \min_{k\in\mathbb{N}_0}\gamma_k)\max(L_{\text{loc}}, G)$. Further, we have

$$\nu_k = \frac{\eta_k^{-1}}{S_K} = \frac{k}{\sum_{i=1}^K i} = \frac{k}{K(K+1)} \leq \frac{1}{K},$$

for all $k \in \{1, \dots K\}$. Let $R \in (0,\infty)$ be large enough such that $\sup_{k\in\mathbb{N}}\|x_k - x^\star\| \leq R$

almost surely. We obtain

$$c_k = \nu_k \|z_k\| \left\|x_k - x^\star\right\| \leq \frac{MR\log^e(K)}{K},$$

which implies

$$\sum_{k=1}^{K} c_k^2 \leq \sum_{k=1}^{K} \frac{M^2R^2\log^{2e}(K)}{K^2} = \frac{M^2R^2\log^{2e}(K)}{K}$$

almost surely. Hence, it holds that

$$\mathbb{P}\Big(\sum_{k=1}^{K} \nu_k\langle z_k, x_k - x^\star\rangle \geq t\Big) \leq \exp\Big(\frac{-t^2K}{2M^2R^2\log^{2e}(K)}\Big),$$

for all $t \in (0,\infty)$. Substituting $t := \delta MR\log^e(K)/\sqrt{K}$ for arbitrary $\delta \in (0,1)$, we obtain

$$\mathbb{P}\Big(\sum_{k=1}^{K} \nu_k\langle z_k, x_k - x^\star\rangle \geq \delta\frac{MR\log^e(K)}{\sqrt{K}}\Big) \leq \exp\Big(-\frac{1}{2}\delta^2\Big).$$

Therefore,

$$f(\bar{x}_K) - f(x^\star) \leq O\Big(\frac{\log^{2e}(K)}{K}\Big) + \delta\frac{MR\log^e(K)}{\sqrt{K}} \tag{3.19}$$

with probability at least $1-\exp(-\delta^2/2)$. To derive a lower bound, we will first prove the second claim, which follows from essentially the same argument. For $k \in \mathbb{N}$, define

$$\tilde{\nu}_k := \frac{\gamma_k^{-1}\eta_k^{-1}}{\tau S_K}$$

and

$$\tilde{X}_k := \sum_{k=1}^{K} \tilde{\nu}_k\langle z_k, x_k - \Pi_{\mathcal{X}}(x_k)\rangle,$$

as well as $\tilde{X}_0 := 0$. The process $(x_k - \Pi_{\mathcal{X}}(x_k))_{k\in\mathbb{N}}$ is a function of x_k, and bounded by theorem 3.8. Hence, applying lemma 3.18, we find that $(\tilde{X}_k)_{k\in\mathbb{N}_0}$ is a martingale. Further, we have

$$|\tilde{X}_k - \tilde{X}_{k-1}| = \tilde{\nu}_k|\langle z_k, x_k - \Pi_{\mathcal{X}}(x_k)\rangle| \leq \tilde{\nu}_k \|z_k\| \operatorname{dist}(x_k, \mathcal{X}) =: d_k,$$

for all $k \in \mathbb{N}$. It holds that

$$\tilde{\nu}_k = \frac{\gamma_k^{-1}\eta_k^{-1}}{\tau S_K} \leq (\tau\gamma_0)^{-1}\frac{\eta_k^{-1}}{S_K} = (\tau\gamma_0)^{-1}\nu_k \leq (\tau\gamma_0)^{-1}\frac{1}{K},$$

for all $k \in \mathbb{N}$. Letting $\tilde{R} \in (0, \infty)$ be large enough such that $\sup_{k\in\mathbb{N}} \operatorname{dist}(x_k, \mathcal{X}) \leq \tilde{R}$ a. s., we thus have

$$d_k \leq \frac{M\tilde{R}\log^e(K)}{\tau\gamma_0 K},$$

for all $k \in \mathbb{N}$. Applying the Azuma-Hoeffding inequality (proposition 2.34), we obtain

$$\mathbb{P}\Big(\sum_{k=1}^{K} \tilde{v}_k\langle z_k, x_k - \Pi_{\mathcal{X}}(x_k)\rangle \geq t\Big) \leq \exp\Big(\frac{-t^2}{2\sum_{k=1}^{K} d_k^2}\Big) \leq \exp\Big(\frac{-t^2\tau^2\gamma_0^2 K}{2M^2\tilde{R}^2\log^{2e}(K)}\Big),$$

for all $t \in (0, \infty)$ and $K \in \mathbb{N}$. Substituting $t := \delta M\tilde{R}\log^e(K)/\tau\gamma_0\sqrt{K}$ for arbitrary $\delta \in (0, \infty)$, we obtain

$$\mathbb{P}\Big(\sum_{k=1}^{K} \tilde{v}_k\langle z_k, x_k - \Pi_{\mathcal{X}}(x_k)\rangle \geq \delta(\tau\gamma_0)^{-1}\frac{M\tilde{R}\log^e(K)}{\sqrt{K}}\Big) \leq \exp\Big(-\frac{1}{2}\delta^2\Big).$$

for all $K \in \mathbb{N}$. We therefore have, for any $\delta \in (0, \infty)$ and $K \in \mathbb{N}$,

$$\operatorname{dist}(\bar{x}_K, \mathcal{X}) \leq \tau^{-1}\mathcal{O}\Big(\frac{\log^e(K)}{K}\Big) + 2\delta(\tau\gamma_0)^{-1}\frac{M\tilde{R}\log^e(K)}{\sqrt{K}} \tag{3.20}$$

with probability at least $1 - \exp(-\delta^2/2)$. Note that for any valid choice of R, we can choose $\tilde{R}$ such that $\tilde{R} \leq R$, which brings us to the desired bound. We can derive a lower bound on $f(\bar{x}_K) - f(x^\star)$ via

$$\begin{aligned} f(\bar{x}_K) - f(x^\star) &\geq \langle\tilde{\nabla}f(x^\star), \bar{x}_K - x^\star\rangle \\ &= \langle\tilde{\nabla}f(x^\star), \bar{x}_K - \Pi_{\mathcal{X}}(\bar{x}_K)\rangle + \langle\tilde{\nabla}f(x^\star), \Pi_{\mathcal{X}}(\bar{x}_K) - x^\star\rangle \\ &\geq \langle\tilde{\nabla}f(x^\star), \bar{x}_K - \Pi_{\mathcal{X}}(\bar{x}_K)\rangle, \end{aligned}$$

where the first step follows from convexity, and the second from optimality of $x^\star$ on $\mathcal{X}$. Cauchy-Schwarz now implies

$$f(\bar{x}_K) - f(x^\star) \geq -\big\|\tilde{\nabla}f(x^\star)\big\|\operatorname{dist}(\bar{x}_K, \mathcal{X}).$$

Combining with the high-probability bound (3.20), we have, for any $\delta \in (0, \infty)$,

$$f(\bar{x}_K) - f(x^\star) \geq -\tau^{-1}\big\|\tilde{\nabla}f(x^\star)\big\|\Big(\mathcal{O}\Big(\frac{\log^e(K)}{K}\Big) + 2\delta\gamma_0^{-1}\frac{M\tilde{R}\log^e(K)}{\sqrt{K}}\Big)$$

with probability at least $1 - \exp(-\delta^2/2)$. Note that we can choose $\tilde{R} \leq R$ without loss

of generality. Hence, by combining with the upper bound (3.19), we have for any $\delta \in (0, \infty)$ and all $K \in \mathbb{N}$,

$$|f(\bar{x}_K) - f(x^\star)| \leq \tau^{-1} O\left(\frac{\log^{2e}(K)}{K}\right) + \delta \max\left(2\left\|\tilde{\nabla} f(x^\star)\right\|(\tau\gamma_0)^{-1}, 1\right)\frac{MR\log^e(K)}{\sqrt{K}}$$

with probability at least $1 - \exp(-\delta^2/2)$, as desired. □

Corollary 3.22. Let assumptions 1 to 3 hold. Let the parameters $(\eta_k)_{k\in\mathbb{N}_0}, (\gamma_k)_{k\in\mathbb{N}_0}$, $(h_k)_{k\in\mathbb{N}_0}$ be the same as in lemma 3.11. Further, let the iterate weights $(w_k)_{k\in\mathbb{N}}$ be defined as $w_k := \eta_k^{-1}$, for all $k \in \mathbb{N}$. Then, there exists a constant $c \in (0, \infty)$ such that, for any $\epsilon \in (0, \infty)$, $\delta \in (0, 1)$, and all $K \in \mathbb{N}$ with

$$\frac{K}{\log^{2e}(K)} \geq c\frac{2\log(\delta^{-1})}{\epsilon^2\tau^2},$$

the output $\bar{x}_K$ of algorithm 2 is an (ϵ, δ)-solution of (P).

3.4. Infeasible problems

As we have seen in the SVM example (TODO), some problems of interest may not be feasible. Yet, our methods can still be applied in those cases. The question is then: What do the iterates converge to, if anything?

Definition 3.23. Let $\delta \in [0, 1]$. A point $x \in \mathbb{R}^n$ is called δ**-feasible**, if

$$\mathbb{P}(A(\xi)x - b(\xi) > 0) \leq \delta.$$

The δ**-set**, denoted $\mathcal{X}_\delta$, is the set of all δ-feasible points. A point $x \in \mathbb{R}^n$ is called **maximally feasible**, if there exists $\delta \in [0, 1]$ such that (x, δ) solves

$$\min_{(x,\delta)\in\mathbb{R}^n\times[0,1]} \delta \quad \text{s.t. } x \in \mathcal{X}_\delta.$$

Conjecture: Consider the following two statements:

1. For any $\delta \in (0, 1]$, the sequence of iterates $(x_k)_{k\in\mathbb{N}}$ is eventually contained in $\mathcal{X}_\delta$ in probability.

2. The sequence of iterates $(x_k)_{k\in\mathbb{N}}$ converges to a maximally feasible point in probability.

At least one of these two statements must hold, regardless of whether (P) is feasible or not. Both statements hold iff. problem (P) is feasible.

4

Numerical Examples

5

Summary and Outlook

Future research directions:

1. Acceleration a la https://proceedings.mlr.press/v54/lei17a/lei17a.pdf;
2. Non-convex settings;
3. Functional constraints;
4. Non-stationary distribution of ζ and ξ (online setting);
5. Adaptive sampling to reduce effective support of constraints.

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Amir Miri Lavasani